Parity equidistribution of nested floor powers, with descent applications to the Juggler map

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Abstract

For odd n set $m = \lfloor n^{3/2} \rfloor$, $v = \lfloor m^{3/2} \rfloor$, and so on: the integer chains obtained by iterating $x \mapsto \lfloor x^{3/2} \rfloor$ and $x \mapsto \lfloor \sqrt{x} \rfloor$ in a prescribed pattern. We prove complete depth-4 parity equidistribution for odd-rooted itineraries, with power savings: every pattern of depth at most four over odd starts receives its expected share. The obstacle at depth two and beyond is that the naive expansion of $m^{3/2}$ leaves the sawtooth $\{n^{3/2}\}$ with an amplitude that grows like $n^{3/4}$, which defeats the classical van der Corput method. The proof reorganizes the phase by an exact linearization — $m^{3/2} = \frac{3}{2}mn^{3/4} - \frac{1}{2}n^{9/4} + O(n^{-3/4})$ with a one-signed remainder, a Taylor rewrite that keeps the integer m linear with a smooth coefficient — and funnels the deeper patterns into a single *kernel*: the exponential sum of the level-2 floor defect $\{\lfloor n^{3/2} \rfloor^{3/2}\}$ against the monomial weight $c = \frac{3k}{4}n^{9/8}$. The fundamental obstruction inside that kernel is the exact level-2 wave $e(q\lfloor n^{3/2} \rfloor^{3/2})$ riding a frozen floor; Lemma 5.2 bounds these mixed pieces by $|q|^{-1/6}P^{23/24+\varepsilon}$, uniformly enough for depth four, and the central result (Theorem 5.3), $K_c \ll P^{1-1/96+\varepsilon}$, follows by double Weyl differencing over an exact carry-branch decomposition and master identity. The depth-≤ 3 discrepancy estimates (Theorems 4.4 and 4.7) are also proved on sub-dyadic intervals of length $\geq P^{1/2}$, with a slowly varying twist attached (Section 3.5), which is the form a companion paper needs. The kernel theorem itself remains a dyadic-block statement.

The sequences arise as the itineraries of the Juggler map $J(n) = \lfloor \sqrt{n} \rfloor$ (n even), $\lfloor n^{3/2} \rfloor$ (n odd), whose finite exact theory is developed in a companion manuscript [22]. As a corollary, the class of starts carrying a uniform power-envelope descent certificate of length at most five has certificate density 7/8; the four-step subclass has certificate density 13/16. An unconditional counting argument shows that parity equidistribution at *all* depths would give density one to the set of starts with some finite descent certificate. None of these is a density of starts that reach 1, and no statement about the Juggler conjecture itself is claimed. What is solved here is the first genuinely nested layers of the parity process, not the infinite-depth problem: the remaining obstacle is the level-3 kernel, where the weight scale $n^{27/16}$ exceeds n; we state it as an open problem and record precisely what blocks every method of this paper, and what the depth-4 results do and do not buy for the termination problem.

1. Introduction

Equidistribution questions for a *single* floor of a smooth function — the Piatetski–Shapiro tradition — are classical and well developed. This paper concerns the *composition* of floors:

$$m = \lfloor n^{3/2} \rfloor, \qquad v = \lfloor m^{3/2} \rfloor, \qquad w = \lfloor m^{1/2} \rfloor, \qquad z = \lfloor v^{3/2} \rfloor, \ \ldots$$

and, specifically, the joint distribution of the *parities* along such chains. Fix a pattern (an *itinerary word*) $w_1 w_2 \cdots w_d$ over the alphabet $\{E, O\}$, where an O applies $x \mapsto \lfloor x^{3/2} \rfloor$ and an E applies $x \mapsto \lfloor \sqrt{x} \rfloor$. The question is whether the 2^d parity classes of the resulting chain equidistribute over odd starts $n \leq N$, with a power saving. Odd starts have $w_1 = O$; the theorems of this paper concern the O-rooted words. (Even starts are a separate and easier problem: one square root drops the state to scale $\sqrt{N}$, and we make no claims about them.)

The difficulty is genuinely one of composition. Writing $\theta = \{n^{3/2}\}$, the naive expansion $m^{3/2} = n^{9/4} - \frac{3}{2}\theta n^{3/4} + O(n^{-3/4})$ leaves a sawtooth of *growing* amplitude $n^{3/4}$ inside the phase. No estimate for exponential sums with smooth monomial phases applies; the phase is not smooth, and the sawtooth cannot be discarded. All results in the single-floor literature stop before this point (Section 1.2).

Two ideas carry the paper.

1. **Exact linearization** (Lemma 4.3). The identity $m^{3/2} = \frac{3}{2}mn^{3/4} - \frac{1}{2}n^{9/4} + E(n)$ with $0 \le E(n) \le \frac{1}{2}n^{-3/4}$ eliminates the inner floor *exactly*: the integer m enters linearly, multiplied by a smooth coefficient, and the only non-smooth object left is m itself, which is handled by differencing and by an exact gap-cell decomposition. The identity is a Taylor rewrite, not a discovery; what carries the paper is the package built on it — one-signed remainders at every level, gap cells, the carry-branch decomposition, and a kernel with weight $n^{9/8}$. Iterated along the chain, this is what makes depth ≥ 2 accessible.

2. **The level-2 wave and the kernel theorem** (Lemma 5.2, Theorem 5.3). Every reorganization of the depth-4 pattern $OOO*$ funnels into one object, the exponential sum of the level-2 floor defect

$$K_c(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e\big(c(n)\,\{\lfloor n^{3/2} \rfloor^{3/2}\}\big), \qquad c(n) = \tfrac{3k}{4} n^{9/8}.$$

   After two Weyl differencings over an exact carry-branch decomposition, the master identity (Lemma 5.1) leaves no growing smooth part; what remains is bounded carries and, hardest, exact level-2 waves $e(qY)$, $Y = \lfloor n^{3/2} \rfloor^{3/2}$, possibly riding a frozen floor. The depth-2 nested-floor wave is the fundamental obstruction of the whole paper, and Lemma 5.2 is the result that overcomes it: a bound $|q|^{-1/6} P^{23/24+\varepsilon}$ for these mixed pieces by a targeted third differencing, exactly depth-2 strength, which is what pins the kernel saving at 1/96. We then prove $K_c(P) \ll P^{1-1/96+\varepsilon}$ for $c = \frac{3k}{4} n^{9/8}$, uniformly for $1 \le k \le P^{1/24}$ (Theorem 5.3). This is the hardest result of the paper, and Section 4 is written at full length — every estimate displayed with its constant — so that it can be checked without reference to anything outside this manuscript.

With the kernel theorem, depth-4 parity equidistribution for odd-rooted itineraries is complete (Theorem 6.1): the eight O-rooted length-4 words each receive their expected share with a power saving. The same estimates, applied to one further letter, count the two length-five contractors $OOOEE$ and $OOEOE$ (Theorem 6.3), so the certified descent class has certificate density 7/8 (Corollary 6.4). The leftover eighth is the expanding length-five tree $OOEOO \cup OOOEO \cup OOOO*$: the first two of those itineraries are counted and do not contract, and $OOOO*$ is the level-3 kernel. Section 3.5 proves the depth-≤ 3 theorems on sub-dyadic intervals of length $\ge P^{1/2}$ with a slowly varying twist attached, the form in which the companion paper [24] uses them. Theorem 5.3 is not localized.

The logical dependence of the counting theorems is

Lemma 5.2(i) $\implies$ Lemma 5.2(ii) $\implies$ Theorem 5.3 $\implies$ Theorem 6.1 $\implies$ Theorem 6.3 and Corollary 6.4.

The localization used by the companion [24] is a separate chain and does not include Theorem 5.3:

Theorems 4.11–4.12 and Corollary 4.13 (Section 3.5) $\implies$ companion [24].

The chains are not chosen at random: they are the itineraries of the Juggler map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even}, \\ \lfloor n^{3/2} \rfloor, & n \text{ odd}, \end{cases}$$

introduced by Pickover [1] (OEIS A094683 [2]); universal arrival at 1 is open. The map is niche; the nested-floor sums must stand on their own, and the paper is organized so that they do. The exact finite-itinerary calculus of J — the power envelope $J^{|w|}(n)^{2^{|w|}} \le n^{3^{\#O(w)}}$, the defect identity, and a small-cycle census — is a companion manuscript [22]; here we use only the contraction criterion (Proposition 3.1), whose short induction is written out below. The dynamical payoff of the counting theorems is a pair of *certified-descent densities*: the set of starts guaranteed to drop below their starting value within four steps has certificate density 13/16 (Corollary 4.9), and the two length-five contractors raise that class to certificate density 7/8 (Corollary 6.4). Equidistribution at all depths would give the set of starts with *some* finite descent certificate density one (Proposition 7.1). We state plainly what these corollaries are not: they are not densities of starts that reach 1, and they do not touch the Juggler conjecture, whose analogue of Terras's almost-all theorem for Collatz [4, 5, 6] remains open; see Lagarias [3] for the Collatz survey. A second companion paper [24] shows what the termination problem needs from parity statistics — control at depth of order $\log\log n$, not at any fixed depth — and Section 8 records what the fixed-depth results of this paper do and do not contribute to it.

Section 7 states the frontier precisely. The uncounted piece of the leftover eighth is the *OOOO*$*$ split, one nesting deeper than Theorem 5.3, where the same kernel reappears with weight scale $n^{27/16} > n$ (Conjecture 7.3). Every method of this paper stops below that scale. What survives is a clean model problem — the amplitude-product sums $\sum e(A(t)\{B(t)\})$ with $1 \ll A' \ll A$ — for which a direct L^2 computation in the shift of the fractional argument gives square-root cancellation, up to a $\sqrt{\log}$ factor, for almost every shift (Proposition 7.4). That average says nothing about the single deterministic shift; the deterministic instance (Conjecture 7.5) is open, and we record in three sentences why the obvious shortcuts fail.

1.1 Verification

Every theorem below is a human proof from the stated classical inequalities. No numerical computation is a proof step. A companion repository records machine checks of the exact identities and a row-by-row audit ledger (paper_b_audit_ledger.md); nothing in this paper depends on it. Certificate densities are never densities of starts that reach 1. The reconstruction of Lemma 5.2 is the author's, not an independent verification.

1.2 Related work

Single floors. The distribution of $\lfloor n^c \rfloor$ ($c > 1$, $c \notin \mathbb{N}$) in arithmetic progressions and among primes goes back to Piatetski–Shapiro [12]; Leitmann [13] treated general smooth $\lfloor f(n) \rfloor$, and the prime exponent for $\lfloor n^c \rfloor$ has a long refinement history, e.g. Rivat–Sargos [14] and Rivat–Wu [15]. All of these concern a single floor of a smooth function; the analytic core is an exponential sum with smooth monomial phase, estimated by van der Corput's method in the form presented by Graham–Kolesnik [11].

Digital and parity properties of $\lfloor n^c \rfloor$. Mauduit–Rivat [16] established q-multiplicative equidistribution along $\lfloor n^c \rfloor$ for c in an explicit range, Morgenbesser [17] proved the analogous statements for the sum of digits of $\lfloor n^c \rfloor$ in residue classes, and Müllner–Spiegelhofer [23] proved normality of the Thue–Morse sequence along $\lfloor n^c \rfloor$ for $1 < c < 3/2$. All of these estimate sums of the shape $\sum e(f(\lfloor n^c \rfloor))$ where f is a *digital* (automatic or q-additive) function: the outer function is handled by its own carry structure, and the analytic core is again a smooth-phase sum over the single floor. Parity of $\lfloor n^{3/2} \rfloor$ — our depth-1 statement, Theorem 4.1 — is the simplest instance of this circle and is classical; we include the short proof to fix constants and notation. Our outer function is not digital: it is a second convex power, and its fractional defect enters later phases with polynomially growing amplitude, which is a different regime.

Compositions and generalized polynomials. Two adjacent theories compose floors and do not cover our sums. For compositions of *Beatty* sequences $\lfloor \alpha \lfloor \beta n \rfloor \rfloor$, the inner floor error is bounded and the composition is again a bounded-perturbation linear sequence, so the linear theory applies. Bergelson–Leibman's generalized polynomials [7] — expressions built from polynomials by iterated floors, sums, and products — admit a complete distributional theory via nilmanifolds; but that theory is confined to maps generated by *polynomials*, and $n^{3/2}$ is not one: no nilsystem models $\{f(\lfloor g(n) \rfloor)\}$ for convex non-polynomial powers f, g, and the growing-amplitude sawtooth that drives Sections 3–6 has no analogue there. We are unaware of a published power-saving estimate for $\sum e(\xi\{f(\lfloor g(n) \rfloor)\})$-type sums, or for the fractional parts $\{f(\lfloor g(n) \rfloor)\}$, with f, g convex powers and growing ξ.

Arithmetic of Piatetski–Shapiro sequences. Baker, Banks, Brüdern, Shparlinski, and Weingartner [18] study squarefree values, prime factors, and Carmichael numbers along $\lfloor n^c \rfloor$; Glasscock [21] studies linear equations whose unknowns range over Piatetski–Shapiro sequences. These results intersect a Piatetski–Shapiro sequence with other arithmetic sets; none composes two floors.

Beatty sequences. For the linear case $\lfloor \alpha n \rfloor$ the discrepancy theory is governed by the continued fraction of α (three-distance/Ostrowski structure); see Abercrombie [19] and, for character sums along Beatty sequences, Banks–Shparlinski [20]. The linear case admits exact self-similar structure that the convex case $n^{3/2}$ does not.

What is not covered. We are unaware of a published equidistribution or parity result for *nested* floor powers $\lfloor \lfloor n^c \rfloor^d \rfloor$ with $c, d > 1$, or of a treatment of the level-2 defect sums $K_c(P)$ of Section 4. We do not make the novelty of this paper depend on that survey. What is new here, independently of the literature, is a specific object and a specific bound: the level-2 defect kernel $K_c(P) = \sum_{n \sim P \text{ odd}} e(c(n)\{\lfloor n^{3/2} \rfloor^{3/2}\})$ with smooth weights of scale $kP^{9/8}$, and the estimate $K_c(P) \ll P^{1-1/96+\varepsilon}$ uniformly for $k \le P^{1/24}$ (Theorem 5.3), resting on the mixed-piece bound for exact level-2 waves (Lemma 5.2). Both are falsifiable statements about explicit sums. The obstruction they address is structural, not incremental: after one floor the argument of the second floor is an integer sequence, not a smooth function, and its fractional defect enters later phases with growing amplitude. The individual devices are not new — the linearization of Lemma 4.3 is a Taylor rewrite, and the carry bookkeeping of Lemma 5.1 will be recognized by anyone who has differenced $\lfloor f(n) \rfloor$; what we believe is new is the package (one-signed remainders, gap cells, the master identity, and the kernel bound) and the results it yields. A literature check was last refreshed in September 2026; we would welcome pointers to anything missed.

Dynamical context. For the Collatz map, Terras [4] and Everett [5] proved that almost every start has finite stopping time, and Tao [6] proved that almost all orbits attain almost bounded values; Lagarias [3] surveys the problem. These are methodological cousins — density statements through parity-word counting — and motivate Proposition 7.1; they prove nothing about J, whose branches are floor powers rather than affine maps.

Standard tools. The van der Corput estimates are used in the form presented by Graham–Kolesnik [11]; see also Iwaniec–Kowalski [9, Ch. 8]. The indicator expansions use Vaaler's extremal functions [10], and the discrepancy bridge is the Erdős–Turán inequality in the form of Kuipers–Niederreiter [8].

2. Exact linearization and the parity bridge

Throughout, n denotes an odd integer, $e(t) = e^{2\pi i t}$, $\{x\}$ the fractional part, and $\psi(x) = (-1)^{\lfloor x \rfloor}$. On a dyadic block $n \in (P, 2P]$ we write $n \sim P$. For the chains,

$$X = n^{3/2}, \quad m = \lfloor X \rfloor, \quad \theta = X - m; \qquad Y = m^{3/2}, \quad v = \lfloor Y \rfloor, \quad \theta_2 = Y - v;$$

and on even m, $U = m^{1/2}$, $w = \lfloor U \rfloor$, $\theta_w = U - w$. The same three letters are reused at later even states (Theorem 4.8 at m, Lemma 6.2 at v). An itinerary word of depth d at n is the sequence of parities $\mathrm{word}_d(n)$ of $n, J(n), \dots, J^{d-1}(n)$. We write $\mathrm{word}(n)$ for the infinite itinerary, and say it has prefix w when $\mathrm{word}_{|w|}(n) = w$.

Proposition 3.1 (power-envelope contraction; companion [22]). If the itinerary w is realized at $n \ge 2$ (the orbit's parities are the letters of w) and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Write $k = |w|$ and $m = J^k(n)$. We first prove the envelope $m^{2^k} \le n^{3^{\#O(w)}}$, by induction on prefix length. The empty prefix is $n \le n$. Suppose a realized prefix of length ℓ with odd count o ends at x and satisfies $x^{2^\ell} \le n^{3^o}$, and the next letter is realized. If that letter is even, then $J(x)^2 \le x$, so $J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \le x^{2^\ell} \le n^{3^o}$. If it is odd, then $J(x)^2 \le x^3$, so $J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \le x^{3 \cdot 2^\ell} = (x^{2^\ell})^3 \le (n^{3^o})^3 = n^{3^{o+1}}$. This is the envelope at w. The exponent gap and $n \ge 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, hence $m^{2^k} < n^{2^k}$. Since $m \ge 1$, one has $m < n$. $\square$

The companion [22] develops the same envelope as a finite-itinerary identity with an exact defect; only the contraction criterion is used below. The induction is recorded here so that the criterion does not depend on an unpublished text.

An itinerary w with $3^{\#O(w)} < 2^{|w|}$ is *contracting*; realizing a contracting itinerary is a *descent certificate* of length $|w|$. Contracting words used below: E, OE, $OOEE$, and the two length-five words $OOOEE$ and $OOEOE$. Every even start realizes E, and every odd start with even image realizes OE; those two certificates cover all starts except the odd-to-odd class, which is where the counting problem lives. Longer contracting itineraries exist (length seven and eight); this paper makes no counting claims about them.

Lemma 3.2 (parity bridge). $\lfloor x\rfloor$ is odd if and only if $\{x/2\} \geq \frac{1}{2}$.

Proof. Write $x = \lfloor x\rfloor + \{x\}$ and split by the parity of $\lfloor x\rfloor$. $\square$

Lemma 3.3 (van der Corput, second-derivative form [11]). Let f be twice differentiable on an interval of length M, with $\lambda \leq |f''| \leq \alpha\lambda$ for some $\alpha \geq 1$. Then

$$\Big|\sum e(f)\Big| \ll_\alpha M\lambda^{1/2} + \lambda^{-1/2},$$

the sum over the integers of the interval.

The Weyl A-process used in Theorems 4.4 and 5.3 and in Lemma 5.2 is the following form, applied to a unimodular sequence on the odd integers of a block of length $\asymp P$ (equivalently, in the r-variable of Lemma 3.10, with shift h in r equal to shift $2h$ in n). For $1 \leq H \leq P$,

$$\Big|\sum a_n\Big|^2 \leq \frac{2P^2}{H} + \frac{4P}{H}\sum_{1\leq h<H}\Big|\sum a_{n+2h}\overline{a_n}\Big|.$$

This is the classical inequality $|\sum a_n|^2 \leq \frac{P+2H}{H}\sum_{|h|<H}(1-\frac{|h|}{H})\sum a_{n+h}\overline{a_n}$ with the even-shift restriction and with the weights $1-|h|/H \leq 1$ absorbed in the absolute constants. Every later appeal to "the classical inequality" is this display.

Lemma 3.4 (Erdős–Turán [8]). The discrepancy of R points $x_j \in \mathbb{R}/\mathbb{Z}$ against an interval satisfies, for every $H \geq 1$,

$$D \ll \frac{R}{H} + \sum_{h=1}^{H}\frac{1}{h}\Big|\sum_{j\leq R} e(hx_j)\Big|.$$

Lemma 3.5 (Vaaler [10]). For every $J \geq 1$ there are trigonometric polynomials $V_J(t) = \sum_{0<|q|\leq J} a_q e(qt)$ with $|a_q| \leq \min(1, \frac{2}{|q|})$ and a nonnegative trigonometric polynomial $\Delta_J \geq 0$ of degree J with constant term and coefficients at most $1/(J+1)$, such that the period-2 square wave satisfies $\psi(x) = V_J(x/2) + O(\Delta_J(x/2))$. For products, $|\psi_1\psi_2 - V^{(1)}V^{(2)}| \leq \Delta^{(1)} + \Delta^{(2)} + \Delta^{(1)}\Delta^{(2)}$, and each error term is again a nonnegative trigonometric polynomial of the same shape.

The class indicators below are products of half-wave factors $\frac{1}{2}(1\pm\psi)$ evaluated along *formal* chains — chains whose branches are dictated by the target itinerary, not by the orbit. The next lemma is the exact identity that justifies this; it replaces any appeal to sampled verification.

Lemma 3.6 (branch consistency). Let $d \geq 1$ and let $w = w_1 \cdots w_d$ be an itinerary with $w_1 = O$. For odd n, define the formal chain $x_1 = n$ and, for $1 \leq t < d$,

$$\xi_t = \begin{cases} x_t^{3/2}, & w_t = O,\\ x_t^{1/2}, & w_t = E,\end{cases} \qquad x_{t+1} = \lfloor \xi_t\rfloor,$$

and set $\sigma_t = +1$ if $w_t = E$, $\sigma_t = -1$ if $w_t = O$. Then, for every odd n,

$$\prod_{t=1}^{d-1}\tfrac{1}{2}\big(1+\sigma_{t+1}\,\psi(\xi_t)\big) = [\mathrm{word}_d(n) = w].$$

Proof. Induct on d. For $d = 1$ both sides are 1. Let Π_{d-1} be the product over $t \leq d-2$; by induction $\Pi_{d-1} = [\mathrm{word}_{d-1}(n) = w_1 \cdots w_{d-1}]$. If $\Pi_{d-1} = 0$, both sides vanish, since every factor lies in $[0,1]$. If

$\Pi_{d-1} = 1$, then the orbit realizes $w_1 \cdots w_{d-1}$, and a second induction along the chain gives $x_t = J^{t-1}(n)$ for $t \le d-1$: assuming $x_t = J^{t-1}(n)$, the true parity of x_t is w_t, so J applies exactly the w_t-branch and $J^t(n) = \lfloor \xi_t \rfloor = x_{t+1}$. Hence ξ_{d-1} is the true real image with $\lfloor \xi_{d-1} \rfloor = J^{d-1}(n)$, and the last factor $\frac{1}{2}(1 + \sigma_d \psi(\xi_{d-1}))$ equals 1 exactly when the parity of $J^{d-1}(n)$ is w_d, else 0. □

Each $\psi(\xi_t)$ is defined for *every* odd n, which is what permits the unconditional Vaaler expansion of the product: off the cylinder, the vanishing of an earlier factor kills the term, and on the cylinder every factor computes the true letter.

The last three preliminaries are the remaining analytic tools of Section 4: a finite Fourier expansion for sawtooth phases whose coefficient may exceed 1; a second-derivative test for two-term monomial phases in which the curvature transition — the short set where the two curvatures cancel — receives a trivial bound; and a sublevel splitting for three-term monomial phases, again with the transition set destined for a trivial bound. No third- or fourth-derivative exponential-sum test is used anywhere in this paper: on the short cells of Section 4 those tests do not sum, and the transition sets are cheaper to measure than to oscillate over. A final remark fixes the parity reindexing once and for all.

Lemma 3.7 (finite Fourier expansion with a shifted window). Let $B \in \mathbb{R}$, and let T, J be integers with $J \ge 1$ and $T \ge 8(1 + |B|)$. There exist coefficients $(b_u)_{|u| \le T}$ with

$$|b_u| \le \min\Big(2, \tfrac{1}{\pi|u+B|}\Big) + \min\Big(2, \tfrac{1}{\pi|u|}\Big), \qquad \sum_{|u| \le T} |b_u| \le 8 + 2\log(2 + |B| + T),$$

and coefficients $(v_q)_{0<|q| \le J}$ with $|v_q| \le \min(2, 2\pi|B|)/|q|$, such that for every $t \in \mathbb{R}$

$$\Big| e(-B\{t\}) - \sum_{|u| \le T} b_u e(ut) - \sum_{0<|q| \le J} v_q e(qt) \Big| \le \min(2, 2\pi|B|)\, \Delta_J(t) + \frac{8(1+|B|)}{T},$$

where $\Delta_J \ge 0$ is a trigonometric polynomial of degree J with constant term and coefficients at most $1/(J+1)$.

Proof. Write $f(t) = e(-B\{t\})$ and $\sigma(t) = \frac{1}{2} - \{t\}$, the sawtooth with jump $+1$ at the integers. The jump of f at the integers is $1 - e(-B)$, so $g := f - (1 - e(-B))\,\sigma$ is continuous, 1-periodic, and piecewise C^1. Since $|f| = 1$, $|\sigma| \le \frac{1}{2}$, and $|1 - e(-B)| = 2|\sin \pi B| \le \min(2, 2\pi|B|) \le 2$, we have $|g| \le 2$, hence $|\widehat{g}(0)| \le 2$. For $u \ne 0$, direct integration gives $\widehat{f}(u) = (1 - e(-B))/(2\pi i(u + B))$ when $u + B \ne 0$ and $\widehat{\sigma}(u) = 1/(2\pi i u)$, so

$$\widehat{g}(u) = \frac{1 - e(-B)}{2\pi i}\Big(\frac{1}{u+B} - \frac{1}{u}\Big) = -\frac{(1 - e(-B))\, B}{2\pi i\, u(u+B)}.$$

Two bounds follow. If $\pi|u+B| \ge \frac{1}{2}$, the first form gives $|\widehat{g}(u)| \le \frac{2}{2\pi}\big(\frac{1}{|u+B|} + \frac{1}{|u|}\big) = \frac{1}{\pi|u+B|} + \frac{1}{\pi|u|}$; if $\pi|u+B| < \frac{1}{2}$, then $|\widehat{g}(u)| \le |\widehat{f}(u)| + |1 - e(-B)||\widehat{\sigma}(u)| \le 1 + \frac{1}{\pi|u|} \le 2 + \frac{1}{\pi|u|}$. In both cases (and when $u + B = 0$, where $|\widehat{f}(u)| \le 1$ directly)

$$|\widehat{g}(u)| \le \min\Big(2, \tfrac{1}{\pi|u+B|}\Big) + \min\Big(2, \tfrac{1}{\pi|u|}\Big),$$

and summing this over $|u| \le T$ — at most four terms hit each of the two caps, and the harmonic tails integrate to logarithms — gives $\sum_{|u| \le T} |\widehat{g}(u)| \le 8 + 2\log(2 + |B| + T)$. The second form gives, for $|u| > T \ge 8(1 + |B|)$ (so $|u + B| \ge |u| - |B| \ge |u|/2$),

$$|\widehat{g}(u)| \le \frac{2|B|}{2\pi} \cdot \frac{2}{u^2} = \frac{2|B|}{\pi u^2}, \qquad \sum_{|u| > T} |\widehat{g}(u)| \le \frac{4|B|}{\pi T} \le \frac{8(1 + |B|)}{T}.$$

The full series $\sum_u |\widehat{g}(u)|$ therefore converges, and since g is continuous its Fourier series converges to g everywhere; take $b_u = \widehat{g}(u)$ for $|u| \le T$ and charge the tail to the flat error. For the sawtooth part, Vaaler's theorem [10] provides a trigonometric polynomial $V_J^*(t) = \sum_{0<|q| \le J} a_q^* e(qt)$ with $|a_q^*| \le 1/|q|$ and $|\sigma(t) - V_J^*(t)| \le \Delta_J(t)$ with Δ_J as stated; multiply by $1 - e(-B)$ and set $v_q = (1 - e(-B))a_q^*$. □

When the lemma is applied with $t = G(n)$ along a block, the Δ_J-term is a nonnegative majorant: its sum over the block is $P/(J+1)$ plus J-truncated mode sums of G with coefficients $\le 1/(J+1)$, and the flat term contributes $8(1+|B|)P/T$. Both costs are displayed at each application below.

Lemma 3.8 (two-term monomial test with a trivial transition bound). Let $E \subset \mathbb{Q} \cap [-4,4]$ be a fixed finite set disjoint from $\{0,1,2,3\}$, let $\alpha \ne \beta$ lie in E, let $I \subseteq (P,2P]$, and let

$$f(n) = a\,n^{\alpha} + b\,n^{\beta} + g(n), \qquad M := \max(|a|P^{\alpha-2},\, |b|P^{\beta-2}) \le 1,$$

where the perturbation satisfies, for every $n \in I$,

$$|g''(n)| \le \rho\,(|a|n^{\alpha-2} + |b|n^{\beta-2}), \qquad |g'''(n)| \le \rho\,(|a|n^{\alpha-3} + |b|n^{\beta-3}),$$

with $\rho \le \rho_0(E)$ sufficiently small in terms of E alone. Then

$$\Big|\sum_{n\in I} e(f(n))\Big| \ll_E |I|\,M^{1/2} + M^{-1/2} + (P/M)^{1/3}.$$

(If $M \le P^{-2}$ the third term exceeds $|I|$ and the bound is trivial; the content is the range $M > P^{-2}$.)

Proof. Write $A(n) = a\,\alpha(\alpha-1)\,n^{\alpha-2}$ and $B(n) = b\,\beta(\beta-1)\,n^{\beta-2}$, so that

$$f'' = A + B + g'', \qquad n\,f''' = (\alpha-2)A + (\beta-2)B + n\,g'''.$$

On the block, $|A(n)| \asymp_E |a|P^{\alpha-2}$ and $|B(n)| \asymp_E |b|P^{\beta-2}$. The ratio $|A/B| = |a\alpha(\alpha-1)/(b\beta(\beta-1))|\,n^{\alpha-\beta}$ is strictly monotone in n, so for any threshold $K = K(E) \ge 3$ the block splits into at most three consecutive intervals: I_1 where $|A/B| \ge K$, I_2 where $|A/B| \le 1/K$, and a middle I_0. (If a or b vanishes, only I_1 or I_2 is present and the argument shortens accordingly.)

On I_1: $|f''| \ge (1-\frac{1}{K})|A| - |g''| \ge \frac{1}{2}|A| \gg_E M$-scale, with $\sup|f''|/\inf|f''| \le C(E)$; Lemma 3.3 gives $\ll_E |I|M^{1/2} + M^{-1/2}$. Symmetrically on I_2.

On I_0, $|A| \asymp_K |B|$. If A and B have the same sign there, $|f''| \ge |A|+|B|-|g''| \ge |A|$ and Lemma 3.3 applies as before. If they have opposite signs, write $B = -sA$ with $s = s(n) \in [1/K, K]$, monotone in n. Then

$$f'' = A\,(1-s) + g'', \qquad n\,f''' = A\,((\alpha-2) - s(\beta-2)) + n\,g'''.$$

The two affine functions $s \mapsto 1-s$ and $s \mapsto (\alpha-2) - s(\beta-2)$ have distinct zeros: $s = 1$ and $s = (\alpha-2)/(\beta-2) \ne 1$ (here $\alpha \ne \beta$ and $\beta \ne 2$ are used). Hence

$$c_6(E) := \min_{s>0}\ \max(|1-s|,\ |(\alpha-2) - s(\beta-2)|) > 0.$$

Since $s(n)$ is monotone, the set where $|1-s(n)| \ge c_6$ is the complement of a single interval, so I_0 splits into at most three consecutive intervals: two on which $|f''| \ge \frac{1}{2}c_6|A|$ (after absorbing g by $\rho_0(E) \le c_6/8$) — Lemma 3.3 as before — and one transition interval J on which $|nf'''| \ge \frac{1}{2}c_6|A|$ throughout, i.e. $|f'''| \ge c_8(E)\,M/P$ in normalized scale. On J the second derivative is strictly monotone. Let $V \in (0,M]$ be a parameter: the set $\{n \in J : |f''(n)| < V\}$ is a single interval of length at most $2V/(c_8M/P)$, which receives the trivial bound, and its at most two flanking intervals carry $|f''| \ge V$ of a single sign with $\sup|f''| \le C(E)M$, so Lemma 3.3 (applied at the piece's own infimum) gives $\ll_E |I|M^{1/2} + V^{-1/2}$ for each. Choosing $V = (c_8M/P)^{2/3}$ equalizes $V^{-1/2}$ and the trivial cost at $\ll_E (P/M)^{1/3}$. Summing the $O_E(1)$ contributions proves the lemma. □

The proof's only content is that the second and third derivatives of a two-term monomial phase cannot both be small on the same subinterval — the linear system with matrix $\left(\begin{smallmatrix}1 & 1\\ \alpha-2 & \beta-2\end{smallmatrix}\right)$ is invertible — so the transition set where the second-derivative test fails is short, and *measuring* it costs less than any attempt to oscillate over it: crucially, the transition cost $(P/M)^{1/3}$ carries no factor $|I|$, so it sums over many short cells. All applications in Section 4 use exponent pairs from $E = \{\frac{3}{4}, \frac{5}{4}, \frac{11}{8}, \frac{3}{2}, \frac{15}{8}\}$. One corner of Section 4 (three curvature scales meeting on a zero-offset branch) needs the three-term analogue, which we state directly as a sublevel bound: there the transition set is measured once against a global smooth model, and only the second-derivative test is ever applied.

Lemma 3.9 (three-term monomial sublevel splitting). Let E be as in Lemma 3.8 and let $\alpha, \beta, \gamma \in E$ be pairwise distinct, $I \subseteq (P, 2P]$, and

$$f(n) = a\, n^{\alpha} + b\, n^{\beta} + c\, n^{\gamma} + g(n), \qquad S := \max(|a|P^{\alpha-2}, |b|P^{\beta-2}, |c|P^{\gamma-2}) > 0,$$

with g satisfying the analogues of the perturbation bounds of Lemma 3.8 for the second, third, *and fourth* derivatives, at some $\rho \leq \rho_0(E)$. There are constants $c_7(E) > 0$ and $C(E)$ such that for every V with $0 < V \leq c_7 S/2$:

(i) the sublevel set $\Omega_V = \{n \in I : |f''(n)| \leq V\}$ is a union of at most $C(E)$ intervals of total length

$$|\Omega_V| \leq C(E) \Big(\frac{PV}{S} + P\Big(\frac{V}{S}\Big)^{1/2} \Big);$$

(ii) $I \setminus \Omega_V$ is a union of at most $C(E)$ intervals, on each of which f'' is single-signed with $V \leq |f''| \leq C(E)\, S$. If one of a, b, c vanishes the claim reduces to Lemma 3.8; if two vanish, to Lemma 3.3. The application in Theorem 5.3, Step 5b, uses the three-term statement when a window mode is present and the two-term reduction when $w = 0$.

Proof. Write A, B, C for the three curvature terms $a\alpha(\alpha-1)n^{\alpha-2}$ etc., so that

$$f'' = A + B + C + g'', \quad n f''' = (\alpha{-}2)A + (\beta{-}2)B + (\gamma{-}2)C + n g''',$$

$$n^2 f'''' = (\alpha{-}2)(\alpha{-}3)A + (\beta{-}2)(\beta{-}3)B + (\gamma{-}2)(\gamma{-}3)C + n^2 g''''.$$

The coefficient matrix is the Vandermonde-type matrix in the distinct values $\alpha{-}2, \beta{-}2, \gamma{-}2$ (rows 1, x, $x(x{-}1)$), hence invertible with inverse bounded in terms of E: at every point of I,

$$\max(|f''|, |nf'''|, |n^2 f''''|) \geq c_7(E) \max(|A|, |B|, |C|) \geq c_7(E)\, \tilde{S}, \qquad \tilde{S} \asymp_E S,$$

after absorbing g by $\rho_0 \leq c_7/8$ (zero coefficients only shorten the argument). The ratios A/B, B/C, A/C are monotone, so I splits into $O_E(1)$ consecutive intervals on each of which one fixed derivative order $r \in \{2, 3, 4\}$ is good at full scale throughout. On an $r = 2$ piece, Ω_V is empty (as $V \leq c_7 S/2$). On an $r = 3$ piece, $|f'''| \geq c_7 S/(2P)$ makes f'' strictly monotone, so Ω_V meets it in a single interval of length $\leq 4PV/(c_7 S)$. On an $r = 4$ piece, $|f''''| \geq c_7 S/(4P^2)$ makes f''' strictly monotone, so f'' has at most one interior critical point and Ω_V meets the piece in at most two intervals; if $[x, y] \subseteq \Omega_V$, the second-difference identity $f''(x) - 2f''(\frac{x+y}{2}) + f''(y) = f''''(\xi)\,(y-x)^2/4$ forces $(y-x)^2 \leq 16V \cdot 4P^2/(c_7 S)$, i.e. $y - x \leq 8P\,(V/(c_7 S))^{1/2}$. Summing the $O_E(1)$ pieces gives (i). For (ii): on the complement, f'' is continuous with $|f''| \geq V$, hence single-signed on each maximal interval, and $|f''| \leq |A| + |B| + |C| + |g''| \leq C(E)S$. For the triple $(\frac{5}{4}, \frac{11}{8}, \frac{3}{2})$ of Theorem 5.3, Step 5b, the inverse matrix is $\left(\begin{smallmatrix} 10 & 68 & 32 \\ -24 & -144 & -64 \\ 15 & 76 & 32 \end{smallmatrix}\right)$; the norm the argument needs — the ℓ^∞ operator norm, i.e. the maximal absolute row sum, since $(A, B, C)^{\mathsf{T}} = M^{-1}(f'', nf''', n^2 f'''')^{\mathsf{T}}$ — is 232, and its ℓ^1 operator norm (maximal absolute column sum) is 288. One may therefore take $c_7 = 1/232$; we keep the weaker value $c_7 = 1/288$ used in Step 5b, which remains valid. $\square$

The lemma is a measure statement, not an exponential-sum estimate: in its one application (Theorem 5.3, Step 5b) the sublevel set receives the trivial bound and the complement the second-derivative test, with the parameter V chosen there.

Lemma 3.10 (parity reindexing). All counting sums below run over odd n. The substitution $n = 2r + 1$ maps a block interval of length L onto an interval of length $L/2$ and a phase f to $f^*(r) = f(2r + 1)$ with $(f^*)^{(k)}(r) = 2^k f^{(k)}(2r + 1)$. Consequently:

(a) any two terms of a composite phase have the same curvature *ratios* and the same *signs* in the r-variable as in the n-variable, so every dominance margin and single-sign check displayed below is invariant under the substitution;

(b) if $\lambda \le |f''| \le \alpha\lambda$ on an interval of length L, then Lemma 3.3 applied in the r-variable (length $L/2$, curvature window $[4\lambda, 4\alpha\lambda]$) gives

$$\Big|\sum_{n\in I,\ n \text{ odd}} e(f(n))\Big| \ll_\alpha \tfrac{L}{2}\,(4\lambda)^{1/2} + (4\lambda)^{-1/2} \le L\lambda^{1/2} + \lambda^{-1/2}:$$

the n-variable display dominates the true reindexed bound, and the same check for Lemma 3.8 replaces its conclusion by a smaller quantity;

(c) sublevel and transition sets (Lemmas 3.8–3.9) are measured in the n-line, and a set of total length Λ contains at most $\Lambda/2 + C(E)$ odd integers, so trivial bounds displayed in n-length dominate.

Every application of Lemmas 3.3 and 3.7–3.9 in Sections 3–6 is to be read through this substitution; no displayed constant below needs adjustment.

3. Depths one to three

For odd n write $s(n) = \psi(n^{3/2}) = (-1)^m$ and $S_O(N) = \sum_{n\le N,\ n \text{ odd}} s(n)$. Let $M(N)$ be the number of odd $n \le N$ and $\mathrm{OO}(N) = \#\{n \le N : n \text{ odd},\ J(n) \text{ odd}\}$, so $S_O(N) = M(N) - 2\,\mathrm{OO}(N)$. By Lemma 3.2, with $g(r) = \frac{1}{2}(2r+1)^{3/2}$, $s(2r+1) = -1$ iff $\{g(r)\} \ge \frac{1}{2}$, and $S_O(N)$ is (twice) an interval discrepancy of the sequence $\{g(r)\}$.

Theorem 4.1 (depth one; classical). $|S_O(N)| \ll N^{5/6}$.

Proof. This is the standard van der Corput estimate for the monomial g, included for completeness. $g''(r) = \frac{3}{2}(2r+1)^{-1/2}$ is positive and decreasing. Partition $1 \le r < R$ into dyadic blocks $[M, \min(2M, R))$; on a block $g'' \asymp M^{-1/2}$, so for the h-th mode $f = hg$ has $\lambda \asymp hM^{-1/2}$ and Lemma 3.3 gives $|\sum_{r\asymp M} e(hg(r))| \ll h^{1/2}M^{3/4} + h^{-1/2}M^{1/4}$. Lemma 3.4 with $H = \lfloor M^{1/6}\rfloor$ bounds the block's discrepancy by $\ll M^{5/6}$, and the dyadic sum is $O(R^{5/6}) = O(N^{5/6})$. $\square$

Corollary 4.2 (density of the odd-to-odd class). $\big|\mathrm{OO}(N) - N/4\big| \ll N^{5/6}$. Equivalently, the starts covered by the uniform one- or two-step certificates (E and OE) have natural density $3/4$.

This is a density of a certificate class, not of starts that reach 1. Theorem 4.1 concerns consecutive source intervals; it supplies no transfer theorem for orbit samples or sparse image sets, so it cannot by itself control the odd-to-odd dynamics after the first step. The rest of the paper does not transfer it; it attacks the nested sums directly, by removing the inner floors exactly before any analytic step.

Lemma 4.3 (exact linearization and gap cells). (i) For every odd $n \ge 3$,

$$m^{3/2} = \tfrac{3}{2}\,m\,n^{3/4} - \tfrac{1}{2}\,n^{9/4} + E(n), \qquad 0 \le E(n) \le \tfrac{3}{8}\,(n^{3/2}-1)^{-1/2} \le \tfrac{1}{2}\,n^{-3/4}.$$

(ii) For $h \ge 1$ and odd n, let $g(n) = m(n+2h) - m(n)$ and $\delta(n) = (n+2h)^{3/2} - n^{3/2}$. Then

$$g(n) = \lfloor \delta(n)\rfloor + \kappa(n), \qquad \kappa(n) = \big[\{n^{3/2}\} \ge 1 - \{\delta(n)\}\big] \in \{0, 1\}.$$

Proof. (i) Let $f(t) = (X-t)^{3/2}$ on $[0,\theta]$, so $f(\theta) = m^{3/2}$. Then $f'(t) = -\frac{3}{2}(X-t)^{1/2}$ and $f''(t) = \frac{3}{4}(X-t)^{-1/2}$. Taylor with Lagrange remainder at 0 gives $f(\theta) = f(0) + f'(0)\theta + \frac{1}{2}f''(\xi)\theta^2 = X^{3/2} - \frac{3}{2}X^{1/2}\theta + \frac{3}{8}(X-\xi)^{-1/2}\theta^2$ for some $\xi \in (0,\theta)$. Substituting $\theta = X - m$ in the linear term yields $-\frac{1}{2}X^{3/2} + \frac{3}{2}mX^{1/2} = -\frac{1}{2}n^{9/4} + \frac{3}{2}mn^{3/4}$. Since $\theta \in [0,1)$ and $f'' > 0$, the remainder $E(n) = \frac{3}{8}(X-\xi)^{-1/2}\theta^2$ lies in $\big[0, \frac{3}{8}(X-1)^{-1/2}\big]$, which is the bound of the statement; and $(X-1)^{-1/2} \le \frac{4}{3}X^{-1/2}$ already for $n \ge 3$, whence $E(n) \le \frac{1}{2}n^{-3/4}$. (ii) $g = \lfloor X+\delta\rfloor - \lfloor X\rfloor = \lfloor\delta\rfloor + \lfloor\{X\} + \{\delta\}\rfloor$, and the last floor is 1 precisely when $\{X\} + \{\delta\} \ge 1$. $\square$

The point of (i): the non-smooth integer m enters the phase *linearly*, multiplied by a smooth coefficient; no fractional part of growing amplitude survives. The naive expansion $m^{3/2} = n^{9/4} - \frac{3}{2}\theta n^{3/4} + O(n^{-3/4})$ instead leaves the sawtooth θ with the growing amplitude $n^{3/4}$, which defeats the van der Corput method directly. For (ii), since δ is smooth and increasing with $\delta' \asymp hn^{-1/2}$, the level sets of $\lfloor\delta\rfloor$ partition a dyadic block $n \in (P, 2P]$ into $O(hP^{1/2})$ cells of length $\asymp P^{1/2}/h$ on which the integer part of the gap is constant.

Theorem 4.4 (nested parity discrepancy). For every $\varepsilon > 0$ and every $(a,b) \in \{0,1\}^2 \setminus \{(0,0)\}$,

$$\Big|\sum_{n\le N,\ n \text{ odd}} \psi(n^{3/2})^a\,\psi(m(n)^{3/2})^b\Big| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Consequently each of the four joint parity classes of $(m, \lfloor m^{3/2}\rfloor)$ on odd $n \le N$ has cardinality $\frac{N}{8} + O(N^{23/24+\varepsilon})$; in particular the odd-to-odd cylinder refines one letter deeper than Theorem 4.1.

Proof. Work on dyadic blocks $n \in (P, 2P]$, odd, with truncations $J_1 = J_2 = P^{1/24}$ (wave modes), $H = P^{1/12}$ (differencing), and $R = P^{1/4}$ (cell modes). All derivative tests are read through the parity reindexing of Lemma 3.10; every constant below already dominates the reindexed bound.

Step 1 (wave expansion). By Lemma 3.5 applied to each present factor, $\psi_1^a\psi_2^b$ differs from the product of the truncated polynomials $V^{(1)}V^{(2)}$ by at most $\Delta^{(1)} + \Delta^{(2)} + \Delta^{(1)}\Delta^{(2)}$, and each error layer is a non-negative trigonometric polynomial of degree J_1 resp. J_2 with constant term and coefficients $\le 1/(J+1)$. Summed over the block, a majorant layer costs its constant term $\le P/(2(J_1{+}1)) \le P^{23/24}$ plus mode sums $\sum_{0<r\le 2J_1}(J_1{+}1)^{-1}|\sum_n e(\frac{r}{2}n^{3/2})|$: each inner sum has $|(\frac{r}{2}X)''| \in [0.26, 0.75]\, rP^{-1/2}$, single sign, so Lemma 3.3 gives $\le 1.4r^{1/2}P^{3/4} + 2P^{1/4}$, and the weighted total is $\le 2.4J_1^{1/2}P^{3/4} + 2P^{1/4} \le 3P^{19/24}$. The theorem is thus reduced to bounds, uniform in the mode pair, for

$$S_{i,j}(P) = \sum_{n\in(P,2P],\ n \text{ odd}} e\big(\tfrac{i}{2}\, n^{3/2} + \tfrac{j}{2}\, m^{3/2}\big), \qquad |i| \le 2J_1,\ 1 \le |j| \le 2J_2,$$

with mode weights of total mass $O(\log^2 P)$; the $j = 0$ sums are the depth-1 sums of Theorem 4.1. Replacing the summand by its conjugate we may take $j \ge 1$.

Step 2 (linearization). By Lemma 4.3(i), replacing $\frac{j}{2}m^{3/2}$ by $\frac{3j}{4}mn^{3/4} - \frac{j}{4}n^{9/4}$ changes each summand by a phase of modulus $\le \frac{j}{2}E(n) \le \frac{j}{4}n^{-3/4}$, hence changes $S_{i,j}$ by at most $2\pi\frac{j}{4}\sum_{n\sim P} n^{-3/4} \le 2jP^{1/4} \le 4P^{7/24}$. Call the linearized phase Φ.

Step 3 (Weyl differencing). For $H = P^{1/12}$, the A-process recorded after Lemma 3.3 gives

$$|S_{i,j}|^2 \le \frac{2P^2}{H} + \frac{4P}{H}\sum_{1\le h<H}|T_h|, \qquad T_h = \sum_{\substack{n,\, n+2h\in(P,2P]\\ n \text{ odd}}} e\big(\Phi(n{+}2h) - \Phi(n)\big).$$

Step 4 (the exact differenced phase). Write $\Delta f = f(\cdot{+}2h) - f(\cdot)$, $g(n) = m(n{+}2h) - m(n)$, and $m = n^{3/2} - \theta_n$. Exactly,

$$\Phi(n{+}2h) - \Phi(n) = \tfrac{3j}{4}\, g(n)\,(n{+}2h)^{3/4} + \tfrac{j}{2}\, A_h(n) - B_h(n)\,\theta_n + \tfrac{i}{2}\,\Delta(n^{3/2}),$$

with $A_h = \frac{3}{2}n^{3/2}\Delta(n^{3/4}) - \frac{1}{2}\Delta(n^{9/4})$ and $B_h = \frac{3j}{4}\Delta(n^{3/4})$. Taylor with third-order remainders gives, for $n \sim P$ and $h \le H$,

$$\Delta(n^{3/4}) = \tfrac{3}{2}hn^{-1/4} - \tfrac{3}{8}h^2n^{-5/4} + O(h^3P^{-9/4}), \qquad \Delta(n^{9/4}) = \tfrac{9}{2}hn^{5/4} + \tfrac{45}{8}h^2n^{1/4} + O(h^3P^{-3/4}),$$

so the two $n^{5/4}$-scale contributions of A_h cancel exactly at leading order (both equal $\frac{9}{4}hn^{5/4}$) and

$$A_h = -\tfrac{27}{8}\, h^2n^{1/4}(1 + O(hP^{-1})), \qquad |(\tfrac{j}{2}A_h)''| \le 0.32\, jh^2P^{-7/4},$$

$$B_h \in [0.95,\ 1.13]\, jh\, P^{-1/4}, \qquad |(\tfrac{i}{2}\Delta(n^{3/2}))''| = \tfrac{i}{2}\, 2h|X'''(\xi)| \le 0.38\, ihP^{-3/2}.$$

The sawtooth term is *discarded*: $|e(-B_h\theta_n) - 1| \le 2\pi B_h \le 7.1\, jhP^{-1/4}$ per term, so deleting it changes T_h by at most $3.6\, jhP^{3/4}$. This cost is charged in Step 7; it is admissible only because $jh \le 2P^{1/24+1/12} = 2P^{1/8}$, which is the reason the truncation J_2 and the differencing length H cannot be enlarged in this argument.

Step 5 (cells and the exact shift device). Split T_h over the level sets of $\lfloor\delta\rfloor$, $\delta(n) = (n{+}2h)^{3/2} - n^{3/2}$: by (the argument of) Lemma 4.3(ii) and $\delta' \in (1.5, 2.2)\, hP^{-1/2}$, at most $1.5hP^{1/2} + 1$ cells of length in $[\frac{2}{3}, 0.95]P^{1/2}/h$. On a cell $g = G + \kappa$ with $G = \lfloor\delta\rfloor$ frozen and $\kappa(n) = [\{n^{3/2}\} \ge \rho(n)]$, $\rho = 1 - \{\delta(n)\}$ smooth and monotone there. Vaaler-expand κ (one sawtooth layer, Lemma 3.5) with modes $e(rn^{3/2})$, $0 < |r| \le R$, weights $\le 1/|r|$, majorant cost $\le P/(R{+}1)+(R$-mode sums at coefficients $\le 1/(R{+}1))\le P^{3/4}+P^{7/8}$. The moving endpoint costs nothing: since $\rho(n) = 1 + G - \delta(n)$, the coefficient's n-dependent piece contributes the exact smooth phase $r\delta(n)$, and $e(rn^{3/2} + r\delta(n)) = e(r(n{+}2h)^{3/2})$ up to a unimodular constant, so every r-term falls into one of the two smooth families $e(rn^{3/2})$, $e(r(n{+}2h)^{3/2})$; the remaining in-cell coefficient variation is exactly 1 per cell and one Abel summation absorbs it at a factor $\le 2{+}2\pi$.

Step 6 (second-derivative test per cell). After Steps 4–5 the phase on a cell is $\frac{3j}{4}G(n{+}2h)^{3/4} + \frac{j}{2}A_h + \frac{i}{2}\Delta(n^{3/2})$ plus, for the r-pieces, $rn^{3/2}$ or $r(n{+}2h)^{3/2}$.

For $r = 0$: the main curvature is the single monomial term

$$\left(\tfrac{3j}{4}G(n{+}2h)^{3/4}\right)'' = -\tfrac{9j}{64}\,G\,(n{+}2h)^{-5/4} \in -[0.15,\, 0.68]\,jh\,P^{-3/4},$$

using $G \in (3hP^{1/2}{-}1,\, 4.3hP^{1/2})$ from (the mean value bound for) δ and $(n{+}2h)^{-5/4} \in (2^{-5/4}, 1]P^{-5/4}$. Its competitors are dominated at displayed margins: $|(\frac{j}{2}A_h)''|/0.15jhP^{-3/4} \le 2.2\,hP^{-1} \le 2.2P^{-11/12}$ and $|(\frac{i}{2}\Delta(n^{3/2}))''|/0.15jhP^{-3/4} \le 2.5\,(i/j)P^{-3/4} \le 5P^{1/24-3/4}$, so the composite second derivative is single-signed with in-cell ratio ≤ 4.6. Lemma 3.3 per cell (through Lemma 3.10) and summation over cells give

$$\sum_{\text{cells}}(\ell_i\lambda^{1/2} + \lambda^{-1/2}) \le 0.83\,(jh)^{1/2}P^{5/8} + 3.9\,(h/j)^{1/2}P^{7/8}.$$

For $r \ne 0$: the mode curvature $|(rX)''| \ge 0.53|r|P^{-1/2}$ dominates the full cell curvature $\le 0.68jhP^{-3/4}$ at ratio $\ge 0.78\,|r|P^{1/4}/(jh) \ge 0.39\,P^{1/8} \ge 4$ for $P \ge P_0$ (here $jh \le 2P^{1/8}$ is used again), so the composite keeps the mode's sign and, up to a factor $\frac{3}{4}$, its size. Lemma 3.3 per cell and the $1/|r|$ weights give

$$\sum_{0<|r|\le R}\frac{1}{|r|}\Big(1.3|r|^{1/2}P^{3/4} + 1.5hP^{1/2}\cdot 1.6|r|^{-1/2}P^{1/4}\Big) \le 5.2\,R^{1/2}P^{3/4} + 13\,hP^{3/4} \le 6P^{7/8} + 13P^{5/6}.$$

Step 7 (totals and assembly). Collecting Steps 4–6,

$$|T_h| \le C((jh)^{1/2}P^{5/8} + (h/j)^{1/2}P^{7/8} + P^{7/8} + jhP^{3/4})\log P \le C'\,P^{7/8}(1 + h^{1/2})\log P$$

uniformly in the modes (for the last step: $(jh)^{1/2}P^{5/8} \le 2^{1/2}P^{1/16+5/8} \le P^{3/4}$ and $jhP^{3/4} \le 2h^{1/2}\cdot(jh^{1/2})P^{3/4} \le 2h^{1/2}P^{1/24+1/24+3/4} \le 2h^{1/2}P^{7/8}$, using $j \le 2P^{1/24}$, $h^{1/2} \le P^{1/24}$). Hence

$$|S_{i,j}|^2 \le \frac{2P^2}{H} + \frac{4P}{H}\cdot C'\log P\sum_{h<H}P^{7/8}(1+h^{1/2}) \le 2P^{23/12} + 6C'P^{15/8}H^{1/2}\log P \ll P^{23/12+\varepsilon}$$

at $H = P^{1/12}$, so $|S_{i,j}| \ll P^{23/24+\varepsilon}$. Summing mode weights ($O(\log^2 P)$), the majorant and truncation costs of Steps 1 and 5 ($\le P^{23/24} + 3P^{19/24} + P^{7/8}$ per layer), and dyadic blocks gives the theorem. □

The exponent 23/24 is deliberately unoptimized; every stage has slack. The proof above is the undecorated instance of Lemma 5.2(i) below, which reruns the same six stages with three classes of decorations riding along; the reader who has checked this proof has checked the skeleton of that one.

Proposition 4.5 (OE-branch third letter). For $a \in \{0, 1\}$,

$$\Big|\sum_{n\le N,\ n\ \text{odd}}\left((-1)^m\right)^a\psi(m^{1/2})\Big| \ll_\varepsilon N^{7/8+\varepsilon},$$

and consequently $\#\mathrm{OEO}(N),\ \#\mathrm{OEE}(N) = N/8 + O(N^{7/8+\varepsilon})$. Together with Theorem 4.4 this completes depth 3 for odd-rooted itineraries: each of *OOO*, *OOE*, *OEO*, *OEE* has density 1/8 among odd starts with a power saving.

Proof. Taylor expansion of $(X-\theta)^{1/2}$ with both correction terms one-signed gives

$$m^{1/2} = n^{3/4} + D_1(n), \qquad -\tfrac{1}{2}X^{-1/2} - \tfrac{1}{8}(X-1)^{-3/2} \le D_1 \le 0.$$

D_1 is decaying, so replacing $\frac{l}{2}m^{1/2}$ by $\frac{l}{2}n^{3/4}$ inside a Vaaler mode costs $\ll l\sum_{n\sim P} n^{-3/4} \ll lP^{1/4}$, absorbable at $l \le 2P^{1/24}$. The mode sums are then pure: $\sum_{n\sim P} e(\frac{i}{2}n^{3/2} + \frac{l}{2}n^{3/4})$ with $l \neq 0$. For $i \neq 0$, $\varphi'' \asymp |i|P^{-1/2}$ is single-signed and Lemma 3.3 gives $\ll i^{1/2}P^{3/4} + i^{-1/2}P^{1/4}$; for $i=0$, $\varphi'' \asymp lP^{-5/4}$ gives $\ll l^{1/2}P^{3/8} + l^{-1/2}P^{5/8}$. Vaaler majorants, truncation tails, and dyadic assembly are as in Theorem 4.4, and every bound is $\ll P^{7/8}$ after summing mode weights. Branch consistency is Lemma 3.6: the $(1+(-1)^m)/2$ factor vanishes exactly on odd m, where the true chain takes the 3/2-power branch. □

Lemma 4.6 (fourth-letter linearization). For every odd $n \ge 3$,

$$v^{1/2} = n^{9/8} + D(n), \qquad -\tfrac{3}{4}\,n^{-3/8} - n^{-9/8} \le D(n) \le 0.$$

Proof. Two applications of the Lemma 4.3(i) pattern. With $f(t) = (Y-t)^{1/2}$ on $[0,\theta_2]$: $v^{1/2} = m^{3/4} - \frac{1}{2}\theta_2 m^{-3/4} - E_2$, $0 \le E_2 \le \frac{1}{8}(Y-1)^{-3/2}$. With $f(t) = (X-t)^{3/4}$ on $[0,\theta]$: $m^{3/4} = n^{9/8} - \frac{3}{4}\theta n^{-3/8} - E_3$, $0 \le E_3 \le \frac{3}{32}(X-1)^{-5/4}$. Every term is nonpositive after the leading one, and $\frac{1}{2}m^{-3/4} + E_3 + E_2 \le n^{-9/8}$ for $n \ge 3$. □

Unlike the depth-2 layer, *every* non-smooth term now has decaying amplitude: the cumulative phase cost of replacing $\frac{k}{2}v^{1/2}$ by the smooth $\frac{k}{2}n^{9/8}$ over a dyadic block is $\ll kP^{5/8}$, negligible against the target.

Theorem 4.7 (triple parity discrepancy). For every $\varepsilon > 0$ and every $(a,b,c) \in \{0,1\}^3 \setminus \{(0,0,0)\}$,

$$\Big|\sum_{n\le N,\ n\ \mathrm{odd}} \psi(n^{3/2})^a\, \psi(m^{3/2})^b\, \psi(v^{1/2})^c\Big| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Consequently each of the eight sign classes of $(\psi(n^{3/2}), \psi(m^{3/2}), \psi(v^{1/2}))$ on odd $n \le N$ has cardinality $\frac{N}{16} + O(N^{23/24+\varepsilon})$.

Proof. The $c=0$ cases are Theorem 4.4 and Theorem 4.1. For $c=1$, wave-expand all present sign factors as in Step 1 of Theorem 4.4, with truncations $J_1 = J_2 = J_3 = P^{1/24}$, and bound

$$S_{i,j,k}(P) = \sum_{n\in(P,2P],\ n\ \mathrm{odd}} e\big(\tfrac{i}{2}n^{3/2} + \tfrac{j}{2}m^{3/2} + \tfrac{k}{2}v^{1/2}\big)$$

uniformly for $|i| \le 2J_1$, $|j| \le 2J_2$, $1 \le |k| \le 2J_3$. By Lemma 4.6, replacing $\frac{k}{2}v^{1/2}$ by $\frac{k}{2}n^{9/8}$ costs $\ll |k|P^{5/8} \le P^{2/3}$, before any differencing.

If $j=0$, the remaining sum is a single smooth exponential sum with phase $\frac{i}{2}n^{3/2} + \frac{k}{2}n^{9/8}$. Both curvature terms are positive for positive modes, and for mixed signs with $|i| \ge 1$ the first dominates by $P^{3/8-1/24}$; Lemma 3.3 gives $\ll P^{5/8}$ for $i=0$ and $\ll |i|^{1/2}P^{3/4} + P^{1/4}$ otherwise, both $\ll P^{23/24}$.

If $j \neq 0$, the phase is exactly the Theorem 4.4 phase plus the smooth passenger $\frac{k}{2}n^{9/8}$. Steps 2–6 run verbatim; the passenger modifies the smooth part of the differenced phase by $\frac{k}{2}[(n+2h)^{9/8} - n^{9/8}]$, whose second derivative $\ll |k|hP^{-15/8}$ is smaller than the retained cell curvature $jhP^{-3/4}$ and the r-mode curvature $|r|P^{-1/2}$ by powers of P. Every sign-dominance check of Theorem 4.4 holds with these margins. □

Theorem 4.8 (the OE splits).**

$$\#\mathrm{OEOE}(N),\ \#\mathrm{OEOO}(N) = \tfrac{N}{16} + O(N^{7/8+\varepsilon}), \qquad \#\mathrm{OEEE}(N),\ \#\mathrm{OEEO}(N) = \tfrac{N}{16} + O(N^{13/16+\varepsilon}).$$

Together with Theorems 4.4 and 4.7 this proves every depth-4 itinerary word class except *OOO*∗.

Proof. On the *OE* branch m is even and $w = \lfloor U \rfloor$, $U = m^{1/2}$. The w-level linearization is Lemma 4.3(i) verbatim with base U: since $Um^{1/4} = m^{3/4}$ exactly,

$$w^{3/2} = m^{3/4} - \tfrac{3}{2}\,m^{1/4}\,\theta_w + E, \qquad 0 \le E \le \tfrac{3}{8}(U-1)^{-1/2},$$

and $m^{3/4} = n^{9/8}$ up to one-signed corrections of scale $n^{-3/8}$. For a Vaaler mode $k \neq 0$, the phase is therefore $\varphi_1(n) - B(n)\theta_w(n)$ up to absorbable errors, with $\varphi_1 = \frac{i}{2}n^{3/2} + \frac{j}{2}n^{3/4} + \frac{k}{2}n^{9/8}$ and $B = \frac{3k}{4}n^{3/8}(1 + O(n^{-3/2}))$: a *single growing sawtooth* of amplitude $\asymp kn^{3/8}$ riding $\theta_w = \{m^{1/2}\}$, whose underlying integer increments once every $\asymp \frac{4}{3}n^{1/4}$ steps. Truncations: $J_1 = J_2 = J_3 = P^{1/8}$.

Drift-1 intervals. $B' \asymp kn^{-5/8}$, so B drifts by at most 1 on intervals I of length $L_0 = P^{5/8}/k$; there are $\asymp kP^{3/8}$ of them. On I, expand $e(-B\{U\}) = \sum_r a_r(B)e(rU)$ (Fourier in U, Vaaler truncation $|r + B| \leq T = P^{5/16}$); the coefficients satisfy $|a_r(B)| \leq \min(1, |r + B|^{-1})$ with window mass $O(\log T)$, and their n-dependence through $B(n)$ has total variation $O(1)$ per interval, removed by one partial summation per mode.

Mode sums. Smoothing $rU \to rn^{3/4}$ costs $kP^{5/8}$ in total. Writing $r = -B_0 + t$, $|t| \leq T$, the curvature of the mode phase is

$$\varphi'' = \tfrac{27k}{128}n^{-7/8} - \tfrac{3}{16}(t + \tfrac{j}{2})n^{-5/4} + \tfrac{3i}{8}n^{-1/2}.$$

For $i \neq 0$ the last term dominates and Lemma 3.3 gives $\ll i^{1/2}P^{3/4} + i^{-1/2}kP^{5/8}$ after summing intervals. For $i = 0$, $(T + J_2)P^{-5/4} \ll kP^{-7/8}$ because $T = P^{5/16} \ll kP^{3/8}$ for every $k \geq 1$, so $\lambda \asymp kn^{-7/8}$ is single-signed; Lemma 3.3 per interval and summation over $\asymp kP^{3/8}$ intervals give $S_k \ll k^{1/2}P^{13/16+\varepsilon}$. With Vaaler weights $1/k$, $\sum_{k \leq J_3} k^{-1/2}P^{13/16} = J_3^{1/2}P^{13/16} = P^{7/8}$, balanced against the truncation error $P/J_3 = P^{7/8}$ at $J_3 = P^{1/8}$. Total $\ll P^{7/8+\varepsilon}$.

OEE branch. The fourth letter needs $\psi(w^{1/2})$, and $w^{1/2} = U^{1/2} - \frac{1}{2}\theta_w U^{-1/2} - \frac{1}{8}\theta_w^2(U - \xi)^{-3/2}$ has *decaying* amplitudes only, with $U^{1/2} = m^{1/4} \to n^{3/8}$ likewise. The pure modes $e(\frac{l}{2}n^{3/8})$ have $\lambda \asymp lP^{-13/8}$, giving $l^{1/2}P^{3/16} + l^{-1/2}P^{13/16}$; the mixed cases sit in the dominance hierarchy $iP^{-1/2} \gg jP^{-5/4} \gg lP^{-13/8}$ with all three second derivatives of the same sign. Total $\ll N^{13/16+\varepsilon}$. $\square$

Corollary 4.9 (certified-descent density 13/16**).** The three uniform certificate classes

$$E, \qquad OE, \qquad OOEE$$

are disjoint, and

$$\#\{n \leq N : \mathrm{word}(n) \text{ has prefix } E\} = \lfloor \tfrac{N}{2} \rfloor,$$

$$\left|\#\{n \leq N : \mathrm{word}(n) \text{ has prefix } OE\} - \tfrac{N}{4}\right| \ll N^{5/6},$$

$$\left|\#\{n \leq N : \mathrm{word}(n) \text{ has prefix } OOEE\} - \tfrac{N}{16}\right| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Hence the class of starts with a certified descent within four steps has cardinality

$$\tfrac{N}{2} + \tfrac{N}{4} + \tfrac{N}{16} + O(N^{23/24+\varepsilon}) = \tfrac{13N}{16} + O(N^{23/24+\varepsilon}).$$

No other depth-≤ 4 word class is used. In particular this is not a census of E-rooted itineraries of length ≥ 2, and it is not a census of every O-rooted itinerary of length four (the classes $OOO*$ are Theorem 6.1, and they are non-contracting at this depth).

Proof. Every even start realizes E, so the first count is elementary. The OE count is Corollary 4.2. For $OOEE$, Lemma 3.6 gives $\#OOEE = \frac{1}{8}\sum_{n \leq N \text{ odd}}(1 - \psi_1)(1 + \psi_2)(1 + \psi_3)$ with $\psi_1 = \psi(n^{3/2})$, $\psi_2 = \psi(m^{3/2})$, $\psi_3 = \psi(v^{1/2})$; expanding gives the main term $N/16$ and seven sign sums bounded by Theorem 4.7. Every $OOEE$ start descends within four steps by Proposition 3.1: $3^2 < 2^4$. The three classes are disjoint because they are distinct prefixes. $\square$

The certificate density 13/16 is the exact ceiling of this one-growing-layer machinery: an itinerary contracts iff $3^o < 2^\ell$, the method so far proves letters at positions 1–3 of any itinerary plus further letters along even branches only, and the contracting minimal words with all odd letters at positions ≤ 2 are exactly E, OE, and $OOEE$. Completing depth 4 over odd starts requires the $OOO*$ split — a second growing layer, where the fourth-letter phase coefficient $W \asymp kn^{9/8}$ crosses integers within single steps and no drift-1 interval exists. Sections 5 and 6 close that split. The $OOO*$ words are non-contracting at depth 4, so the four-step certified density remains 13/16; the next increment is the two length-five contractors of Theorem 6.3.

3.5 Localization to sub-dyadic intervals and slow twists

Theorems 4.4 and 4.7 are stated on dyadic blocks. Two applications outside this paper need them on shorter intervals and with a slowly varying phase attached: the contagion recursion of the companion paper [24] counts the *OOEEE* starts landing in a prescribed even block, which is a statement about the interval $I(m') = [m'^{32/9}, (m'+1)^{32/9})$ of length $\asymp P^{23/32}$ at scale $P = m'^{32/9}$, and the fifth letter enters through a Fourier expansion whose modes $e(\frac{\ell}{2}n^{9/16})$ multiply the depth-4 sign products. This subsection proves both theorems in that generality. The proofs are the proofs of Theorems 4.4 and 4.7 with the number of summands Y in place of P wherever the number of summands enters, and the point of writing them out is to exhibit the terms that do *not* scale with Y — there are three, all of size at most $P^{7/16}$ — and to show that the twist is removed by one partial summation after the differencing.

Throughout, $I \subseteq (P, 2P]$ is an interval of length Y (so I contains $Y/2 + O(1)$ odd integers), and a *slow twist* is $g(n) = \frac{\ell}{2}n^{9/16}$ with an integer ℓ, $|\ell| \le P^{1/24}$. For $n \in (P, 2P]$,

$$|g'(n)| \le \tfrac{9}{32}|\ell|P^{-7/16}, \qquad |g''(n)| \le \tfrac{63}{512}|\ell|P^{-23/16} \le 0.13\, P^{1/24-23/16}. \tag{4.5}$$

Lemma 4.10 (removing a slow twist after differencing). Let a_n be complex numbers on an interval I and $w(n) = e(\gamma(n))$ with γ real and differentiable on I. Then

$$\Big|\sum_{n\in I} a_n w(n)\Big| \le (1 + 2\pi\, \mathrm{TV}_I(\gamma)) \max_{I'} \Big|\sum_{n\in I'} a_n\Big|,$$

the maximum over the initial sub-intervals I' of I, where $\mathrm{TV}_I(\gamma) = \int_I |\gamma'|$. In particular, for $\gamma = \Delta_{2h} g = g(\cdot + 2h) - g$ with g a slow twist, $h \le P^{1/12}$ and $|I| \le P$, $\mathrm{TV}_I(\gamma) \le 2h\,|I| \sup|g''| \le 0.26\, P^{1/24+1/12+1-23/16} = 0.26\, P^{-5/16}$.

Proof. Abel summation: with $A(x) = \sum_{n\in I,\, n\le x} a_n$, $\sum_{n\in I} a_n w(n) = \sum_n A(n)(w(n) - w(n+1)) + A(\max I) w(\max I)$, and $\sum_n |w(n) - w(n+1)| \le 2\pi \sum_n |\gamma(n+1) - \gamma(n)| \le 2\pi\, \mathrm{TV}_I(\gamma)$. The second claim is $|(\Delta_{2h} g)'| \le 2h \sup|g''|$ and (4.5). □

Theorem 4.11 (nested parity discrepancy on sub-dyadic intervals, with a slow twist). For every $\varepsilon > 0$ there is P_0 such that for all $P \ge P_0$, every interval $I \subseteq (P, 2P]$ of length $Y \ge P^{1/2}$, every $(a, b) \in \{0,1\}^2 \setminus \{(0,0)\}$ and every slow twist g,

$$\Big|\sum_{n\in I,\ n\ \mathrm{odd}} \psi(n^{3/2})^a\, \psi(m^{3/2})^b\, e(g(n))\Big| \ \le\ Y\, P^{-1/24+\varepsilon}.$$

Proof. We run Steps 1–7 of Theorem 4.4 over I and record each cost as *proportional* (to the number of summands, hence to Y) or *absolute* (independent of it).

Step 1. The majorant layers are nonnegative and the twist is unimodular, so the expansion is unchanged: each layer costs its constant term $\le Y/(2(J_1+1)) \le \frac{1}{2} Y P^{-1/24}$ (proportional) plus the mode sums $\sum_{0<r\le 2J_1} (J_1 + 1)^{-1} |\sum_{n\in I} e(\frac{r}{2} n^{3/2})|$; Lemma 3.3 on I gives each inner sum $\le 1.4 r^{1/2} Y P^{-1/4} + 2P^{1/4}$, and the weighted total is $\le 2.6 J_1^{1/2} Y P^{-1/4} + 4P^{1/4} = 2.6\, Y P^{-11/48} + 4P^{1/4}$ (proportional plus absolute). The theorem reduces to the twisted mode sums

$$S^g_{i,j}(I) = \sum_{n\in I,\ n\ \mathrm{odd}} e(\tfrac{i}{2} n^{3/2} + \tfrac{j}{2} m^{3/2} + g(n)), \qquad |i| \le 2J_1,\ 1 \le |j| \le 2J_2,$$

with mode weights of total mass $O(\log^2 P)$; the $j = 0$ sums are the depth-1 sums $\sum_{n\in I} e(\frac{i}{2} n^{3/2} + g(n))$, $i \ne 0$, whose phase has curvature in $[0.53, 0.75]|i|P^{-1/2}$ up to the twist's $\le 0.13 P^{1/24-23/16}$ — a relative perturbation $\le 0.25 P^{1/24-15/16}$ — so Lemma 3.3 gives $\le 1.4|i|^{1/2} Y P^{-1/4} + 2P^{1/4}$.

Step 2. Linearization changes each summand by a phase of modulus $\le \frac{j}{4} n^{-3/4}$: total $\le 2\pi \cdot \frac{j}{4} \cdot \frac{Y}{2} P^{-3/4} \le jYP^{-3/4} \le 2YP^{1/24-3/4}$ (proportional).

Step 3. The A-process for a sum of $Y/2+O(1)$ unimodular terms with $H = P^{1/12} \le Y$:

$$|S^g_{i,j}(I)|^2 \le \frac{2Y^2}{H} + \frac{4Y}{H}\sum_{1\le h<H}|T^g_h|, \qquad T^g_h = \sum_{\substack{n,\,n+2h\in I\\ n \text{ odd}}} e\big(\Phi(n{+}2h) - \Phi(n) + \Delta_{2h} g(n)\big).$$

By Lemma 4.10, $|T^g_h| \le (1+1.7P^{-5/16}) \max_{I'} |T_h(I')|$, where $T_h(I')$ is the *untwisted* differenced sum over an initial sub-interval I' of I. The twist has now been removed; it remains to bound $T_h(I')$ for every sub-interval $I' \subseteq I$ by an expression that is nondecreasing in $|I'|$, and to evaluate it at $|I'| = Y$.

Step 4. The exact differenced phase is unchanged. Discarding the sawtooth term costs $\le 7.1jhP^{-1/4}$ per summand, i.e. $\le 3.6jhY'P^{-1/4}$ on I' (proportional).

Step 5. The level sets of $\lfloor\delta\rfloor$ cut I' into at most $1.5hY'P^{-1/2}+2$ cells: the full cells have length in $[\frac{2}{3}, 0.95]P^{1/2}/h$ as before, and at most two end cells are partial. The Vaaler layer for κ costs its constant term $\le Y'/(R+1) \le Y'P^{-1/4}$ plus the R-mode sums at coefficients $\le 1/(R+1)$; per cell and mode Lemma 3.3 gives $\ell\lambda_r^{1/2} + \lambda_r^{-1/2}$ with $\lambda_r \ge 0.53|r|P^{-1/2}$, and summing over cells and modes,

$$\le 1.2\,R^{1/2}Y'P^{-1/4} + (1.5hY'P^{-1/2}+2)\cdot 5.5\,P^{1/4}R^{-1/2} = 1.2\,Y'P^{-1/8} + 8.3\,hY'P^{-3/8} + 11\,P^{1/8}$$

(two proportional terms and one absolute). The exact shift device is pointwise and unchanged.

Step 6. For $r=0$ the per-cell test gives, over the cells meeting I',

$$\sum_{\text{cells}}(\ell_i\lambda^{1/2} + \lambda^{-1/2}) \le 0.83\,(jh)^{1/2}Y'P^{-3/8} + 3.9\,(h/j)^{1/2}Y'P^{-1/8} + 5.2\,(jh)^{-1/2}P^{3/8},$$

the last term being the two partial end cells. For $r \ne 0$ the same per-cell bounds with the mode curvature, summed against the $1/|r|$ weights, give $\le 5.2\,Y'P^{-1/8} + 13\,hY'P^{-1/4} + 17\,P^{1/4}$, the last term again from the end cells. All sign-dominance checks are pointwise and unchanged.

Step 7. Collecting, for every sub-interval I' of length Y',

$$|T_h(I')| \le C\,Y'P^{-1/8}(1+h^{1/2})\log P + C'P^{3/8},$$

nondecreasing in Y' (the absolute constant C' collects $5.2(jh)^{-1/2}P^{3/8}$, $17P^{1/4}$, $11P^{1/8}$ and the Step-1 term $4P^{1/4}$). Hence

$$|S^g_{i,j}(I)|^2 \le \frac{2Y^2}{H} + \frac{4Y}{H}\sum_{h<H}\Big(CYP^{-1/8}(1{+}h^{1/2})\log P + C'P^{3/8}\Big)(1{+}1.7P^{-5/16}) \le Y^2P^{-1/12}(3{+}8C\log P) + 5C'YP^{3/8}.$$

For $Y \ge P^{1/2}$, $YP^{3/8} \le Y^2P^{-1/8} \le Y^2P^{-1/12}$, so $|S^g_{i,j}(I)| \ll YP^{-1/24}\log^{1/2} P$. Summing the mode weights $(O(\log^2 P))$ and adding the proportional and absolute costs of Steps 1–2 gives the theorem, the absolute costs being $\le 4P^{1/4} \le YP^{-1/4}$. □

Theorem 4.12 (triple parity discrepancy on sub-dyadic intervals, with a slow twist). For every $\varepsilon > 0$ there is P_0 such that for all $P \ge P_0$, every interval $I \subseteq (P, 2P]$ of length $Y \ge P^{1/2}$, every $(a,b,c) \in \{0,1\}^3 \setminus \{(0,0,0)\}$ and every slow twist g,

$$\Big|\sum_{n\in I,\ n \text{ odd}} \psi(n^{3/2})^a\,\psi(m^{3/2})^b\,\psi(v^{1/2})^c\,e(g(n))\Big| \;\le\; Y\,P^{-1/24+\varepsilon}.$$

Proof. The $c=0$ cases are Theorem 4.11 and its depth-1 case. For $c=1$, wave-expand as in Theorem 4.7 and bound the twisted mode sums $S^g_{i,j,k}(I)$, $1 \le |k| \le 2J_3$. Lemma 4.6 replaces $\frac{k}{2}v^{1/2}$ by $\frac{k}{2}n^{9/8}$ at a per-summand cost $\le \frac{k}{2}(\frac{3}{4}n^{-3/8} + n^{-9/8})$, total $\le 2.4\,|k|YP^{-3/8} \le 5\,YP^{1/24-3/8}$ (proportional).

If $j=0$ the sum is a single smooth exponential sum with phase $\frac{i}{2}n^{3/2} + \frac{k}{2}n^{9/8} + g(n)$. For $i \ne 0$ the first term's curvature $\ge 0.53|i|P^{-1/2}$ dominates the second $(\le 0.14|k|P^{-7/8} \le 0.14P^{1/24-7/8})$ and the twist

($\le 0.13P^{1/24-23/16}$) by powers of P, and Lemma 3.3 gives $\le 1.4|i|^{1/2}YP^{-1/4} + 2P^{1/4}$. For $i = 0$ the passenger's curvature $\ge 0.07|k|P^{-7/8}$ dominates the twist by $P^{-23/16+7/8-1/24} = P^{-7/12}$, and Lemma 3.3 gives $\le 0.27|k|^{1/2}YP^{-7/16} + 3.8|k|^{-1/2}P^{7/16}$: one proportional term and the absolute *pure passenger* term $3.8P^{7/16}$, which is $\le 3.8\,YP^{-1/16}$ for $Y \ge P^{1/2}$.

If $j \ne 0$, the phase is the Theorem 4.11 phase plus the smooth passenger $\frac{k}{2}n^{9/8}$. After the differencing of Step 3 and the removal of the twist by Lemma 4.10, the passenger contributes $\frac{k}{2}[(n+2h)^{9/8} - n^{9/8}]$ to the smooth part of the differenced phase, with second derivative $\le 0.15|k|hP^{-15/8}$, smaller than the retained cell curvature $0.15jhP^{-3/4}$ and the r-mode curvature $0.53|r|P^{-1/2}$ by the factors $P^{1/24-9/8}$ and $P^{1/24+1/12-11/8}$; every sign-dominance check of Steps 4–6 holds with these margins, and Step 7 is unchanged. $\square$

The threshold $Y \ge P^{1/2}$ is where the three absolute terms — the two partial end cells ($5.2P^{3/8}$), the pure passenger ($3.8P^{7/16}$) and the majorant end cells ($17P^{1/4}$) — fall below the main saving $YP^{-1/24}$; nothing else in either proof uses the length of the block except through the number of summands. The twist enters only through Lemma 4.10 (after differencing) and through curvature comparisons in which it loses by a power of P; any g with $|g''| \le P^{-4/3}$ on the block would do, and the class $\frac{\ell}{2}n^{9/16}$, $|\ell| \le P^{1/24}$, is the one the application needs.

Corollary 4.13 (the *OOEEE* production on even blocks). For $m' \ge 2$ let $I(m') = [m'^{32/9}, (m'+1)^{32/9})$ and

$$\mathcal{O}(m') = \{n \text{ odd} \in I(m') :\ \mathrm{word}_5(n) = OOEEE,\ J^5(n) = m'\},$$

where $\mathrm{word}_5(n)$ is the itinerary of the first five steps of n under J. Then every $n \in \mathcal{O}(m')$ has $J^4(n) \in [m'^2, (m'+1)^2)$ even, and

$$|\mathcal{O}(m')| = \tfrac{1}{16}\#\{n \text{ odd} \in I(m')\} + O_\varepsilon(|I(m')|\, m'^{-4/27+\varepsilon}).$$

Proof. Put $P = m'^{32/9}$ and $Y = |I(m')| = \frac{32}{9}m'^{23/9}(1 + O(1/m')) \asymp P^{23/32} \ge P^{1/2}$. Along *OOEEE* the states are n (odd), m (odd), v (even), $\lfloor v^{1/2}\rfloor$ (even), $J^4(n) = \lfloor\sqrt{\lfloor v^{1/2}\rfloor}\rfloor = \lfloor v^{1/4}\rfloor$ (even), and $J^5(n) = \lfloor\sqrt{J^4(n)}\rfloor$; so $J^5(n) = m'$ forces $J^4(n) \in [m'^2, (m'+1)^2)$. Three elementary facts:

(a) *Nesting at the fifth letter.* For odd $n \ge 3$, $0 \le n^{9/16} - v^{1/4} \le n^{-15/16}$: the upper bound is $v \le n^{9/4}$; the lower one follows from $v \ge (n^{3/2}-1)^{3/2} - 1 \ge n^{9/4} - \frac{3}{2}n^{3/4} - 1$ and the mean value theorem for $u \mapsto u^{1/4}$, using $v \ge \frac{1}{2}n^{9/4}$. Hence $\lfloor v^{1/4}\rfloor = \lfloor n^{9/16}\rfloor$ unless $\{n^{9/16}\} < n^{-15/16}$.

(b) *The exceptional set is small.* By Lemma 3.4 (Erdős–Turán) with $H = \frac{4}{9}P^{7/16}$ and the Kusmin–Landau bound $|\sum_{n\in I \text{ odd}} e(hn^{9/16})| \ll P^{7/16}/h$ (the phase $h(2k+1)^{9/16}$ has monotone derivative $\frac{9}{8}h(2k+1)^{-7/16} \in (0, \frac{1}{2}]$),

$$\#\{n \text{ odd} \in I :\ \{n^{9/16}\} < n^{-15/16}\} \ll YP^{-15/16} + YP^{-7/16} + P^{7/16} \ll YP^{-1/16}.$$

(c) *The fifth letter as a Fourier series.* With $J = \lfloor P^{1/24}\rfloor$, Lemma 3.5 gives coefficients b_q, $0 < |q| \le J$, $|b_q| \ll 1/|q|$, and a majorant Δ_J of degree J with coefficients $\le 1/(J+1)$, such that $|\psi(y) - \sum_q b_q e(qy/2)| \le 2\Delta_J(y/2)$ for all real y; and by Kusmin–Landau, $|\sum_{n\in I \text{ odd}} e(\frac{q}{2}n^{9/16})| \ll P^{7/16}/|q|$ for $0 < |q| \le J$, so $\sum_{n\in I} \Delta_J(\frac{1}{2}n^{9/16}) \ll YP^{-1/24} + P^{7/16}\log P$.

Now, off the exceptional set of (a), $n \in I(m')$ gives $\lfloor n^{9/16}\rfloor \in [m'^2, (m'+1)^2)$, so for such n with $\mathrm{word}_5(n) = OOEEE$ the value $J^5(n) = m'$ is automatic; therefore $|\mathcal{O}(m')|$ differs from $\#\{n \text{ odd} \in I :\ \mathrm{word}_5(n) = OOEEE\}$ by at most the exceptional count of (b). By Lemma 3.6 the indicator of *OOEEE* on odd n is $\frac{1}{16}(1-\psi_1)(1+\psi_2)(1+\psi_3)(1+\psi_4)$ with $\psi_4 = \psi(v^{1/4})$, each factor enforcing the branch on which the next is the true letter. Expanding, the main term is $\frac{1}{16}\#\{n \text{ odd} \in I\}$; the seven sign sums without ψ_4 are Theorem 4.12 with $g = 0$; for the eight with ψ_4, replace $\psi(v^{1/4})$ by $\psi(n^{9/16})$ (they agree off the exceptional set, by (a)), expand by (c), and bound each twisted sum $\sum_n \psi_1^a\psi_2^b\psi_3^c\, e(\frac{q}{2}n^{9/16})$ by Theorem 4.12 when $(a,b,c) \ne 0$ and by the Kusmin–Landau bound of (c) when $(a,b,c) = 0$; the coefficient sums are $O(\log P)$. Every error is $\ll YP^{-1/24+\varepsilon} + P^{7/16}\log P \ll YP^{-1/24+\varepsilon}$, and $P^{-1/24} = m'^{-4/27}$. $\square$

The corollary is what the contagion recursion of [24] consumes: for a backward-closed set $A \ni m'$, every member of $\mathcal{O}(m')$ lies in A, with log-mass at least $\frac{1}{9m'}(1 - O(m'^{-4/27+\varepsilon}))$, and the recursion gains the term $\frac{1}{9}g(9t/32)$, raising the contagion exponent from 0.4050 to 0.4922. Nothing in this subsection is used in Sections 5–7.

4. The level-2 wave and the kernel theorem

The reader should hold one object in mind through this section: the exact level-2 wave $e(qY)$, $Y = \lfloor n^{3/2} \rfloor^{3/2}$, a depth-2 nested-floor phase, possibly multiplied by a frozen floor of the same depth. It is the fundamental obstruction of the paper. The kernel theorem (Theorem 5.3) is organized so that, after two differencings and an exact bookkeeping of carries, *everything* that is not a standard second-derivative test is such a wave, and Lemma 5.2 — stated and proved first, as a standalone result with its own differencing — bounds it with exactly depth-2 strength, $|q|^{-1/6}P^{23/24+\varepsilon}$. The exponent $1 - 1/96$ of the kernel theorem is nothing but that depth-2 strength propagated through two differencings; any improvement of Lemma 5.2 improves the kernel theorem proportionally, and no other part of the argument is close to its limit.

Every reorganization of the $OOO*$ phase funnels into one object: for the monomial weight $c(n) = \frac{3k}{4}n^{9/8}$ on $n \sim P$,

$$K_c(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e(c(n)\,\{\lfloor n^{3/2} \rfloor^{3/2}\}).$$

This section proves the kernel bound at full length: every estimate that the proof uses is displayed, with its dependence on the mode k, the differencing shifts h_1, h_2, h_3, the branch offset j, and the expansion frequencies made explicit, and with numerical constants displayed at each of the sign-critical dominance checks. Two structural facts shape the proof and are worth stating in advance. First, the hardest pieces of the decomposition are exact level-2 waves $e(qY)$, possibly riding a frozen floor; a frozen-coefficient model $e(sX)$ with $s \asymp qP^{3/4}$ would silently discard the sawtooth $-\frac{3}{2}qX^{1/2}\theta$ of the same amplitude hidden in $Y = (X - \theta)^{3/2}$, so the waves are treated exactly — by the same exact linearization that proves Theorem 4.4 — and the honest bound is $q^{-1/6}P^{23/24+\varepsilon}$ (Lemma 5.2): the wave is a depth-2 object and receives exactly depth-2 strength, which is what pins the kernel saving at $\delta = \frac{1}{96}$. Second, the offset-branch pieces of Step 5 carry a factor $(k|j|)^{1/2}$, and $(k|j|)^{1/2}P^{15/16} \le P^{23/24}$ exactly on the standing range $k \le P^{1/24}$: the two bottlenecks meet at the endpoint, which is why the theorem is uniform in k on that range and why the range cannot be enlarged by this assembly.

The next lemma collects the exact reductions behind the treatment: the kernel phase is the level-2 local floor defect; the second difference of the level-2 integer obeys an exact floor identity; on an explicit carry-branch decomposition that second difference is a smooth function with a frozen floor; and the doubly differenced kernel phase decomposes exactly into four bounded pieces — the master identity — with no growing smooth part left to estimate.

Lemma 5.1 (level-2 defect, double gap, branch freeze, and the master identity). Fix shifts $d_1 = 2h_1$, $d_2 = 2h_2$ and write Δ_1, Δ_2, $\Delta\Delta$ for the corresponding difference operators in n, and $W = \Delta_1 Y$, $W' = \Delta_2 Y$.

(i) For odd $n \ge 3$,

$$\tfrac{3}{4}\,v^{1/2}\theta_2 = \tfrac{1}{2}(m^{9/4} - v^{3/2}) - R, \qquad 0 \le R \le \tfrac{3}{16}\,v^{-1/2}.$$

Hence for $c = \frac{3k}{4}v^{1/2}$ the kernel phase $c\,\theta_2$ equals $\frac{k}{2}(Y^{3/2} - v^{3/2})$ up to $kR \ll kP^{-9/8}$ per term: the kernel is the exponential sum of the level-2 local floor defect.

(ii) Exactly, with $g_2 = \Delta_1 v$,

$$\Delta_2 g_2 = \lfloor \Delta\Delta Y \rfloor + \kappa'' + \Delta_2\kappa_2, \qquad \kappa_2 = [\theta_2 \ge 1 - \{W\}], \quad \kappa'' = [\{W\} \ge 1 - \{\Delta\Delta Y\}],$$

and every carry is a difference of unit sawtooths: $[\{A\} + \{B\} \ge 1] = \{A\} + \{B\} - \{A{+}B\}$ for all reals A, B (both sides equal $\{A\} + \{B\} - \{A + B\}$: the left side because $\{A\} + \{B\} \in [0, 2)$ and $\{A + B\} = \{A\} + \{B\} - \lfloor \{A\} + \{B\} \rfloor$).

(iii) Write $b_1 = \lfloor \Delta_1 X \rfloor$, $b_2 = \lfloor \Delta_2 X \rfloor$, $b_{12} = \lfloor \Delta_{12} X \rfloor$ (shift $d_1{+}d_2$) — each constant on runs of length $\asymp P^{1/2}/h$ — and let $\kappa = (\kappa_1, \kappa_2, \kappa_{12}) \in \{0,1\}^3$ be the level-1 carries, so that the shifted values of m are $m + \beta_1$, $m + \beta_2$, $m + \beta_{12}$ with $\beta_i = b_i + \kappa_i$. On each b-run intersection and carry branch,

$$\Delta\Delta Y = F_\kappa(m), \qquad F_\kappa(m) = (m{+}\beta_{12})^{3/2} - (m{+}\beta_1)^{3/2} - (m{+}\beta_2)^{3/2} + m^{3/2}$$

exactly. The net offset $j = \beta_{12} - \beta_1 - \beta_2$ satisfies $|j| \le 3$ for $h_1 h_2 \le P^{1/2}/3$, and F splits exactly into an offset term and a genuine second difference: for some ξ_1 between 0 and j and some $\xi_2 \in (0, \beta_1 + \beta_2)$,

$$F_\kappa(m) = \tfrac{3}{2}\, j\,(m + \beta_1 + \beta_2 + \xi_1)^{1/2} + \tfrac{3}{4}\,\beta_1\beta_2\,(m + \xi_2)^{-1/2}.$$

Consequently, with $G := F_\kappa \circ X$,

$$\tfrac{3}{2}|j|P^{3/4} \le \left|\tfrac{3}{2}j(\cdot)^{1/2}\right| \le 2.6\,|j|P^{3/4}, \qquad 1.4\,h_1h_2P^{1/4} \le \tfrac{3}{4}\beta_1\beta_2(\cdot)^{-1/2} \le 15\,h_1h_2P^{1/4},$$

$$|G'(n)| \le 2|j|P^{-1/4} + 20\,h_1h_2P^{-3/4} < 1, \qquad |G''(n)| \le 2|j|P^{-5/4} + 25\,h_1h_2P^{-7/4},$$

so $\lfloor G(n) \rfloor$ is constant on runs of length $\ge \frac{1}{22}\min\!\big(P^{1/4}/(|j|{+}1),\, P^{3/4}/(h_1h_2)\big)$. The branch indicator is a finite union of arcs in the single variable θ with slowly moving endpoints (drifts $\le 1.5hP^{-1/2}$ per step): exactly the moving-endpoint pattern of Step 4 of Theorem 4.4, absorbed by the same exact shift device at $O(\log P)$ mode-mass cost.

(iv) *(Master identity.)* With $\kappa_2 = [\theta_2 \ge 1 - \{W\}]$, $\kappa_2' = [\theta_2 \ge 1 - \{W'\}]$ the level-2 carries at the two shifts, and κ'', $\Delta_2\kappa_2$ as in (ii), the doubly differenced kernel phase decomposes exactly:

$$\Delta\Delta(c\,\theta_2) = (\Delta\Delta c)\,\theta_2 + (\Delta_2 c)(n{+}d_1)\,(\{W\}{-}\kappa_2) + (\Delta_1 c)(n{+}d_2)\,(\{W'\}{-}\kappa_2') + c_{11}\,(\{\Delta\Delta Y\}{-}\kappa''{-}\Delta_2\kappa_2),$$

where c_{11} is the doubly shifted weight. Every bracket is bounded by 2 in absolute value: no unbounded smooth part survives the differencing.

Proof. (i) Taylor of $(v + \theta_2)^{3/2}$ at v, with $Y^{3/2} = m^{9/4}$. (ii) The gap identity of Lemma 4.3(ii) applied twice — to Y at shift d_1, then to the real sequence $n \mapsto W(n)$ at shift d_2, with $\Delta_2 W = \Delta\Delta Y$; the sawtooth form of the carry is the displayed elementary identity. (iii) The corner identities are Lemma 4.3(ii) at the three shifts, and the offset bound follows from $|\Delta\Delta X| \le 4h_1h_2 \sup|X''| = 3h_1h_2P^{-1/2} < 1$. For the split, group $F = \big[(m{+}\beta_1{+}\beta_2{+}j)^{3/2} {-} (m{+}\beta_1{+}\beta_2)^{3/2}\big] + \big[(m{+}\beta_1{+}\beta_2)^{3/2} {-} (m{+}\beta_1)^{3/2} {-} (m{+}\beta_2)^{3/2} {+} m^{3/2}\big]$: the first bracket is $\frac{3}{2}j(\cdot)^{1/2}$ by the mean value theorem, the second is $\beta_1\beta_2\, f''(m + \xi_2)$ for $f = x^{3/2}$ by the double mean value theorem. The displayed numerical bounds use $m \in (P^{3/2}{-}1, 2^{3/2}P^{3/2}]$, $\beta_i \in [3h_iP^{1/2}{-}1,\, 3\sqrt{2}\,h_iP^{1/2}{+}1]$ (from $\Delta_i X = 2h_iX'(\xi)$ and (E1) below), and $G' = F'(X)X'$, $G'' = F''(X)X'^2 + F'(X)X''$ with $F' = \frac{3}{4}j(\cdot)^{-1/2} - \frac{3}{8}\beta_1\beta_2(\cdot)^{-3/2}$, $F'' = -\frac{3}{8}j(\cdot)^{-3/2} + \frac{9}{16}\beta_1\beta_2(\cdot)^{-5/2}$. (iv) By (ii) and the definitions: $\Delta_1\theta_2 = \Delta_1 Y - \Delta_1 v = W - (\lfloor W \rfloor + \kappa_2) = \{W\} - \kappa_2$ (Lemma 4.3(ii) applied to Y), likewise $\Delta_2\theta_2 = \{W'\} - \kappa_2'$, and $\Delta\Delta\theta_2 = \Delta\Delta Y - \Delta_2 g_2 = \{\Delta\Delta Y\} - \kappa'' - \Delta_2\kappa_2$ by (ii). Substitute these into the exact product rule

$$\Delta\Delta(cf) = c_{11}\,\Delta\Delta f + (\Delta_2 c)(n{+}d_1)\,\Delta_1 f + (\Delta_1 c)(n{+}d_2)\,\Delta_2 f + (\Delta\Delta c)\,f,$$

which is verified by expanding both sides over the four base points n, $n{+}d_1$, $n{+}d_2$, $n{+}d_1{+}d_2$. $\square$

One warning, essential to the organization: $\lfloor \Delta\Delta Y \rfloor$ itself is *not* frozen. The level-1 carry toggles at essentially every step and shifts $\Delta\Delta Y$ by $\frac{3}{2}j_1m^{1/2}$, a jump of size $\asymp P^{3/4}$. The branch decomposition of (iii), which carries the flicker inside the indicator over the eight carry branches, is therefore forced: no decomposition conditioning on the *values* of the level-1 gaps produces sets on which the second difference is smooth.

Standing estimates

All estimates of this section take place on a dyadic block $n \in (P, 2P]$, n odd, under the standing constraints

$$\text{(C1)}\quad kh_1h_2 \le P^{1/8}, \qquad \text{(C2)}\quad h_1h_2 \le P^{1/2}/3, \qquad \text{(C3)}\quad 1 \le k \le P^{1/24}, \qquad \text{(C4)}\quad h_1,\, h_2 \le P^{1/24}.$$

(C1)–(C3) do not by themselves bound the individual shifts: the product constraints permit one of h_1, h_2 to be as large as $\asymp P^{1/2}$. The decoration class (D1) and the third-differencing reduction of Lemma 5.2(ii) need $h_1 + h_2 \le 2P^{1/24}$, which is (C4). Theorem 5.3 takes $H_1 = P^{1/48}$ and $H_2 = P^{1/24}$, so (C4) holds there. Every displayed constant below is valid for $P \ge P_0$ with an absolute P_0. All sums run over odd n, and every derivative test below is read through the parity reindexing of Lemma 3.10: margins and signs are invariant, and each displayed test bound dominates its reindexed value, so no constant needs adjustment. The base derivatives are

$$\text{(E1)}\quad X' = \tfrac{3}{2}n^{1/2} \in (1.5, 2.13]\,P^{1/2},\quad X'' = \tfrac{3}{4}n^{-1/2} \in [0.53, 0.75)\,P^{-1/2},$$
$$-X''' = \tfrac{3}{8}n^{-3/2} \in [0.13, 0.375)\,P^{-3/2},\quad X'''' = \tfrac{9}{16}n^{-5/2} \le 0.57\,P^{-5/2},$$

and $m = X - \theta \in (P^{3/2} - 1,\ 2.83\,P^{3/2}]$. Every gap is a mean value: for any shift $2h$,

$$\text{(E2)}\quad \Delta X = 2hX'(\xi) \in (3hP^{1/2},\ 4.3hP^{1/2}),\qquad \beta := \lfloor \Delta X \rfloor + \kappa \in (3hP^{1/2} - 1,\ 4.3hP^{1/2} + 1),$$

so that every level-1 gap integer at shift $2h$ is $\asymp hP^{1/2}$ with the displayed constants. For the monomial weight $c = \frac{3k}{4}n^{9/8}$ of Theorem 5.3,

$$\text{(E3)}\quad c' = \tfrac{27}{32}kn^{1/8},\quad c'' = \tfrac{27}{256}kn^{-7/8},\quad c''' = -\tfrac{189}{2048}kn^{-15/8},\quad c'''' = \tfrac{2835}{16384}kn^{-23/8},$$

$$\text{(E4)}\quad \Delta_i c = 2h_i c'(\xi) \in (1.68, 1.85)\,kh_iP^{1/8},\qquad \Delta\Delta c = 4h_1h_2c''(\xi) \in (0.22, 0.43)\,kh_1h_2P^{-7/8}.$$

On the level-2 side, with β_i as in Lemma 5.1(iii),

$$\text{(E5)}\quad W = \Delta_1 Y \in (4.4, 11)\,h_1P^{5/4},\qquad \big|(W \circ \mathrm{cell})'\big| = \big|W_1'(m)\,X'\big| \in (2.2, 10.4)\,h_1P^{1/4},$$

where $W_1(m) = (m+\beta_1)^{3/2} - m^{3/2}$, so $\{W\}$ is a *fast* sawtooth (its argument moves by $\gg 1$ per step), while θ, $\{\Delta_i X\}$, and $\{\Delta\Delta Y\}$ are slow. The branch bounds of Lemma 5.1(iii) give, per branch with offset j,

$$\text{(E6)}\quad \big|(cF)''\big| = \tfrac{945}{512}\,k|j|\,n^{-1/8}\,(1 + O(P^{-1/4})) + O(kh_1h_2P^{-5/8}).$$

The first summand is the offset monomial $c\cdot\frac{3}{2}j\,m^{1/2} = \frac{9}{8}kj\,\nu^{15/8}$, differentiated in ν with j frozen. The second is the zero-offset scale: on a cell the gaps β_i are frozen, and Lemma 5.2b computes the local curvature exactly as $-\frac{135}{1024}\,k\beta_1\beta_2\nu^{-13/8}$, which is $\asymp kh_1h_2\nu^{-5/8}$ and *negative*. (A model that differentiates the moving gaps $\Delta_iX(\nu)$ produces a different, positive leading coefficient $\frac{243}{128}$ and is not the local f''.) Finally, the run and cell inventories: level-1 gap cells at shift $2h$ number at most $1.5hP^{1/2} + 1$ with lengths in $[\frac{2}{3},\ 0.95]\,P^{1/2}/h$, and the frozen-floor runs of $\lfloor F_\kappa(X)\rfloor$ number at most $22(|j| + 1)P^{3/4}$ by the derivative bound of Lemma 5.1(iii).

Throughout, "A is dominated by B at margin $P^{-\rho}$" means $\sup|A|/\inf|B| \le CP^{-\rho}$ with the displayed C: the composite second derivative then keeps the sign and, up to the factor $1 + CP^{-\rho}$, the size of B, so Lemma 3.3 applies at B's scale.

The symbol P^ε is used only for factors of the form $C(\varepsilon)(\log P)^{O(1)}$. The number of Fourier/Vaaler expansion layers that produce those logarithms is three (the pieces M_2, M_3, M_4 of the master identity), independent of P and of k, h_1, h_2, t. Each layer contributes $O(\log P)$ mode mass; there is no further nesting of expansions whose depth would grow with P. Implied constants in every $\ll_\varepsilon$ of Sections 4–6 depend on ε and on (C1)–(C4) only: on those ranges they do not grow with k, h_i, or t. The threshold P_0 depends on ε (and is large for the Lemma 3.9 comparison $V \le c_7S/2$) but is independent of k, h_i, t.

Every numerical margin in this section is claimed only for $P \ge P_0$, with an absolute but ineffective P_0. The comparisons stratify. The Lemma 3.7 / Lemma 5.2 window hypotheses first hold together on the printed majorants at a moderate threshold (of size $3\cdot 10^5$). The interpolant error of Step 5b is $O(P^{-5/6})$ with leading coefficient 0.11, and is not absorbed into a smaller constant. The comparison $V \le c_7S/2$ at $c_7 = 1/288$ for the printed triple of Step 5b forces P_0 to be large (of size 10^{24}): until then a three-term zero of Φ'' can keep the sublevel Ω_V of length $\Theta(P)$ on a dyadic block, and the printed length $P^{89/96}$ is the bound of Lemma 3.9, not a claim that the sublevel is short at small P. The constants are not asserted to be sharp.

The proof of the kernel theorem classifies the differenced pieces into four classes; three are handled by standard tests, and the fourth — an exact level-2 wave $e(qY)$, possibly riding a frozen floor — is the genuinely new difficulty. We isolate it as a standalone lemma with its own differencing, so that it can be checked independently. The checkable objects, in order, are: Lemma 5.1 (exact identities); Lemma 5.2(i) (differenced wave); Lemma 5.2(ii) from (i) (mixed pieces); Lemma 5.2b (frozen-shape interpolant: local f'' versus global Λ); Theorem 5.3 Step 5a (offset composite); Theorem 5.3 Step 5b (three regimes, the middle band citing Lemma 5.2b and Lemma 3.9); Theorem 6.1 Step E (frozen-shape total phase $\Delta\Delta(\frac{k}{2}m^{9/4}) - \Delta\Delta(c\theta_2)$, composites $\frac{243}{512}$ and $\frac{1095}{1024}$). No later section re-derives these.

Lemma 5.2 (level-2 waves: the mixed-piece bound). Assume (C1)–(C4), write $\mathcal{D} = \{0, d_1, d_2, d_1{+}d_2\}$, and call a *decoration* any sum ρ of at most nine terms of the classes

- (D1) $q'\,\Delta_{2h}\Delta_{2h'}Y(n{+}d')$ with $|q'| \le 4P^{1/24}$, $1 \le h' \le 2P^{1/24}$, $d' \in \mathcal{D}$;
- (D2) $-\Delta_{2h}(c\,\lfloor F_\kappa(X)\rfloor)(n)$, with F_κ a branch function of Lemma 5.1(iii), offset $|j| \le 3$;
- (D3) a smooth φ with $|\varphi''| \le 3kh_1h_2P^{-5/8}$ and $|\varphi'''| \le 3kh_1h_2P^{-13/8}$;

here $2h$ is the shift of part (i), and in part (ii) the decorations are read after the differencing there. Then, for sums over $n \in (P, 2P]$, n odd:

(i) *(differenced wave)* for all integers $u, h \ge 1$ with $h \le P^{1/8}$, $uh \le P^{1/2}$, and $d \in \mathcal{D}$,

$$V := \Big|\sum_n e\big(u\,\Delta_{2h}Y(n{+}d) + \rho(n)\big)\Big| \ll \Big((uh)^{1/2}P^{5/8} + (h/u)^{1/2}P^{7/8} + P^{7/8}\Big)P^{\varepsilon};$$

(ii) *(waves)* for all integer coefficients $(q_d)_{d\in\mathcal{D}}$ with $|q_d| \le 4P^{1/24}$ and $t := \sum_{d\in\mathcal{D}} q_d \ne 0$, all $\varepsilon_0 \in \{0,1\}$, and φ of class (D3),

$$U := \Big|\sum_n e\Big(\sum_{d\in\mathcal{D}} q_d\,Y(n{+}d) - \varepsilon_0\,c(n)\lfloor F_\kappa(X(n))\rfloor + \varphi(n)\Big)\Big| \ll |t|^{-1/6}\,P^{23/24+\varepsilon}.$$

Part (ii) is the mixed-piece bound proper: level-2 waves $e(qY)$, possibly riding a frozen floor. Part (i) is the engine that (ii)'s targeted differencing feeds, and is also used directly for the $\{W\}$-content of the kernel proof. The exponent $\frac{23}{24}$ is not an accident: written exactly, the wave $e(qY)$ is the depth-2 object of Theorem 4.4, and part (ii) recovers exactly the depth-2 strength — no more. A model of U as $e(sX)$ with a frozen real coefficient $s \asymp tP^{3/4}$ would discard the sawtooth $-\frac{3}{2}tX^{1/2}\theta$ of amplitude $\asymp tP^{3/4}$ inside $tY = t(X-\theta)^{3/2}$; no such shortcut is taken anywhere below.

Proof of (ii) from (i). Replacing the summand by its conjugate we may take $t \ge 1$. Fix $e_1 \in \mathcal{D}$ with $q_{e_1} \ne 0$ and write, exactly (telescoping onto the base e_1),

$$\sum_{d\in\mathcal{D}} q_d\,Y(n{+}d) = t\,Y(n{+}e_1) + \sum_{d\ne e_1} q_d\,\Delta_{|d-e_1|}Y(n{+}\min(d, e_1)),$$

and the second sum is a set of at most three decoration seeds with shifts $|d - e_1|/2 \le h_1{+}h_2 \le 2P^{1/24}$ by (C4). Weyl differencing at shift $2h_3$, $1 \le h_3 \le H_3 := \lceil t^{1/3}P^{1/12}\rceil$, by the A-process recorded after Lemma 3.3:

$$|U|^2 \le \frac{P^2}{H_3} + \frac{2P}{H_3}\sum_{h_3=1}^{H_3}|V_{h_3}|, \qquad V_{h_3} = \sum_n e(t\,\Delta_{2h_3}Y(n{+}e_1) + \rho_{h_3}(n)),$$

where ρ_{h_3} collects $q_d\Delta_{2h_3}\Delta_{|d-e_1|}Y$ (class (D1) with $h' = |d{-}e_1|/2 \le 2P^{1/24}$), $-\varepsilon_0\Delta_{2h_3}(c\lfloor F(X)\rfloor)$ (class (D2)), and $\Delta_{2h_3}\varphi$ (class (D3): $|(\Delta_{2h_3}\varphi)''| \le 2h_3\sup|\varphi'''| \le 6kh_1h_2h_3P^{-13/8} \le 3kh_1h_2P^{-5/8}$ since $h_3 \le P^{1/4}$). The hypothesis of (i) holds: $th_3 \le t\cdot 2t^{1/3}P^{1/12} = 2t^{4/3}P^{1/12}$. With $|q_d| \le 4P^{1/24}$ one has $t \le 16P^{1/24}$, so $2t^{4/3}P^{1/12} \le 81P^{5/36} \le P^{1/2}$ for $P \ge P_0$, and $H_3 \le 3P^{7/72} \le P^{1/4}$. Part (i) gives

$$|U|^2 \le \frac{P^2}{H_3} + \Big(4t^{1/2}H_3^{1/2}P^{13/8} + 4t^{-1/2}H_3^{1/2}P^{15/8} + 4P^{15/8}\Big)P^{\varepsilon},$$

and at $H_3 = \lceil t^{1/3}P^{1/12}\rceil$ the four terms are, in order, $\le t^{-1/3}P^{23/12}$; $\le 6t^{2/3}P^{5/3} = 6\,(tP^{-1/4})\,t^{-1/3}P^{23/12} \le 96P^{1/24-1/4}\cdot t^{-1/3}P^{23/12}$; $\le 6t^{-1/3}P^{1/24}P^{15/8} = 6t^{-1/3}P^{23/12}$; and $\le 4\,(t^{1/3}P^{-1/24})\,t^{-1/3}P^{23/12} \le 11P^{1/72-1/24}\,t^{-1/3}P^{23/12}$. Hence $|U|^2 \ll t^{-1/3}P^{23/12+\varepsilon}$, which is (ii).

Proof of (i). Six stages; write $\nu = n + d$ (a shift of the block by at most $d_1{+}d_2 \le P^{1/2}$, harmless in every estimate) and $\Delta = \Delta_{2h}$.

Stage 1 (the m-linear form). By Lemma 4.3(i) at the bases ν and $\nu{+}2h$,

$$u\,\Delta Y = u\,A_h(\nu) + \tfrac{3u}{2}\,g(\nu)\,(\nu{+}2h)^{3/4} - \tfrac{3u}{2}\,\theta_\nu\,\Delta(\nu^{3/4}) + u\,\Delta E,$$

with $g = m(\nu{+}2h) - m(\nu)$, $|u\Delta E| \le uP^{-3/4}$ (total cost over the block $\le 2\pi u P^{1/4} \le 2\pi P^{3/4}$), and

$$A_h(\nu) = \tfrac{3}{2}\nu^{3/2}\Delta(\nu^{3/4}) - \tfrac{1}{2}\Delta(\nu^{9/4}) = -\tfrac{27}{8}h^2\nu^{1/4}(1 + O(hP^{-1})):$$

the two $\nu^{5/4}$-scale contributions cancel exactly at leading order (both equal $\frac{9}{4}h\nu^{5/4}$), leaving $|uA_h''| \le 0.64\,uh^2P^{-7/4}$. The sawtooth coefficient is

$$B(\nu) := \tfrac{3u}{2}\,\Delta(\nu^{3/4}) = \tfrac{9}{4}\,uh\,\xi^{-1/4} \in (1.89, 2.25]\,uhP^{-1/4}.$$

Stage 2 (gap cells and the exact shift device). Split into the level sets of $\lfloor \delta_h \rfloor$, $\delta_h(\nu) = (\nu{+}2h)^{3/2} - \nu^{3/2}$: at most $1.5hP^{1/2} + 1$ cells of length in $[\frac{2}{3}, 0.95]\,P^{1/2}/h$ (E1). On a cell $g = G + \kappa$ with $G = \lfloor \delta_h \rfloor$ frozen and $\kappa = [\theta_\nu \ge 1 - \{\delta_h\}]$; Vaaler-expand κ at truncation $R_0 = P^{1/4}$ (majorant cost $\le 4P/R_0 = 4P^{3/4}$). The moving endpoint costs nothing: as in Theorem 4.4, Step 4, the endpoint's smooth part contributes the exact phase $r\delta_h(\nu)$, and $e(r\nu^{3/2} + r\delta_h(\nu)) = e(r(\nu{+}2h)^{3/2})$, so every r-term falls into the two smooth families $e(r\nu^{3/2})$, $e(r(\nu{+}2h)^{3/2})$, $0 < |r| \le R_0$, with weights $\le 1/|r|$; the in-cell variation of the coefficient is exactly 1 and one Abel summation absorbs it at a factor $\le 2{+}2\pi$.

Stage 3 (the θ-sawtooth). Two regimes. (s1) If $uh \le P^{3/16}$ then $|B| \le 2.25P^{-1/16} < \frac{1}{2}$ for $P \ge P_0$: Lemma 3.7 with $T = P^{1/2}$, $J = R_0$ (hypothesis $T \ge 8(1 + |B|)$: $P^{1/2} \ge 9$ for $P \ge P_0$) expands $e(-B_0\{\nu^{3/2}\}\dots)$ per window (here one window: the total drift of B is $\le 0.6uhP^{-1/4} \le 0.6P^{-1/16} < 1$) into the same two mode families at coefficient factor $\min(2, 2\pi|B|) \le 14P^{-1/16}$, mass $O(\log P)$, flat cost $\le 12P^{1/2}$, majorant $\le 14P^{-1/16} \cdot 4P^{3/4}$. (s2) If $P^{3/16} < uh \le P^{1/2}$ then $|B| \le 2.25P^{1/4}$ and B drifts by at most 1 on windows of length $\ge P^{5/4}/(0.6uh) \ge 1.2P^{3/4}$: at most $0.6P^{1/4} + 1$ windows. Per window Lemma 3.7 at the centre B_0 ($T = P^{1/2} \ge 8(1 + 2.25P^{1/4})$ for $P \ge P_0$, since $P^{1/4} \ge 19$; flat cost $\le 8(1{+}2.25P^{1/4})P^{1/2} \le 27P^{3/4}$ in total) produces modes $e(w\nu^{3/2})$ whose coefficients decay as $\min(2, \frac{1}{\pi|w+B_0|}) + \min(2, \frac{1}{\pi|w|})$; the window-boundary cost is, using $uh > P^{3/16}$, $\le (0.6P^{1/4}{+}1) \cdot 1.83\,(0.30\,uh)^{-1/2}P^{3/8} \le 2.1\,P^{1/4-3/32+3/8} = 2.1\,P^{17/32} \le P^{5/8}$. Modes with $|w|$ in the collision window are treated in Stage 5.

Stage 4 (the main curvature, $r = w = 0$). On a cell the remaining phase is $\frac{3u}{2}G(\nu{+}2h)^{3/4} + uA_h +$ (smooth decorations), with second derivative

$$-\tfrac{9}{32}\,uG\,(\nu{+}2h)^{-5/4} \in -[0.30,\ 1.35]\,uh\,P^{-3/4} \qquad \text{(E2)},$$

single-signed with in-cell ratio $\sup/\inf \le 4.5$: the second derivative of $\frac{3u}{2}G(\nu{+}2h)^{3/4}$ is the single term $\frac{3u}{2}G \cdot \frac{3}{4} \cdot (-\frac{1}{4})(\nu{+}2h)^{-5/4}$, with G frozen and bounded by (E2). The competitor uA_h'' has ratio $\le 0.64uh^2P^{-7/4}/(0.30uhP^{-3/4}) \le 2.2hP^{-1} \le 2.2P^{-3/4}$; decoration competitors are bounded in Stage 6. Lemma 3.3 per cell and summation give

$$\sum_{\text{cells}} \Big(\ell_i\lambda^{1/2} + \lambda^{-1/2}\Big) \le 2.3\,(uh)^{1/2}P^{5/8} + 2.8\,(h/u)^{1/2}P^{7/8}.$$

Stage 5 (nonzero modes and collisions). A mode $w \ne 0$ (or $r \ne 0$; identical treatment with weight $1/|r|$) adds the phase $w\nu^{3/2}$ of curvature $\ge 0.53|w|P^{-1/2}$. If $0.53|w|P^{-1/2} \ge 4 \cdot 1.35uhP^{-3/4}$, i.e. $|w| \ge 10.2\,uhP^{-1/4}$, the mode curvature dominates at margin ≥ 4: Lemma 3.3 over the full block gives $\le 1.4|w|^{1/2}P^{3/4} + 1.4|w|^{-1/2}P^{1/4}$, and the coefficient-weighted sums are $\le 3R_0^{1/2}P^{3/4}\log P = 3P^{7/8}\log P$ (families of Stage 2, regime (s1)) and $\le CP^{7/8}\log P$ (regime (s2) tails). If $|w| \le 0.1\,uhP^{-1/4}$ the main curvature dominates at margin ≥ 4 and the mode rides along at Stage 4's scale, weight summable to $O(\log P)$. In the remaining collision window $0.1\,uhP^{-1/4} \le |w| \le 10.2\,uhP^{-1/4}$ (nonempty only in regime (s2)), the two curvatures can cancel: there the phase is the two-term monomial $a\nu^{3/4}+w\nu^{3/2}$ with $a=\frac{3u}{2}G \asymp uhP^{1/2}$ plus perturbations already shown ρ_0-small, and Lemma 3.8 with $(\alpha,\beta)=(\frac{3}{4},\frac{3}{2})$ applies per window with $M\in[0.03,\,11]\,uhP^{-3/4}$ (both curvature scales lie in this window on the collision band, and $M\le 11P^{-1/4}$):

$$\le C_E(|I_w|M^{1/2}+M^{-1/2}+(P/M)^{1/3})\quad \text{per window.}$$

Summing over the at most $0.6P^{1/4}{+}1$ windows — the transition cost $(P/M)^{1/3}$ carries no window-length factor, which is why it sums — with the $O(\log P)$ collision-band coefficient mass:

$$\sum_w |I_w|M^{1/2}\le 3.4\,(uh)^{1/2}P^{5/8},\qquad \sum_w M^{-1/2}\le 4.5\,(uh)^{-1/2}P^{5/8}\le 4.5\,P^{5/8},$$

$$\sum_w (P/M)^{1/3}\le 0.77P^{1/4}\cdot 3.3\,(uh)^{-1/3}P^{7/12}=2.5\,(uh)^{-1/3}P^{5/6}\le 2.5\,P^{5/6-1/16}=2.5\,P^{37/48},$$

the last using $uh>P^{3/16}$ in regime (s2). The collision-band total is $\le C((uh)^{1/2}P^{5/8}+P^{5/6})\log P\le CP^{7/8}\log P$.

Stage 6 (decorations). Each class is dominated at a displayed margin against the Stage-4 curvature $\ge 0.30uhP^{-3/4}$.

- (D1). On the branch decomposition of Lemma 5.1(iii) at the shift pair (h_3-role h,h'): the arcs are absorbed by the shift device (log mass); the b-run boundaries add $\le 1.5(h{+}2h')P^{1/2}+2$ cells, at boundary cost $\le 1.5(h{+}2h')P^{1/2}\cdot 1.83(0.30uh)^{-1/2}P^{3/8}\le 5.1(h/u)^{1/2}P^{7/8}+10.3\,h'(uh)^{-1/2}P^{7/8}$ (the second term is what the H_3-averaging of part (ii) absorbs: it contributes $\le 2h't^{-2/3}P^{43/24}\le P^{1/24-1/8}\,t^{-1/3}P^{23/12}$ there). The θ-coefficient of the decoration is $\le |q'|(2|j'|P^{-1/4}+20hh'P^{-3/4})\le 24P^{1/24-1/4}+160P^{1/24+1/8+1/24-3/4}=24P^{-5/24}+160P^{-13/24}$. This is $O(P^{-5/24})$, hence $|B|\le 1$ for $P\ge P_0$; Lemma 3.7 at $T=P^{1/2}$ applies either way (even a constant-size leftover would satisfy $T\ge 8(1+|B|)$). The modes are expanded in the Stage-2 families. Its smooth curvature has ratio $\le |q'|(2|j'|P^{-5/4}+25hh'P^{-7/4})/(0.30uhP^{-3/4})\le 80P^{1/24-1/2}+672P^{1/24+1/24-1}\le P^{-1/4}$: dominated.
- (D2). Write $\Delta(c\lfloor F\rfloor)=(\Delta c)\lfloor F\rfloor+c_+\Delta\lfloor F\rfloor$ with $c_+=c(\cdot{+}2h)$.
  - (a) $(\Delta c)\lfloor F\rfloor=(\Delta c)F-(\Delta c)\{F\}$: the smooth part has curvature $\le ((\Delta c)F)''\le 7kh|j|P^{-9/8}+28khh_1h_2P^{-13/8}$, ratio to the main curvature $\le 24k|j|P^{-3/8}/u\le 72P^{1/24-3/8}$: dominated. The sawtooth part has coefficient $\Delta c\in(1.68,1.85)\,khP^{1/8}$, handled per windows of drift ≤ 1: at most $2khP^{1/8}{+}1$ windows, at boundary cost $\le 2khP^{1/8}\cdot 1.83\,(0.30\,uh)^{-1/2}P^{3/8}\le 7k\,(h/u)^{1/2}P^{1/2}$. Per window, Lemma 3.7 at $T=P^{1/2}$ yields modes $e(q''(F\circ X))$ of curvature $\le (2khP^{1/8}{+}P^{1/2})\cdot 3|j|P^{-5/4}\le 18P^{-3/4}\cdot P^{-1/16}$, ratio to the main curvature $\le 60P^{-1/16}$: dominated. The flat cost is, per point, $8(1{+}B_0)/T$, so in total $\le 8(1{+}1.85khP^{1/8})P^{1/2}\le 8P^{1/2}+15\,khP^{5/8}\le 23\,P^{1/24+1/8+5/8}=23\,P^{19/24}\le P^{7/8}$, using $h\le P^{1/8}$ and (C3).
  - (b) $c_+\Delta\lfloor F\rfloor$: by the gap identity applied to the sequence $F\circ X$, $\Delta\lfloor F\rfloor=\lfloor\Delta F\rfloor+\kappa_F$. ΔF has total drift $\le P\cdot 2h\sup|(F\circ X)''|\le P\cdot 2h(2|j|P^{-5/4}+25h_1h_2P^{-7/4})\le 13hP^{-1/4}+50h\,h_1h_2P^{-3/4}<1$, so $\lfloor\Delta F\rfloor$ takes at most two values, on at most two intervals; per value the phase $-q_0c_+$ ($|q_0|\le 2{+}6hP^{-1/4}\le 3$) has curvature $\le 0.4kP^{-7/8}$, ratio $\le 1.4kP^{-1/8}/(uh)\le 1.4P^{1/24-1/8}$: dominated. The carry $\kappa_F=\{F\}+\{\Delta F\}-\{F{+}\Delta F\}$ is a sum of unit sawtooths in slow variables (all drifts <1 per step): Lemma 3.5 at truncation $P^{1/8}$ (majorant $\le 3\cdot 4P^{7/8}$) gives modes $e(q''(F\circ X))$-type of curvature $\le P^{1/8}\cdot 3|j|P^{-5/4}\le 9P^{-9/8}$, ratio $\le 30P^{-3/8}$: dominated.
- (D3). Ratio $\le 2h\sup|\varphi'''|/(0.30uhP^{-3/4})\le 20kh_1h_2P^{-7/8}/u\le 20P^{1/8-7/8}$: dominated.

Totals. Stages 1–6 bound V by

$$C\Big((uh)^{1/2}P^{5/8}+(h/u)^{1/2}P^{7/8}+P^{7/8}+k\,(h/u)^{1/2}P^{1/2}\Big)\log^3 P,$$

and the fourth term is $\le P^{1/24}\cdot(h/u)^{1/2}P^{1/2}$, absorbed by the second. This is (i) up to the stated P^{ε}.

Costs collected. Each class contributes to V as follows, all times P^{ε}.

Class	Bound contributed to V	Absorbed by
Stage 1 remainder $u\Delta E$	$P^{3/4}$	$P^{7/8}$
Stage 2 majorant / shift device	$P^{3/4}$	$P^{7/8}$
Stage 3 (s1) sawtooth	$P^{3/4}$	$P^{7/8}$
Stage 3 (s2) windows / collision	$(uh)^{1/2}P^{5/8} + P^{7/8}$	(i)
Stage 4, $r = w = 0$	$(uh)^{1/2}P^{5/8} + (h/u)^{1/2}P^{7/8}$	(i)
Stage 5, non-collision modes	$P^{7/8}$	(i)
(D1) curvature	dominated at $P^{-1/4}$	Stage 4
(D1) run boundaries	$(h/u)^{1/2}P^{7/8} + h'(uh)^{-1/2}P^{7/8}$	(i); H_3-average
(D2)(a) smooth / sawtooth	dominated; $P^{7/8}$ flat	Stage 4; (i)
(D2)(b) gap / carry	dominated; $P^{7/8}$ majorant	Stage 4; (i)
(D3)	dominated at $P^{-3/4}$	Stage 4

No decoration class is used later unless it appears in this table.

Checklist (author reconstruction, not an independent verification). The third-Weyl parameter is admissible: $1 \le h_3 \le H_3 = \lceil t^{1/3}P^{1/12}\rceil \le 3P^{7/72} \le P^{1/4}$ at $t \le 16P^{1/24}$. After that differencing, (D1) is the telescoped $q_d\Delta_{2h_3}\Delta_{|d-e_1|}Y$ with $h' \le 2P^{1/24}$; (D2) is $-\varepsilon_0\Delta_{2h_3}(c\lfloor F\rfloor)$; (D3) is $\Delta_{2h_3}\varphi$ with $|(\Delta_{2h_3}\varphi)''| \le 6kh_1h_2h_3P^{-13/8} \le 3kh_1h_2P^{-5/8}$. Collision windows are only the band $0.1\,uhP^{-1/4} \le |w| \le 10.2\,uhP^{-1/4}$ of Stage 5, and only in regime (s2). On a cell the integer $G = \lfloor \delta_h \rfloor$ is held frozen; the smooth remainder A_h does not differentiate a frozen gap. Lemma 3.9 is not used in (i) or (ii); it is used only on the global interpolant Φ of Lemma 5.2b. Every k, h_1, h_2 dependence passes through (C1)–(C4). $\square$

The balance to keep in mind: with the trivial bound $|V_{h_3}| \le P$, part (ii)'s differencing returns nothing beyond the choice of H_3; the content of the lemma is that the floor, coefficient, and carry bookkeeping of Stages 3–6 survives the targeted differencing with the Stage-4 curvature intact. What is *not* available is the frozen-coefficient shortcut: per frozen run the run length $P^{1/4}$ is shorter than $\lambda^{-1/2} \asymp (uh)^{-1/2}P^{3/8}$ for small uh, and the run-boundary cost $(h/u)^{1/2}P^{7/8}$, summed against the H_3-average, is exactly what pins the final exponent at $\frac{23}{24}$.

The remaining object that Theorem 5.3 must not confuse with Lemma 5.2 is the *zero-offset anchor*: the phase $c(G_F - J_F)$ on a branch with $j = 0$. On a cell the integers β_1, β_2, J_F are frozen, so $(cG_F)'' = c''G_F + 2c'G_F' + cG_F''$ is computed with those integers held constant while $X(\nu)$ and $c(\nu)$ move. The interpolant used on the whole block is the same frozen-shape formula with the smooth gap values $\Delta_i X(\nu)$ substituted as numbers, not differentiated. The next lemma records that computation, so that Step 5b is only a classification of scales.

Lemma 5.2b (frozen-shape interpolant on a zero-offset piece). Assume (C1)–(C4) and $j = 0$. Work on a common-refinement piece of the gap cells of both shifts, the frozen-floor runs of $\lfloor F_\kappa(X)\rfloor$, and the sawtooth windows already counted, so that the integers G_1, G_2, β_1, β_2, and $J_F = \lfloor F_\kappa(X)\rfloor$ are constant. Write $G_F = F_\kappa \circ X$. The remaining phase on the piece has second derivative

$$f'' = -\tfrac{9}{32}\Big(uG_1(\nu{+}2h_1)^{-5/4} + u'G_2(\nu{+}2h_2)^{-5/4}\Big) + 2c'G_F' + c\,G_F'' + c''(G_F - J_F) + wX'' + O(\rho_0),$$

where the $O(\rho_0)$ collects decorations already shown small against the scales of Lemma 3.8, and the chain rule uses *frozen* β_i:

$$G_F' = F'(X)\,X', \qquad G_F'' = F''(X)\,(X')^2 + F'(X)\,X'',$$

with, from Lemma 5.1(iii) at $j = 0$,

$$F'(m) = -\tfrac{3}{8}\beta_1\beta_2\,m^{-3/2}, \qquad F''(m) = \tfrac{9}{16}\beta_1\beta_2\,m^{-5/2}$$

up to the relative $O(\beta/m) = O(P^{-1})$ of the mean-value ξ. Define the *frozen-shape interpolant* Λ by substituting the smooth gap values $\widetilde{\beta}_i(\nu) = \Delta_i X(\nu)$ and $\delta_{h_i}(\nu) = (\nu{+}2h_i)^{3/2} - \nu^{3/2}$ for those frozen integers *as* *values only.* The ν-derivatives of $\tilde\beta_i$ are not introduced:

$$\begin{aligned}\Lambda(\nu) = &-\tfrac{9}{32}u\,\delta_{h_1}(\nu)\,(\nu{+}2h_1)^{-5/4} - \tfrac{9}{32}u'\,\delta_{h_2}(\nu)\,(\nu{+}2h_2)^{-5/4}\\ &+ 2c'(\nu)\,\widetilde{F}'(X(\nu))\,X'(\nu) + c(\nu)\,\widetilde{G}''(\nu) + \tfrac12 c''(\nu) + wX''(\nu),\end{aligned}$$

where $\widetilde{F}'(m) = -\frac38\tilde\beta_1\tilde\beta_2\,m^{-3/2}$ and $\widetilde{G}''$ is the frozen-shape formula for G_F'' evaluated at $\tilde\beta_i$. Then, for $P \ge P_0$,

$$|f'' - \Lambda| \le \tfrac{9}{32}(u{+}u')P^{-5/4} + |c''| + 8k(h_1{+}h_2)P^{-9/8} \le 219P^{-25/24} + 0.11P^{-5/6}.$$

The second term is the leading interpolant error and is not absorbed into a smaller multiple of $P^{-5/6}$. Moreover $\Lambda = \Phi'' + r$, where

$$\Phi(\nu) = a\,\nu^{5/4} + b\,\nu^{11/8} + w\,\nu^{3/2}, \qquad a = -\tfrac{27}{10}(uh_1 + u'h_2), \quad b = -\tfrac{405}{176}\,kh_1h_2,$$

the exponents $(\frac54, \frac{11}{8}, \frac32)$ lie in E and are pairwise distinct, and r obeys the $\rho_0(E)$ bounds of Lemma 3.9. If $w = 0$, drop the third term. The local frozen curvature of the anchor itself is

$$2c'G_F' + c\,G_F'' = -\tfrac{135}{1024}\,k\beta_1\beta_2\,\nu^{-13/8}(1 + O(P^{-1/4})),$$

and therefore

$$\lambda_0 := \left|\big(c(G_F - J_F)\big)''\right| \in [0.35,\ 2.6]\,kh_1h_2P^{-5/8}$$

on the standing range: the summand $c''(G_F - J_F)$ is $O(kP^{-7/8})$ and sits in the $O(P^{-1/4})$ relative error.

Proof. There are three replacements of a frozen integer by a smooth gap of the same scale, and each moves its argument by at most 1.

(i) $|G_i - \delta_{h_i}| < 2$, so the wave terms differ from Λ by at most $\frac{9}{32}(u{+}u')P^{-5/4}$. In the middle band of Step 5b one has $u, u' \le 360P^{5/24}$, hence this is $\le 219P^{-25/24}$.

(ii) $|\beta_i - \tilde\beta_i| \le 1$, so the product $\beta_1\beta_2$ moves by $O((h_1{+}h_2)P^{1/2})$. The frozen-shape second derivative is linear in that product and of size $k\beta_1\beta_2\nu^{-13/8}$; the difference is therefore $O\big(k(h_1{+}h_2)P^{1/2}P^{-13/8}\big) = O\big(k(h_1{+}h_2)P^{-9/8}\big)$, which is the third displayed term.

(iii) $|G_F - J_F| < 1$, and Λ replaces this fractional part by $\frac12$ only in the already-expanded c''-term. The difference is at most $|c''| \le 0.11kP^{-7/8} \le 0.11P^{-5/6}$. This replacement is *not* made in the phase $c(G_F - J_F)$: substituting $J_F \mapsto G_F - \frac12$ there would collapse the anchor to $c/2$ and destroy the curvature.

These are all the differences between f'' and Λ. In particular Λ is not $(cF_{\mathrm{sm}})''$ for $F_{\mathrm{sm}} = \frac34(\Delta_1X)(\Delta_2X)X^{-1/2}$. That composite differentiates the moving gaps and is a different function.

It remains to match leading monomials. Mean-value expansion gives $\delta_h(\nu) = 3h\,\xi^{1/2}$ with $\xi \in (\nu, \nu{+}2h)$, hence $\delta_h(\nu)\,(\nu{+}2h)^{-5/4} = 3h\,\nu^{-3/4}(1 + O(hP^{-1}))$ and

$$-\tfrac{9}{32}u\,\delta_{h_1}(\nu)\,(\nu{+}2h_1)^{-5/4} = -\tfrac{27}{32}uh_1\,\nu^{-3/4}(1 + O(h_1P^{-1})).$$

This is Φ'' of $a\nu^{5/4}$, because $a \cdot \frac54 \cdot \frac14 = -\frac{27}{32}$.

For the anchor, keep β_1, β_2 frozen and expand the chain rule at $c = \frac{3k}{4}\nu^{9/8}$, $X = \nu^{3/2}$. Then $G_F = \frac34\beta_1\beta_2X^{-1/2} = \frac34\beta_1\beta_2\nu^{-3/4}$ at leading order, $G_F' = -\frac{9}{16}\beta_1\beta_2\nu^{-7/4}$, and $G_F'' = \frac{63}{64}\beta_1\beta_2\nu^{-11/4}$, so

$$\begin{aligned} c''G_F &= \tfrac{81}{1024}\,k\beta_1\beta_2\nu^{-13/8},\\ 2c'G_F' &= -\tfrac{972}{1024}\,k\beta_1\beta_2\nu^{-13/8},\\ c\,G_F'' &= \tfrac{756}{1024}\,k\beta_1\beta_2\nu^{-13/8}.\end{aligned}$$

The sum is $-\frac{135}{1024}\,k\beta_1\beta_2\nu^{-13/8}$. (The term $\frac{1}{2}c''$ in Λ is $O(kP^{-7/8})$ and is absorbed in r.) Substituting the interpolating values $\tilde{\beta}_i = 3h_i\nu^{1/2}(1+O(hP^{-1}))$ converts the product $\beta_1\beta_2$ into $9h_1h_2\nu$, and the curvature becomes $-\frac{1215}{1024}kh_1h_2\nu^{-5/8}$. This is Φ'' of $b\nu^{11/8}$, because

$$b\cdot\tfrac{11}{8}\cdot\tfrac{3}{8} = -\tfrac{405}{176}\cdot\tfrac{33}{64} = -\tfrac{1215}{1024}.$$

The window mode is exactly $wX'' = \frac{3}{4}w\nu^{-1/2}$. Write $r = \Lambda - \Phi''$, so that Lemma 3.9 is applied to $\Phi+g$ with $g'' = r$. The remainder r is the sum of the mean-value errors $O(hP^{-1})$ on the two leading monomials and the leftover $\frac{1}{2}c''$. In the middle band of Step 5b one has $S \ge 0.35\,kh_1h_2P^{-5/8}$ (and $S \ge 0.35P^{-5/8}$ whenever $kh_1h_2 \ge 1$). The three ratios Lemma 3.9 needs are then, for $P \ge P_0$,

$$\frac{|r|}{S} \le C_1P^{-1/4}, \qquad \frac{P|r'|}{S} \le C_2P^{-1/4}, \qquad \frac{P^2|r''|}{S} \le C_3P^{-1/4},$$

with absolute C_1, C_2, C_3 coming from $|\frac{1}{2}c''| \le 0.053kP^{-7/8}$, $|\frac{1}{2}c'''| \le 0.047kP^{-15/8}$, $|\frac{1}{2}c''''| \le 0.044kP^{-23/8}$, and the wave remainders $O(uh^2\nu^{-7/4})$ which are $O(P^{-35/24})$ on the printed inventory. Each ratio is therefore $\le \rho_0(E) = c_7/8 = 1/2304$ for $P \ge P_0$. (The worst standing cell is $kh_1h_2 = 1$; larger products only enlarge S.)

The range of λ_0 is the same leading term against $\beta_i \in (3h_iP^{1/2}-1,\, 4.3h_iP^{1/2}+1)$ and $\nu \in (P, 2P]$: the product $\beta_1\beta_2\nu^{-13/8}$ runs through $[2.92, 18.5]\,h_1h_2P^{-5/8}$, and $\frac{135}{1024} \approx 0.132$ converts this to $[0.38, 2.44]$, opened to $[0.35, 2.6]$ for the $O(P^{-1/4})$ and the ± 1 in the β-bounds. $\square$

The global monomial $\nu^{11/8}$ appears only after $\tilde{\beta}_1\tilde{\beta}_2 \sim \nu$ is substituted. On a single frozen run the curvature is a multiple of $\nu^{-13/8}$, i.e. the second derivative of a $\nu^{3/8}$ phase. Lemma 3.9 is applied to the *global* model Φ, whose second derivative tracks Λ; the local f'' stays within the interpolant error of that model, which is the only comparison the sublevel argument uses.

Theorem 5.3 (kernel cancellation). Let $c(n) = \frac{3k}{4}n^{9/8}$ with $1 \le k \le P^{1/24}$. Then

$$K_c(P) \ll P^{1-1/96+\varepsilon},$$

uniformly in k. The cancellation in Step 5a uses the exact ratio of this monomial's derivatives; a two-sided scale $c^{(r)} \asymp kP^{9/8-r}$ would not determine the composite.

Proof. Work on a dyadic block $n \sim P$, odd.

Step 1 (double Weyl differencing). Set $H_1 = P^{1/48}$, $H_2 = P^{1/24}$. The A-process recorded after Lemma 3.3, applied twice, gives

$$|K_c|^2 \le \frac{2P^2}{H_1} + \frac{4P}{H_1}\sum_{h_1=1}^{H_1}|T_1(h_1)|, \qquad |T_1|^2 \le \frac{2P^2}{H_2} + \frac{4P}{H_2}\sum_{h_2=1}^{H_2}|T_2(h_1,h_2)|,$$

with $T_2 = \sum_n e(\varphi_2)$, $\varphi_2 = \Delta\Delta(c\,\theta_2)$. For all $h_1 \le H_1$, $h_2 \le H_2$, $k \le P^{1/24}$: $kh_1h_2 \le P^{1/24+1/48+1/24} = P^{5/48} \le P^{1/8}$, so (C1)–(C4) hold with room $P^{-1/48}$ on (C1) (and (C4) is $P^{1/48}, P^{1/24} \le P^{1/24}$). We prove

$$|T_2| \ll P^{23/24+\varepsilon} \quad \text{uniformly in } h_1, h_2, k;$$

then $|T_1| \ll P^{1-1/48+\varepsilon}$ and $|K_c| \ll P^{1-1/96+\varepsilon}$, the three savings balancing exactly at the chosen H_1, H_2.

Step 2 (master identity; the trivial piece). By Lemma 5.1(iv), exactly,

$$\varphi_2 = \underbrace{(\Delta\Delta c)\,\theta_2}_{M_1} + \underbrace{(\Delta_2 c)(n{+}d_1)(\{W\}-\kappa_2)}_{M_2} + \underbrace{(\Delta_1 c)(n{+}d_2)(\{W'\}-\kappa_2')}_{M_3} + \underbrace{c_{11}(\{\Delta\Delta Y\}-\kappa''-\Delta_2\kappa_2)}_{M_4}.$$

By (E4), $|M_1| \le 0.43\,kh_1h_2P^{-7/8}$, so $|e(M_1)-1| \le 2.7\,kh_1h_2P^{-7/8}$ and deleting M_1 changes T_2 by at most $2.7\,kh_1h_2P^{1/8} \le 2.7\,P^{1/4}$ by (C1). No growing smooth part remains: every other bracket is bounded by 2.

Step 3 (the expansion inventory). The three remaining pieces are expanded as follows; all truncations, masses, and error costs are displayed here once and summed in Step 6.

(3a) M_2, *sawtooth part* $(\Delta_2 c)(n{+}d_1)\{W\}$. The coefficient $B(n) = \Delta_2 c(n{+}d_1) \in (1.68, 1.85)kh_2P^{1/8}$ drifts by $|B'| \le 0.22kh_2P^{-7/8}$ per step: freeze it on at most $2kh_2P^{1/4}{+}1$ windows on which it moves by $\le P^{-1/8}$ (residual cost $\sum_n |e((B{-}B_0)\{W\}) - 1| \le 6.3P^{-1/8} \cdot P = 6.3P^{7/8}$). Per window, Lemma 3.7 at the centre B_0 with $T = P^{1/2}/(2h_1)$, $J = P^{1/4}$ (hypothesis $T \ge 8(1 + |B|)$: $P^{1/2}/(2h_1) \ge \frac{1}{2}P^{23/48}$ and $8(1 + |B|) \le 15\,kh_2P^{1/8} \le 15P^{10/48}$, so the inequality holds for $P \ge P_0$): flat cost in total $\le 8(1{+}1.85kh_2P^{1/8}){\cdot}2h_1P^{1/2} \le 16h_1P^{1/2} + 30\,kh_1h_2P^{5/8} \le 46P^{3/4}$ by (C1); the majorant Δ_J-part costs $\le 2 \cdot 4P/J = 8P^{3/4}$ plus J-mode sums at coefficients $\le 1/(J{+}1)$; modes $e(uW)$ with mass $\le C\log P$ and $u \le 1.85kh_2P^{1/8} + P^{1/2}/(2h_1)$, so that $uh_1 \le 1.85P^{1/4} + P^{1/2}/2 \le P^{1/2}$: every mode is a Lemma 5.2(i) object at shift $h_1 \le P^{1/48}$. Window boundaries cost $\le 2kh_2P^{1/4} \cdot 3.4P^{3/8} \le 7P^{17/24}$.

(3b) M_2, *carry part* $(\Delta_2 c)(n{+}d_1)\kappa_2$. Here $\kappa_2 \in \{0, 1\}$ is an integer, so exactly $e(-(\Delta_2 c)\kappa_2) = 1 + \kappa_2(e(-\Delta_2 c) - 1)$: the large coefficient never multiplies a sawtooth inside an exponential. The factor $e(-\Delta_2 c(n{+}d_1))$ is a smooth phase with $|(\Delta_2 c)''| \le 2h_2 \sup|c'''| \le 0.19kh_2P^{-15/8}$: class (D3). The weight $\kappa_2 = \{Y\}{+}\{W\}{-}\{Y{+}W\}$ (Lemma 5.1(ii)) is expanded *additively* as real numbers, each unit sawtooth by finite Fourier (Lemma 3.5) at truncation $J_2 = P^{1/24}$, coefficients $\le 1/(\pi|q|)$, majorant cost $\le 4P/J_2 = 4P^{23/24}$ per layer plus J_2-mode averages. Since $Y{+}W = Y(n{+}d_1)$ exactly, the resulting modes are $e(qY(n))$, $e(qY(n{+}d_1))$ — Lemma 5.2(ii) objects — and $e(qW)$ — Lemma 5.2(i) objects.

(3c) M_3, *sawtooth and carry.* The same two expansions as (3a)–(3b), with the roles of the shifts exchanged: $B(n) = \Delta_1 c(n{+}d_2) \in (1.68, 1.85)\,kh_1P^{1/8}$ and Lemma 3.7 at $T = P^{1/2}/(2h_2)$. The window hypothesis is $T \ge \frac{1}{2}P^{11/24} = \frac{1}{2}P^{22/48}$ (since $h_2 \le P^{1/24}$) against $8(1{+}|B|) \le 15\,kh_1P^{1/8} \le 15P^{9/48}$, so it holds for $P \ge P_0$ with more room than (3a). Modes $e(uW')$ satisfy $uh_2 \le 1.85\,kh_1h_2P^{1/8} + P^{1/2}/2 \le P^{1/2}$ by (C1) and are Lemma 5.2(i) objects at shift $h_2 \le P^{1/24}$. Window boundaries cost $\le 2kh_1P^{1/4} \cdot 3.4P^{3/8} \le 7P^{11/16}$. The carry expansion is identical to (3b) at truncation $J_2 = P^{1/24}$, and produces $e(qY(n{+}d_2))$ and $e(qW')$.

(3d) M_4, *carries.* $\kappa'' = \{W\} + \{\Delta\Delta Y\} - \{W{+}\Delta\Delta Y\}$ with $W{+}\Delta\Delta Y = W(n{+}d_2)$ exactly, and $\Delta_2\kappa_2 = \kappa_2(n{+}d_2) - \kappa_2(n)$ with each κ_2 expanded as in (3b) at its base. The smooth factors $e(\mp c_{11})$ have $|c''| \le 0.11kP^{-7/8}$: class (D3). New mode species: $e(q\{\Delta\Delta Y\}\text{-content})$ — slow modes, treated in Step 5 — and $e(qY(n{+}d))$ for $d \in \{0, d_2, d_1{+}d_2\}$, $e(qW(n{+}d_2))$: Lemma 5.2 objects again.

(3e) M_4, *the anchor.* The factor $e(c_{11}\{\Delta\Delta Y\})$ is never expanded away. On the branch decomposition of Lemma 5.1(iii) (arcs absorbed by the exact shift device at $O(\log P)$ mass and $4P^{3/4}$ majorant, as in Theorem 4.4, Step 4), $\{\Delta\Delta Y\} = G - J_F$ with $G = F_\kappa(X{-}\theta)$ and $J_F = \lfloor G \rfloor$ frozen on the runs of (E6). The anchor phase $c_{11}(G - J_F)$ is present in **every** final piece; its curvature scale (E6) is what orders the whole classification of Step 5.

After (3a)–(3e), every final piece of T_2 has the exact form

$$e\Big(\underbrace{c_{11}(G - J_F)}_{\text{anchor}} + \underbrace{\textstyle\sum_i q_i Y(n{+}e_i)}_{\text{waves, } e_i \in \mathcal{D}} + \underbrace{u\,W\text{- and } u'W'\text{-modes}}_{\text{differenced waves}} + \underbrace{\text{slow modes} + \varphi}_{\text{(D3)-smooth}}\Big)$$

with at most one mode from each expansion layer (three layers), and piece weights whose total mass is $O(\log^3 P)$ beyond the displayed majorant and flat costs.

Step 4 (pieces with wave content: Lemma 5.2(ii)). Let $t = \sum_i q_i$ be the total wave frequency, $|t| \le 3J_2 \le 12P^{1/24}$, inside the coefficient budget of Lemma 5.2(ii).

If $t \ne 0$: telescoping, $\sum_i q_i Y(n{+}e_i) = t\,Y(n{+}e_1) + \sum_{i \ge 2} q_i\,\Delta_{e_i - e_1} Y(n{+}\min(e_1, e_i))$, and after the targeted differencing of Lemma 5.2(ii) the differenced-wave remainders and the u, u'-modes become (D1) decorations, the anchor becomes the (D2) decoration, and the smooth content is (D3): Lemma 5.2(ii) applies and gives $\ll |t|^{-1/6}P^{23/24+\varepsilon}$ per piece. The weight sum is

$$\sum_{t \ne 0}\ \sum_{q_1+q_2+q_3=t} \frac{1}{\pi^3 \max(1,|q_1|)\max(1,|q_2|)\max(1,|q_3|)}\,|t|^{-1/6} \ll \sum_{t \ge 1} \frac{\log^2(2{+}t)}{t^{7/6}} \ll 1,$$

so the total wave-piece contribution is $\ll P^{23/24+\varepsilon}$.

If $t = 0$: the wave content collapses exactly to differenced waves — e.g. $q(Y(n+d_2) - Y(n)) = q\,W'(n)$ — i.e. to (D1)-type resonant decorations of the remaining piece; these are handled in Step 5. This exactness is the reason no near-resonant analysis is ever needed for the wave frequencies: the resonant combination is an identity, not an approximation.

Step 5 (pieces without wave content: the anchor classes). The remaining pieces carry the anchor, at most two differenced-wave modes $u\,W$, $u'W'$ (including the resonant remnants of Step 4, which by Lemma 5.1(iii) are smooth-per-branch with θ-coefficients $O(P^{-5/24})$), slow modes, and (D3)-smooth content. Split by the anchor branch type.

(5a) Offset branches ($j \neq 0$). The anchor curvature is, by (E6) and the window-centre composite,

$$\lambda_a = \tfrac{729}{512}\,k|j|\,n^{-1/8}\,(1+O(P^{-1/8})) \;\in\; [1.2,\,1.5]\,k|j|P^{-1/8},$$

where $\frac{729}{512} = \frac{945}{512} - \frac{27}{64}$, by the following algebra. The smooth part of the anchor on an offset branch is $cF = \frac{3k}{4}\nu^{9/8}\cdot\frac{3}{2}j(m+\beta_1+\beta_2+\xi_1)^{1/2} = \frac{9}{8}\,kj\,\nu^{15/8}\,(1+O(P^{-1/4}))$ (Lemma 5.1(iii) and $m = \nu^{3/2}-\theta$), with second derivative $\frac{9}{8}\cdot\frac{15}{8}\cdot\frac{7}{8}\,kj\,\nu^{-1/8} = \frac{945}{512}\,kj\,\nu^{-1/8}$ up to the same relative error. The window-centre mode carries $u = -B(n_0)$, where $B = c\,\partial_m F = \frac{3k}{4}\nu^{9/8}\cdot\frac{3}{4}j\,m^{-1/2} = \frac{9}{16}\,kj\,\nu^{3/8}\,(1+O(P^{-1/4}))$ is the θ-sawtooth coefficient of the anchor (the θ-linear term of $cF(X-\theta)$), so its curvature contribution is $uX'' = -\frac{9}{16}\cdot\frac{3}{4}\,kj\,\nu^{-1/8} = -\frac{27}{64}\,kj\,\nu^{-1/8}$. The two terms have ratio $\frac{945}{512} : \frac{27}{64} = 4.375$, so the composite is single-signed at the displayed scale. Every competitor is dominated at a displayed margin: differenced-wave modes $\leq 0.84\,uh_1P^{-3/4} \leq 0.51P^{-1/4}$ (since $uh_1 \leq 0.6P^{1/2}$), ratio $\leq 0.43P^{-1/8}$ against $\lambda_a \geq 1.2P^{-1/8}$; resonant (D1) content $\leq 6\,|q'|P^{-5/4}$ with $|q'| \leq 4P^{1/24}$, ratio $\leq 20P^{1/24-5/4+1/8} = 20P^{-13/12}$; slow modes $\leq 3J_2|j|P^{-5/4}$, ratio $\leq 8P^{1/24-9/8}$; (D3) content, ratio $\leq 3h_1h_2P^{-1/2} \leq P^{-1/4}$. The θ-sawtooth of the anchor has coefficient $\frac{9}{16}k|j|P^{3/8}$-scale with per-step drift $\leq 0.2k|j|P^{-5/8} < 1$: at most $1.2k|j|P^{3/8}{+}1$ windows of length $\geq 0.8P^{5/8}/(k|j|)$, window-boundary cost $\leq 1.2k|j|P^{3/8}\cdot 0.92(k|j|)^{-1/2}P^{1/16} \leq 1.1(k|j|)^{1/2}P^{7/16}$. Off the collision band, window modes w are dominated at margin ≥ 4 either way and ride along or are estimated by Lemma 3.3 at their own scale with $1/|w{+}B_0|$ weights ($\ll P^{7/8}\log P$ in total); on the collision band $|wX''| \in [\frac{1}{4}, 4]\lambda_a$, Lemma 3.8 with $(\alpha,\beta) = (\frac{15}{8}, \frac{3}{2})$ and $M \in [0.3,\,6]\,k|j|P^{-1/8}$ gives $\leq C_E(|I_w|M^{1/2} + M^{-1/2} + (P/M)^{1/3})$ per window; summed over the at most $1.2k|j|P^{3/8}{+}1$ windows with the $O(\log P)$ band mass,

$$\sum_w |I_w|M^{1/2} \leq 2.5\,(k|j|)^{1/2}P^{15/16}, \qquad \sum_w M^{-1/2} \leq 2.2\,(k|j|)^{1/2}P^{7/16},$$

$$\sum_w (P/M)^{1/3} \leq 1.2k|j|P^{3/8}\cdot 1.5\,(k|j|)^{-1/3}P^{3/8} = 1.8\,(k|j|)^{2/3}P^{3/4} \leq 3.8\,P^{1/36+3/4},$$

so the band total is $\ll (k|j|)^{1/2}P^{15/16}\log P$. The main estimate is Lemma 3.3 per frozen run at scale λ_a: run lengths $\geq \frac{1}{22}P^{1/4}/(|j|{+}1)$ (Lemma 5.1(iii)), and $\lambda_a^{-1/2} \leq 0.92(k|j|)^{-1/2}P^{1/16} \ll$ run length, so

$$\sum_{\text{runs}} (\ell\lambda_a^{1/2} + \lambda_a^{-1/2}) \leq 1.3\,(k|j|)^{1/2}P^{15/16} + 21\,(|j|{+}1)|j|^{-1/2}k^{-1/2}P^{13/16}.$$

Summed over the eight carry branches and $|j| \leq 3$ with the $O(\log^3 P)$ piece masses: $\ll (k)^{1/2}P^{15/16+\varepsilon} \leq 1.8\,P^{1/48}P^{15/16+\varepsilon} = 1.8\,P^{23/24+\varepsilon}$ at $k \leq P^{1/24}$ — the absorbed $(k|j|)^{1/2}$ loss, exactly at the bottleneck.

(5b) Zero-offset branches ($j = 0$). Lemma 5.2b gives the local frozen curvature $\lambda_0 \in [0.35,\,2.6]\,kh_1h_2P^{-5/8}$ (leading coefficient $\frac{135}{1024}$ in $k\beta_1\beta_2\nu^{-13/8}$, converted by $\beta_i \asymp h_iP^{1/2}$). Runs of length $\geq \frac{1}{22}P^{3/4}/(h_1h_2)$. Let $\mu = 0.84\max(uh_1, u'h_2)P^{-3/4}$ be the strongest differenced-wave scale present. Three regimes.

- *Anchor-dominant* ($60\mu \leq \lambda_0$): Lemma 3.3 per run at λ_0, all else dominated at margin ≥ 20: $\leq 1.7(kh_1h_2)^{1/2}P^{11/16} + 40\,(h_1h_2/k)^{1/2}P^{9/16}$.
- *Mode-dominant* ($\mu \geq 60\lambda_0$, i.e. $uh_1 \geq 60\,kh_1h_2P^{1/8}$-form): Lemma 5.2(i) with the undifferenced anchor as decoration: its run boundaries number $\leq 22h_1h_2P^{1/4} \leq 22P^{5/16}$, cost $\leq 22P^{5/16}\cdot 3.4(uh_1)^{-1/2}P^{3/8} \leq 75P^{11/16}$; its smooth part is dominated at margin ≥ 20 by hypothesis. On a zero-offset branch the

anchor's θ-coefficient is $B = c\,\partial_m F = -\frac{3}{8}c\,\beta_1\beta_2\,m^{-3/2} = -\frac{9}{32}k\beta_1\beta_2\nu^{-9/8}(1+O(P^{-1/4}))$. With $\beta_i \in (3h_iP^{1/2}-1,\, 4.3h_iP^{1/2}+1)$ this is $|B| \le 5.3\,kh_1h_2P^{-1/8} \le 5.3$ by (C1), opened to $|B| \le 6$. The sawtooth is of constant size, not sub-unit. Lemma 3.7 still applies in a single window at $T = P^{1/2}$, because $T \ge 8(1+|B|)$ is $P^{1/2} \ge 56$ for $P \ge P_0$; there is no large-B window inventory. (The offset-scale amplitude $kh_1h_2P^{3/8}$ does not occur at $j=0$.) Centre modes w are treated as in Lemma 5.2(i), Stage 5. Total $\ll ((uh_1)^{1/2}P^{5/8} + (h_1/u)^{1/2}P^{7/8} + P^{7/8})P^{\varepsilon}$.

- *Middle band* ($\frac{1}{60} \le \mu/\lambda_0 \le 60$; coefficient mass $O(1)$ per layer): here up to three curvature scales meet, and this is the one place in the paper where the composite second derivative can cross zero *inside* cells. Since the cells are short ($\asymp P^{1/2}/h$) while the third-derivative scale is tiny, no inverse-power van der Corput term may be summed per cell; instead the transition set is measured once, against a global smooth model, and receives the trivial bound (Lemma 3.9).

  Inventory. In this band $0.84\,uh_1P^{-3/4} = \mu \le 60\lambda_0 \le 160\,kh_1h_2P^{-5/8}$ forces $u \le 200\,kh_2P^{1/8} \le 200\,P^{5/24}$, and likewise $u' \le 200\,kh_1P^{1/8} \le 200\,P^{5/24}$. The phase's second derivative f'' is smooth on the common refinement of the gap cells of both shifts (at most $1.5(h_1{+}h_2)P^{1/2}+2 \le 3.1P^{13/24}$), the anchor runs ($\le 22h_1h_2P^{1/4} \le 22P^{3/8}$), and the sawtooth windows already counted: in total at most $N \le 3.5P^{13/24}$ pieces, for $P \ge P_0$. On each piece,

  $$f'' = -\tfrac{9}{32}\Big(uG_1(\nu{+}2h_1)^{-5/4} + u'G_2(\nu{+}2h_2)^{-5/4}\Big) + \big(c_{11}(G_F - J_F)\big)'' + wX'' + O(\rho_0\text{-small}),$$

  with G_1, G_2, J_F (and the β_i inside G_F) frozen integers. This is the local f'' of Lemma 5.2b.

  Interpolant. Invoke Lemma 5.2b: the frozen-shape interpolant Λ (values of $\Delta_i X$ substituted, not differentiated) satisfies $|f''-\Lambda| \le 219P^{-25/24} + 0.11P^{-5/6}$ and $\Lambda = \Phi'' + r$ with $a = -\frac{27}{10}(uh_1 + u'h_2)$ and $b = -\frac{405}{176}\,kh_1h_2$. The comparison $V \ge 10|f''-\Lambda|$ below is read from that two-term majorant.

  The scale is
  $$S = \max(|uh_1 + u'h_2|P^{-3/4},\, kh_1h_2P^{-5/8},\, |w|P^{-1/2}),$$
  and the middle-band constraints give $0.35P^{-5/8} \le S \le 300\,P^{-1/2}$: the lower bound is λ_0 when $kh_1h_2 \ge 1$, and the upper bound uses $kh_1h_2 \le P^{1/8}$ from (C1) together with $\mu \le 60\lambda_0 \le 160\,kh_1h_2P^{-5/8}$ (opened to 300, which only helps) and the collision-band restriction $|wX''| \ll P^{-1/2}$. If $w=0$, drop the third term of Φ and apply Lemma 3.8 (or Lemma 3.3 if only one of a, b is present).

  Splitting. Choose $V := 3S^{1/2}P^{-11/24}$, so that $V/S \le 5.1P^{-7/48}$ (hence $V \le c_7S/2$ at $c_7 = 1/288$ for $P \ge P_0$; the constant 5.1 is $3\cdot(0.35)^{-1/2}$) and $V \ge 1.7P^{-37/48} \ge 10\,|f''-\Lambda|$ against the two-term interpolant majorant of Lemma 5.2b. The comparison $V \le c_7S/2$ is the large ineffective threshold of the standing estimates: until it holds, a three-term zero of Φ'' can keep Ω_V of length $\Theta(P)$ on a dyadic block. The length bound below is that of Lemma 3.9, not a claim that the sublevel is short at small P; interval counts remain $O_E(1)$. By Lemma 3.9 the set $\Omega = \{\nu : |\Lambda(\nu)| \le V\}$ is a union of at most $C(E)$ intervals of total length $\le C(E)\,P(V/S)^{1/2} \le 2.3\,C(E)\,P^{89/96}$, and on its complement f'' is single-signed per interval with $0.9V \le |f''| \le 1.1\,C(E)S$. The three costs:

  $$\text{transition (trivial):}\quad \le 2.3\,C(E)\,P^{89/96};$$

  $$\text{piece boundaries:}\quad \le (N{+}C(E))\,(0.9V)^{-1/2} \le 3.5P^{13/24}\cdot 0.91P^{37/96} + O_E(P^{37/96}) \le 3.2\,P^{89/96};$$

  $$\text{good pieces (Lemma 3.3):}\quad \sum \ell\,(1.1\,C(E)S)^{1/2} \le C'(E)\,P\cdot(300)^{1/2}P^{-1/4} \le 18\,C'(E)\,P^{3/4}.$$

  The middle band therefore totals $\le C(E)\,P^{89/96}\log P \le P^{15/16}$ for $P \ge P_0$.

In all three regimes the $j=0$ pieces total $\ll P^{15/16+\varepsilon} \ll P^{23/24}$.

Step 6 (assembly). Additive costs: the M_1 deletion ($2.7P^{1/4}$); majorants (≤ 3 sawtooth layers at $4P/J_2 = 4P^{23/24}$ each, plus $8P^{3/4}$, $46P^{3/4}$, and the displayed window residuals $\le 7P^{7/8}$); flat costs ($\le 46P^{3/4}$ per layer). Multiplicative costs: mode masses over at most three expansion layers plus the shift devices,

$O(\log^3 P) = P^{\varepsilon}$. Piece totals: Step 4, $\ll P^{23/24+\varepsilon}$; Step 5a, $\le 1.8P^{23/24+\varepsilon}$; Step 5b, $\ll P^{15/16+\varepsilon}$; slow modes and (D3) remnants, $\ll P^{7/8+\varepsilon}$. Hence $|T_2| \ll P^{23/24+\varepsilon}$, and by Step 1

$$|T_1|^2 \le 2P^{2-1/24} + CP^{1+23/24+\varepsilon} \;\Rightarrow\; |T_1| \ll P^{1-1/48+\varepsilon}, \qquad |K_c|^2 \le 2P^{2-1/48} + CP^{2-1/48+\varepsilon},$$

so $|K_c| \ll P^{1-1/96+\varepsilon}$. Exponents deliberately unoptimized. □

Two remarks on the constants. First, the k-range is sharp for this assembly: the class-(5a) loss $(k|j|)^{1/2}P^{15/16}$ meets the wave-piece bottleneck $P^{23/24}$ exactly at $k = P^{1/24}$, which is why the theorem is uniform on the original range with no k-explicit statement needed. Second, the exponent $\frac{1}{96} = \frac{1}{4}\cdot\frac{1}{24}$ traces entirely to the depth-2 strength $P^{23/24}$ of the exact level-2 waves (Lemma 5.2(ii)); any improvement of the wave bound improves δ proportionally.

5. Depth-four equidistribution

Theorem 6.1 (the OOO* splits; complete depth-4 parity equidistribution for odd-rooted itineraries). For $w \in \{OOOE, OOOO\}$,

$$\#\{n \le N \text{ odd}: \ \mathrm{word}_4(n) = w\} = \tfrac{N}{16} + O(N^{1-1/96+\varepsilon}).$$

Hence every length-4 itinerary word class of *odd* starts — the eight words with first letter O — satisfies $\#\{n \le N\} = 2^{-4}N + O(N^{1-\delta_w})$ with explicit $\delta_w > 0$: depth-4 parity equidistribution over odd starts is complete. (E-rooted words, i.e. even starts, are a different and easier problem — one square root drops the state to scale $\sqrt{N}$ — and are not treated in this paper.)

Proof. Step A (expansion; the mode ranges). The class indicator expands (Lemma 3.6, then Lemma 3.5 at truncation $J_3 = P^{1/96}$, majorant cost $4P/J_3 = 4P^{1-1/96}$ per layer) into mode sums

$$S_{ijk} = \sum_{\substack{n\sim P\\ n \text{ odd}}} e\big(\tfrac{i}{2}X + \tfrac{j}{2}Y + \tfrac{k}{2}v^{3/2}\big), \qquad |i| \le 2P^{1/96}, \quad |j| \le 2P^{1/96}, \quad 1 \le |k| \le 2P^{1/96},$$

with weights $\le \min(1, 2/|i|)\min(1, 2/|j|)\min(1, 2/|k|)$ of total mass $O(\log^3 P)$. The $k = 0$ sums are Theorems 4.1 and 4.4. Conjugating, take $k \ge 1$; then every displayed sign below is as written.

Step B (the depth-4 identity). Three applications of the Taylor pattern of Lemma 4.3(i), taken to second order, give the exact identity (odd $n \ge 5$)

$$v^{3/2} = -\tfrac{5}{64}n^{27/8} + \tfrac{9}{32}mn^{15/8} - \tfrac{45}{64}m^2n^{3/8} + \tfrac{15}{64}vn^{9/8} + \tfrac{45}{32}vmn^{-3/8} - \tfrac{9}{64}vm^2n^{-15/8} + \mathrm{err},$$

$|\mathrm{err}| \le \frac{3}{4}n^{-9/8}$ (discarding err costs $\le 2\pi kP^{-1/8}\cdot P \le 7P^{7/8}$): the fourth-letter phase is a polynomial of degree $(2,1)$ in the integer pair (m, v) with smooth coefficients. Substituting $m = X - \theta$ in the v-coefficients, the total smooth v-coefficient is *exactly*

$$c(\nu) := \tfrac{k}{2}\Big(\tfrac{15}{64} + \tfrac{45}{32} - \tfrac{9}{64}\Big)\nu^{9/8} = \tfrac{3k}{4}\,\nu^{9/8},$$

the monomial weight of Theorem 5.3, and the $v\theta$- and $v\theta^2$-cross coefficients are $\le \frac{45}{64}k\nu^{-3/8} + \frac{9}{32}k\nu^{-3/8} \le 1.2k\nu^{-3/8}$ and $\le \frac{9}{128}k\nu^{-15/8}$: sub-unit and decaying. The pure-m part is $\mu_1 m + \mu_2 m^2$ with $\mu_1 = \frac{9}{64}k\nu^{15/8}$, $\mu_2 = -\frac{45}{128}k\nu^{3/8}$, and the pure-smooth part $-\frac{5}{128}k\nu^{27/8} + \mu_1 X + \mu_2 X^2$.

Step C (double differencing; corner exactness). Apply Step 1 of Theorem 5.3 ($H_1 = P^{1/48}$, $H_2 = P^{1/24}$; now $kh_1h_2 \le 2P^{1/96+1/48+1/24} \le P^{1/8}$, so (C1)–(C4) hold) to the whole mode phase. On each level-1 carry branch of Lemma 5.1(iii), the four corner values obey the exact relations

$$m(n{+}d) = m + \beta_d, \qquad \theta(n{+}d) = \theta + \delta_d(\nu) - \beta_d \qquad (d \in \mathcal{D}),$$

with β_d frozen integers and δ_d smooth: the doubly differenced phase on a branch is an exact function $\Phi(\nu, \theta)$ of ν and the *single* sawtooth $\theta = \{X\}$. Two consequences are used repeatedly. First, the θ^2-contents of the four corners cancel exactly in the $(+,-,-,+)$ pattern, so $|\partial_\theta^2\Phi| \le 3kh_1h_2P^{-13/8} + 4|\Delta\Delta\mu_2| \le P^{-1}$: no quadratic sawtooth survives, and $e(\Phi) = e(\Phi(\nu,0) + B(\nu)\theta)(1 + O(P^{-1}))$ with $B = \partial_\theta\Phi(\nu,0)$. Second, the wave and kernel contents are exactly those of Theorem 5.3 with $c = \frac{3k}{4}\nu^{9/8}$: the v-linear part contributes $\Delta\Delta(c\,\theta_2)$ (the master identity, Steps 2–3 of Theorem 5.3) plus $\Delta\Delta(cY)$-content whose wave frequencies join Step 4's bookkeeping.

Step D (passenger inventory). Every phase content beyond the bare kernel is listed here with its range and its check.

- *The $\frac{i}{2}X$-passenger*, $|i| \le 2P^{1/96}$: $\Delta\Delta(\frac{i}{2}X)$ is smooth with second derivative $\le \frac{i}{2}\cdot 4h_1h_2 \sup|X''''| \le 2.3\,ih_1h_2P^{-5/2} \le P^{-9/4}$: class (D3), dominated by every retained scale at margin $\le P^{-3/2}$.
- *The $\frac{j}{2}Y$-passenger*, $|j| \le 2P^{1/96}$: $\Delta\Delta(\frac{j}{2}Y)$ is exactly four waves $\pm\frac{j}{2}Y(n{+}d)$, $d \in \mathcal{D}$, with total frequency 0. In Step 4 of Theorem 5.3 it shifts each wave coefficient q_d by at most $P^{1/96}$, keeping $|q_d| \le 3P^{1/24} + P^{1/96} \le 4P^{1/24}$ — inside Lemma 5.2(ii)'s coefficient budget — and leaves t unchanged; its $t = 0$ remnants are differenced waves that shift the u-coefficients of Step 5 by $\le P^{1/96}$, harmless against the budget $uh_1 \le 0.6P^{1/2}$.
- *The pure-m passengers $\mu_1 m + \mu_2 m^2$*: on a branch, by corner exactness, $\Delta\Delta(\mu m) = (\Delta\Delta\mu)m + (\Delta_2\mu)\beta_1 + (\Delta_1\mu)\beta_2 + \mu_{11}j$ and $\Delta\Delta(\mu m^2) = (\Delta\Delta\mu)m^2 + (\Delta_2\mu)(2\beta_1 m{+}\beta_1^2) + (\Delta_1\mu)(2\beta_2 m{+}\beta_2^2) + \mu_{11}(2jm + \beta_{12}^2{-}\beta_1^2{-}\beta_2^2)$, j the branch offset. The θ-coefficients that arise from $m = X - \theta$ in these terms are $|\Delta\Delta\mu_1| \le 0.92kh_1h_2P^{-1/8} \le P^{-1/48}$, $|2(\Delta_2\mu_2)\beta_1| \le 2.3kh_1h_2P^{-1/8} \le 3P^{-1/48}$, and $|2j\mu_2|$-content, which joins $B(\nu)$ below; all sub-unit ones are expanded in the slow-mode families at $O(\log P)$ mass. The smooth parts join the two anchor families and are accounted in Step E.
- *The $v\theta$-crosses*: writing $v = Y - \theta_2$ and linearizing $Y = m^{3/2}$ once more by Lemma 4.3(i), the crosses decompose into $\nu^{15/8}$-scale θ-sawtooth content (joining $B(\nu)$), a $\theta\theta_2$-product with coefficient $\le 1.2k\nu^{-3/8}$ (deleted at cost $\le 31kP^{5/8} \le P^{2/3}$), and decaying remainders.

Step E (the two sign-critical composites, rerun). The passengers change the two composites of Step 5 of Theorem 5.3. Both are recomputed here from the *frozen-shape* model of Lemma 5.2b, not from the moving-(m, v) interpolant of $\frac{k}{2}\nu^{27/8}$. The identity that organises the difference is Lemma 5.1(i):

$$\tfrac{k}{2}\,v^{3/2} = \tfrac{k}{2}\,m^{9/4} - c\,\theta_2 - kR, \qquad c = \tfrac{3k}{4}\,v^{1/2},$$

with $|kR| \ll kP^{-9/8}$ discarded as in Step B. The differenced total phase is therefore $\Delta\Delta(\frac{k}{2}m^{9/4}) - \Delta\Delta(c\theta_2)$ plus the Step D passengers. On a piece the integers β_i, J_F are held frozen; $X(\nu)$ and the smooth weights move; the values of $\Delta_i X$ may be substituted for the β_i as *values only*.

Offset branches ($j \ne 0$). The frozen-shape offset content of $\Delta\Delta(\frac{k}{2}m^{9/4})$, with $m = X$ as a value, is the same monomial as in Theorem 5.3, $\frac{9}{8}kj\nu^{15/8}$, with curvature $\frac{945}{512}kj\nu^{-1/8}$. The frozen kernel anchor $(c(G_F - J_F))''$, now including the $-J_F c''$ term that Lemma 5.2b isolates as part of $c''(G_F - J_F)$ only after the fractional part is taken, is $\frac{27}{16}kj\nu^{-1/8} = \frac{864}{512}kj\nu^{-1/8}$. The difference — the surviving total-phase offset — is

$$(\tfrac{945}{512} - \tfrac{864}{512})kj\nu^{-1/8} = \tfrac{81}{512}\,kj\nu^{-1/8}.$$

This is $J_F c''$ at leading order: after the master identity cancels $c_{11}F_\kappa$ against $\Delta\Delta(cY)$, what remains of the growing smooth offset is the frozen integer times c''. The total θ-coefficient on the piece is

$$B = \tfrac{27}{32}\,kj\,\nu^{3/8}\,(1 + O(P^{-1/4})),$$

one and a half times the bare-kernel value $\frac{9}{16}kj\nu^{3/8}$ (not the moving-gap value $\frac{45}{32}$). The window-centre mode therefore carries $u_0X'' = -\frac{27}{32}\cdot\frac{3}{4}\,kj\,\nu^{-1/8} = -\frac{81}{128}\,kj\,\nu^{-1/8} = -\frac{324}{512}\,kj\,\nu^{-1/8}$, and the composite is

$$\left(\tfrac{81}{512} - \tfrac{324}{512}\right)kj\,\nu^{-1/8} = -\tfrac{243}{512}\,kj\,\nu^{-1/8},$$

single-signed, with $\lambda_a' \in [0.40,\ 0.52]\,k|j|P^{-1/8}(1 + O(P^{-1/8}))$. The competitors of Step 5a, now read against $\lambda_a' \ge 0.40\,k|j|P^{-1/8} \ge 0.40P^{-1/8}$, remain dominated for $P \ge P_0$: differenced-wave modes $\le 0.84\,uh_1P^{-3/4} \le 0.51P^{-1/4}$, ratio $\le 1.3P^{-1/8}$; resonant (D1) content, ratio $\le 13P^{-9/16}$; slow modes $\le 3J_2|j|P^{-5/4}$, ratio

$\le 9P^{-13/12}$; (D3) content, ratio $\le 4h_1h_2P^{-1/2}/(0.40P^{-1/8}) \le 3P^{-1/8}$. The window count grows by $\frac{27/32}{9/16} = \frac{3}{2}$, so at most $1.8\,k|j|P^{3/8}+1$ windows; the collision-band Lemma 3.8 sum is then $\le 3.0\,(k|j|)^{1/2}P^{15/16}\log P$. Run lengths are unchanged, and

$$\sum_{\text{runs}}(\ell\lambda_a'^{1/2}+\lambda_a'^{-1/2}) \le 1.4\,(k|j|)^{1/2}P^{15/16}+34\,(|j|{+}1)|j|^{-1/2}k^{-1/2}P^{13/16}.$$

Summed over the eight branches and $|j| \le 3$ with the $O(\log^3 P)$ piece masses: $\ll k^{1/2}P^{15/16+\varepsilon} \le P^{1/48}P^{15/16+\varepsilon} = P^{23/24+\varepsilon}$ already at the kernel's $k \le P^{1/24}$; the actual range $k \le 2P^{1/96}$ sits strictly inside. The inverse-power term remains $P^{13/16}$.

Zero-offset branches ($j=0$). The same frozen-shape difference, after $\beta_1\beta_2 \sim 9h_1h_2\nu$, has leading curvature

$$\lambda_0' = \tfrac{1095}{1024}\,kh_1h_2\,\nu^{-5/8} \in [0.60,\ 1.25]\,kh_1h_2P^{-5/8}.$$

The three-regime split of Step 5b is read at this scale. The thresholds (factor 60) are scale-free. In the anchor-dominant regime, Lemma 3.3 per run gives $\le 2.0(kh_1h_2)^{1/2}P^{11/16}+30\,(h_1h_2/k)^{1/2}P^{9/16}$. In the mode-dominant regime the undifferenced-anchor decoration is the same as in Step 5b. The $j=0$ θ-coefficient of the *total* phase is $O(1)$, of the same species as the kernel's zero-offset B (constant size, $|B| \le 6$): Lemma 3.7 applies in a single window at $T = P^{1/2}$ (hypothesis $T \ge 8(1+|B|)$ holds for $|B| \le 6$). In the middle band the frozen-shape interpolant of Lemma 5.2b is reused with the kernel-anchor leading $-\frac{1215}{1024}kh_1h_2\nu^{-5/8}$ replaced by $-\frac{1095}{1024}kh_1h_2\nu^{-5/8}$, so $\Phi = a\nu^{5/4}+b'\nu^{11/8}+w\nu^{3/2}$ with $a = -\frac{27}{10}(uh_1+u'h_2)$ and $b' = -\frac{365}{176}\,kh_1h_2$. The $\rho_0(E)$ bounds are unchanged, and

$$S \le \max(60\lambda_0',\ \lambda_0',\ |w|P^{-1/2}) \le 80\,P^{-1/2}$$

by (C1) and $|w| \le 2$. The balanced choice $V = 3S^{1/2}P^{-11/24}$ still satisfies $V \le c_7S/2$ at $c_7 = 1/288$ and $V \ge 10|f''-\Lambda|$ against the two-term interpolant majorant, for $P \ge P_0$: at the lower end $S \ge 0.60P^{-5/8}$ one has $V \ge 2.3P^{-37/48}$ and the interpolant error is $O(P^{-25/24})+O(P^{-5/6})$. Transition length and piece-boundary costs are as in Step 5b (the same exponents; the smaller S enlarges V/S by a constant, still $O(P^{-7/48})$). Good pieces cost $\le P\cdot(80)^{1/2}P^{-1/4} \le 9P^{3/4}$. The middle band therefore remains $\ll P^{15/16+\varepsilon}$.

The passengers of Step D are inside these estimates: the $\frac{i}{2}X$-term is (D3) as listed; the $\frac{j}{2}Y$-term shifts each q_d by at most $P^{1/96}$, keeping $|q_d| \le 4P^{1/24}$ and $uh_1 \le 0.6P^{1/2}$; the pure-m smooth pieces are already in $\Delta\Delta(\frac{k}{2}m^{9/4})$ and do not add a further relative $O(1)$ to λ_a' or λ_0'; the sub-unit θ-coefficients of those pieces are among the slow modes already bounded.

Step F (assembly). With the inventory of Step D and the composites of Step E, Steps 2–6 of Theorem 5.3 give $|T_2| \ll P^{23/24+\varepsilon}$ uniformly in (i,j,k) on the stated ranges, hence $S_{ijk} \ll P^{1-1/96+\varepsilon}$. Summing the mode weights ($O(\log^3 P)$), the majorant costs ($\le 4P^{1-1/96}$ per layer, three layers), and dyadic blocks gives the theorem. $\square$

The four-step certified class remains 13/16: $OOO*$ does not contract ($3^3 > 2^4$). The next two contracting itineraries are $OOOEE$ and $OOEOE$ ($3^3 < 2^5$). Neither is a third growing layer. After $OOOE$ the fifth letter is a decaying nest plus one slow sawtooth of coefficient $n^{3/16} < n$; after $OOEO$ it is a single growing sawtooth of coefficient $n^{9/16} < n$, of the same species as Theorem 4.8. Theorem 6.3 counts both cylinders.

6. Depth-five contractors

Lemma 6.2 (fifth-letter identities). Let $n \ge 5$ be odd, and write $z = \lfloor v^{3/2}\rfloor$, $U = v^{1/2}$, $w = \lfloor U\rfloor$, and $\theta_w = \{U\}$ (the same letters as in Theorem 4.8, now at the v-level).

(i) *($OOOE*$ smoothing.)*

$$z^{1/2} = n^{27/16} - \tfrac{9}{8}n^{3/16}\,\theta + D_5, \qquad |D_5| \le \tfrac{3}{4}\,m^{-3/8}+\tfrac{1}{2}\,v^{-3/4}+\tfrac{9}{128}\,(X-1)^{-7/8}+\tfrac{3}{32}(Y-1)^{-5/4}+\tfrac{1}{8}(v^{3/2}-1)^{-3/2}.$$

The last three terms are $O(n^{-21/16})$, $O(n^{-45/16})$ and $O(n^{-81/16})$; the leading remainder is $O(n^{-9/16})$.

(ii) *(OOEO∗ linearization.)*

$$w^{3/2} = n^{27/16} - \tfrac{9}{8}n^{3/16}\,\theta - \tfrac{3}{2}\,v^{1/4}\,\theta_w + D_5',$$

with $|D_5'| \le \frac{3}{4}\,m^{-3/8} + \frac{3}{8}(U-1)^{-1/2} + \frac{9}{128}(X-1)^{-7/8} + \frac{3}{32}(Y-1)^{-5/4}$.

Proof. (i) Three applications of the Lemma 4.3(i) pattern. With $f(t) = (v^{3/2}-t)^{1/2}$ on $[0,\theta_z]$, $\theta_z = \{v^{3/2}\}$: $z^{1/2} = v^{3/4} - \frac{1}{2}\theta_z v^{-3/4} - E_z$ and $0 \le E_z \le \frac{1}{8}(v^{3/2}-1)^{-3/2}$. With $f(t) = (m^{3/2}-t)^{3/4}$ on $[0,\theta_2]$: $v^{3/4} = m^{9/8} - \frac{3}{4}\theta_2 m^{-3/8} - E_2$ and $0 \le E_2 \le \frac{3}{32}(Y-1)^{-5/4}$. With $f(t) = (X-t)^{9/8}$ on $[0,\theta]$: $m^{9/8} = n^{27/16} - \frac{9}{8}\theta n^{3/16} + \frac{9}{128}\theta^2(X-\xi)^{-7/8}$ for some $\xi \in (0,\theta)$. Hence $D_5 = \frac{9}{128}\theta^2(X-\xi)^{-7/8} - \frac{3}{4}\theta_2 m^{-3/8} - E_2 - \frac{1}{2}\theta_z v^{-3/4} - E_z$, and the triangle inequality with $\theta, \theta_2, \theta_z < 1$ gives the displayed bound term by term. (The two Lagrange remainders E_2, E_z are of orders $n^{-45/16}$ and $n^{-81/16}$; they are displayed rather than absorbed into the coefficients $\frac{3}{4}$ and $\frac{1}{2}$, which have no slack when θ_2 or θ_z is close to 1.) The leftover sawtooth has coefficient $n^{3/16}$.

(ii) Lemma 4.3(i) at base U, using $U\,v^{1/4} = v^{3/4}$ exactly: $w^{3/2} = v^{3/4} - \frac{3}{2}v^{1/4}\theta_w + E$ with $0 \le E \le \frac{3}{8}(U-1)^{-1/2}$. The chain $v^{3/4} \to n^{27/16}$ is the second and third steps of (i), so $D_5' = E - \frac{3}{4}\theta_2 m^{-3/8} - E_2 + \frac{9}{128}\theta^2(X-\xi)^{-7/8}$, and the bound follows term by term. Two sawtooths remain, of coefficients $n^{3/16}$ on θ and $n^{9/16}$ on θ_w. □

Theorem 6.3 (two length-five splits; not a census of depth five).

$$\#\mathrm{OOOEE}(N),\ \#\mathrm{OOOEO}(N) = \tfrac{N}{32} + O(N^{1-1/96+\varepsilon}), \qquad \#\mathrm{OOEOE}(N),\ \#\mathrm{OOEOO}(N) = \tfrac{N}{32} + O(N^{43/48+\varepsilon}).$$

Of these four words only *OOOEE* and *OOEOE* contract. The expanding tree *OOOO∗* is not estimated: it is the level-3 kernel of Conjecture 7.3. All estimates below are for $P \ge P_0$, with the same ineffective P_0 as in Section 4.

Proof. OOOE∗. By Lemma 3.6 the class indicators are the *OOOE* indicator of Theorem 6.1 times $\frac{1}{2}(1 \pm \psi(z^{1/2}))$. Vaaler-expand the fifth wave at truncation $J_5 = 2P^{1/96}$ (majorant $4P/J_5 = 2P^{1-1/96}$ per block). Lemma 6.2(i) replaces $\frac{l}{2}z^{1/2}$ by $\frac{l}{2}n^{27/16} - \frac{9l}{16}n^{3/16}\theta$. The remainder costs $|l|\cdot P \cdot P^{-9/16} \ll P^{1/96+7/16} = P^{43/96}$, inside $P^{1-1/96}$. Write $C = \frac{9l}{16}n^{3/16}$, so $|C| \le 2P^{19/96}$ on $|l| \le 2P^{1/96}$. Lemma 3.7 with $T = P^{1/4}$ satisfies $T \ge 8(1+|C|)$, since $8|C|/T \le 16P^{-5/96} \to 0$. The produced modes are $e(uX)$ with $|u| \le P^{1/4}$. These are first-letter monomials of the shape already estimated in the proof of Lemma 5.2(i) — the families $e(r\nu^{3/2})$ with truncation $R_0 = P^{1/4}$ — and are not decorations of class (D1). (The coefficient budget $|q'| \le 4P^{1/24}$ of Lemma 5.2(ii) is the kernel's own Y-wave range after differencing; it is not a slot for X-modes.)

Write I_{tot} for the combined first-letter index: Theorem 6.1's own $|i| \le 2P^{1/96}$ plus the fifth-letter $|u| \le P^{1/4}$. Then $|I_{\mathrm{tot}}| \le 2P^{1/4}$. Theorem 6.1 Steps D–E apply at this range.

- *The $\frac{i}{2}X$-passenger.* $\Delta\Delta(\frac{i}{2}X)$ has second derivative $\le 2.3\,|i|h_1h_2P^{-5/2}$. At $|i| \le 2P^{1/4}$ and $h_1h_2 \le P^{1/16}$ this is $O(P^{-35/16})$, inside class (D3) ($|\varphi''| \le 3kh_1h_2P^{-5/8}$) by $P^{-35/16}/P^{-9/16} = P^{-13/8}$. The fifth-letter chirp $\frac{l}{2}n^{27/16}$, after the same double difference, is likewise (D3): $h_1h_2 \cdot l \cdot P^{27/16-4} = O(P^{1/16+1/96-37/16})$.
- *The $\frac{j}{2}Y$-passenger.* The fifth letter does not enlarge j or the Y-wave frequencies. The bound $|q_d| \le 3P^{1/24} + P^{1/96} \le 4P^{1/24}$ is unchanged.
- *The sign-critical composites.* An X-mode is smooth, so it does not enter the frozen θ-coefficient B. The offset composite $\frac{243}{512}$ and the zero-offset curvature $\frac{1095}{1024}kh_1h_2\nu^{-5/8}$ are therefore the values of Theorem 6.1 Step E. The (D3) curvature of the new modes is dominated at the ratios already displayed there ($\le 3P^{-1/8}$ against $\lambda_a' \ge 0.40P^{-1/8}$).
- *The Lemma 5.2(i) budget.* Modes with $|u| \le P^{1/4}$ are a subset of the family already bounded by $3R_0^{1/2}P^{3/4}\log P = 3P^{7/8}\log P$. Collision-band mass stays $O(\log P)$.
- *The flat cost of Lemma 3.7.* Per point $8(1+|C|)/T = O(P^{19/96-1/4}) = O(P^{-5/96})$; over a block $O(P^{1-5/96})$, inside Theorem 6.1's $P^{1-1/96}$ budget.

Theorem 6.1 therefore applies to every fifth-letter decorated *OOOE∗* mode sum, uniformly in $|l| \le 2P^{1/96}$. Hence $\#\mathrm{OOOEE}(N),\ \#\mathrm{OOOEO}(N) = N/32 + O(N^{1-1/96+\varepsilon})$.

OOEO$*$. By Lemma 3.6, the two class indicators are

$$\tfrac{1}{16}(1-\psi(X))(1+\psi(Y))(1-\psi(U))(1\pm\psi(w^{3/2})).$$

Vaaler-expand the four waves. The fifth-letter $k=0$ sums are the depth-4 class *OOEO*, bounded by Theorem 4.7. For $k\neq 0$, Lemma 6.2(ii) writes the fifth-letter phase as $\frac{k}{2}n^{27/16}-C\theta-B\theta_w$ with $B=\frac{3k}{4}v^{1/4}\asymp kn^{9/16}$ and $C=\frac{9k}{16}n^{3/16}$, up to the decaying D_5'. The remainder costs $|k|\cdot P\cdot P^{-9/16}\ll P^{5/48+7/16}=P^{13/24}$ at the fifth-letter truncation $P^{5/48}$ used below, inside $P^{43/48}$. The leading sawtooth is of the same species as Theorem 4.8, now riding $\theta_w=\{v^{1/2}\}$.

Drift-1 intervals. $B'\asymp kn^{-7/16}$, so B drifts by at most 1 on intervals I of length $L_B=P^{7/16}/k$; there are $\asymp kP^{9/16}$ of them. On I, expand $e(-B\{U\})$ by the same shifted-window Fourier expansion as in Theorem 4.8 (Fourier in U, Vaaler truncation $|r+B|\le T_w$ with $T_w=P^{1/4}$): the coefficients satisfy $|a_r(B)|\le\min(1,|r+B|^{-1})$ with window mass $O(\log T_w)$, and their n-dependence through $B(n)$ has total variation $O(1)$ per interval, removed by one partial summation per mode. The window is legal for the sign check below: $T_w=P^{1/4}\ll kP^{9/16}$ for every $k\ge 1$; the majorant is $P/T_w=P^{3/4}$.

At frequency $\ell=-B+t$, $|t|\le T_w$, the combined phase is $-\frac{3k}{4}v^{3/4}+t\,v^{1/2}$ plus the original $\frac{k}{2}n^{27/16}-C\theta$. Linearizing $v^{3/4}$ and $v^{1/2}$ by Lemma 6.2, the θ-coefficients from the two expansions cancel at the window centre up to a residual $C_{\text{net}}=\frac{9k}{32}n^{3/16}$: the contribution of $-\frac{3k}{4}v^{3/4}$ is $\frac{27k}{32}n^{3/16}$, and $\frac{27k}{32}-\frac{9k}{16}=\frac{9k}{32}$. One slow θ-sawtooth remains. Expand $e(-C_{\text{net}}\theta)$ on the same intervals, again by the shifted-window device of Theorem 4.8 (C_{net} drifts by $P^{-3/8}\ll 1$ on each I; window $T_\theta=P^{1/8}\ll kP^{3/16}$; majorant $P^{7/8}$). The window-centre X-mode of this last expansion has curvature $\frac{27k}{128}n^{-5/16}$. The leading $n^{27/16}$ coefficient of the combined phase is $\frac{k}{2}-\frac{3k}{4}=-\frac{k}{4}$, of curvature $-\frac{297k}{1024}n^{-5/16}$. Hence

$$\lambda_2=\Big(-\tfrac{297k}{1024}+\tfrac{27k}{128}+O(T_wn^{-9/16})+O(J_*n^{-3/16})\Big)n^{-5/16},$$

with leading combination $-\frac{297}{1024}+\frac{216}{1024}=-\frac{81}{1024}\neq 0$. The t-error is $O(P^{1/4-9/16})=O(P^{-5/16})$ and the J_*-error, at the truncation $J_*=P^{5/48}$ used below, is $O(P^{5/48-3/16})=O(P^{-1/12})$. Both are smaller than the leading coefficient, so λ_2 is single-signed and $\lambda_2\asymp kn^{-5/16}$. By Lemma 3.10(a) the curvature ratios and signs are invariant under $n=2r+1$; Lemma 3.10(b) says the n-variable van der Corput display dominates the reindexed bound.

Lemma 3.3 on each I: $L_B\lambda_2^{1/2}+\lambda_2^{-1/2}\ll k^{-1/2}P^{9/32}+k^{-1/2}P^{5/32}$. Times $\asymp kP^{9/16}$ intervals: $S_k\ll k^{1/2}P^{27/32+\varepsilon}$. Balance $J_*^{1/2}P^{27/32}=P/J_*$ at $J_*=P^{5/48}$ gives $P^{43/48}$. Dyadic blocks sum to $N^{43/48+\varepsilon}$. □

Corollary 6.4 (certified-descent density $7/8$). The five uniform certificate classes

$$E,\qquad OE,\qquad OOEE,\qquad OOOEE,\qquad OOEOE$$

are disjoint, and

$$\big|\#\{n\le N:\text{word}(n)\text{ has prefix }OOOEE\}-\tfrac{N}{32}\big|\ll_\varepsilon N^{1-1/96+\varepsilon},$$
$$\big|\#\{n\le N:\text{word}(n)\text{ has prefix }OOEOE\}-\tfrac{N}{32}\big|\ll_\varepsilon N^{43/48+\varepsilon}.$$

Hence the class of starts with a certified descent within five steps has cardinality

$$\tfrac{N}{2}+\tfrac{N}{4}+\tfrac{N}{16}+\tfrac{N}{32}+\tfrac{N}{32}+O(N^{1-1/96+\varepsilon})=\tfrac{7N}{8}+O(N^{1-1/96+\varepsilon}).$$

No other depth-≤ 5 word class is used. In particular this is not a census of every O-rooted itinerary of length five: the classes *OOOO$*$* remain open, and *OOEOO*, *OOOEO* are counted by Theorem 6.3 but do not contract.

Proof. The first three counts are Corollary 4.9. The new cylinders are Theorem 6.3. Every *OOOEE* or *OOEOE* start descends within five steps by Proposition 3.1: $3^3<2^5$. The five classes are disjoint because they are distinct prefixes. The error is the worse of the two fifth-letter exponents. □

7. The Terras-style reduction and the frontier

The depth-by-depth counting assembles into a conditional Terras-style statement, and the reduction is unconditional:

Proposition 7.1 (equidistribution implies density-one descent). Let $d \geq 1$ and suppose that for every *O-rooted* itinerary word w of length d,

$$\left|\#\{n \leq N : \mathrm{word}_d(n) = w\} - 2^{-d}N\right| \leq E_d(N).$$

Then the starts with no contracting prefix of length $\leq d$ number at most

$$e^{-cd}\,N + 2^d E_d(N), \qquad c = 2\Big(\tfrac{\log 2}{\log 3} - \tfrac{1}{2}\Big)^2 > 0.0342.$$

Every other start $n \geq 2$ satisfies $J^t(n) < n$ for some $t \leq d$ with the uniform power-envelope certificate of Proposition 3.1. Consequently, if $E_d(N) = O_d(N^{1-\delta_d})$ with $\delta_d > 0$ for every d, the set of starts admitting a finite descent certificate has natural density 1.

Proof. Every E-rooted word has a contracting prefix at length one ($3^0 < 2$), so the starts with no contracting prefix of length $\leq d$ all realize an O-rooted itinerary of length d. An itinerary w of length d has a contracting prefix iff $3^{o_t} < 2^t$ for some $t \leq d$, where o_t counts odd letters among the first t. If w has no contracting prefix then $3^{o_d} \geq 2^d$, i.e. $o_d \geq \beta d$ with $\beta = \log 2/\log 3 = 0.6309\ldots$ The number of such words is at most $2^d \Pr[\mathrm{Bin}(d, \frac{1}{2}) \geq \beta d] \leq 2^d e^{-2(\beta-1/2)^2 d}$ by Hoeffding's inequality. Each such class has at most $2^{-d}N + E_d(N)$ members by the O-rooted hypothesis; summing gives the count. The density-one statement follows by letting $d \to \infty$ slowly with N (any $d(N) \to \infty$ with $2^d E_d(N)/N \to 0$). □

Sections 3–5 prove the hypothesis at every depth $d \leq 4$, so the conclusion of Proposition 7.1 is unconditional for those depths. Corollary 6.4 raises the certified class to certificate density 7/8 without counting every depth-5 word. The first open counting case is the *OOOO*∗ split. It has an exact shape, one nesting deeper than Theorem 5.3. Write $Z = v^{3/2}$, $z = \lfloor Z \rfloor$, $\theta_3 = Z - z$; after four odd letters the fifth is the parity of $\lfloor z^{3/2} \rfloor$.

Lemma 7.2 (level-3 kernel reformulation). For odd $n \geq 5$,

$$\tfrac{1}{2}\big(v^{9/4} - z^{3/2}\big) - \tfrac{3}{4}\,z^{1/2}\theta_3 = R_3, \qquad 0 \leq R_3 \leq \tfrac{3}{16}\,z^{-1/2}.$$

Consequently the fifth-letter phase is, up to $kR_3 \ll kn^{-27/16}$ per term, the exponential sum of the level-3 local floor defect, with coefficient $c = \frac{3k}{4} z^{1/2} \asymp kn^{27/16}$: Lemma 5.1(i) with (m, v) replaced by (v, z).

Proof. Taylor of $(z + \theta_3)^{3/2}$ at z, with $Z^{3/2} = v^{9/4}$. □

For smooth c with $c \asymp kP^{27/16}$ and $c' \asymp kP^{11/16}$ on $n \sim P$ (the $z^{1/2}$-shaped family), define

$$K_3(P) = \sum_{\substack{n \sim P \\ n \text{ odd}}} e\big(c(n)\,\{\lfloor\lfloor n^{3/2}\rfloor^{3/2}\rfloor^{3/2}\}\big).$$

Conjecture 7.3 (level-3 kernel cancellation). $K_3(P) \ll P^{1-\delta}$ for some $\delta > 0$, uniformly over the family above with $k \leq P^{\varepsilon}$.

The boundary between Theorem 5.3 and Conjecture 7.3 is quantitative and sharper than a derivative count. The smooth model iterates: where Theorem 5.3 used two Weyl differencings against $Y'' \asymp P^{1/4} \gg 1 > P^{-3/4} \asymp Y'''$, the level-3 model $n^{27/8}$ has $G''' \asymp P^{3/8} \gg 1 > P^{-5/8} \asymp G^{(4)}$ and predicts three. But the prediction does not descend to the nested floors. The kernel weight now has $c' \asymp kP^{11/16} \gg 1$, so no drift-1 interval exists for any expansion of the weight itself; the branch decomposition of Lemma 5.1(iii) has no analogue at the v-level, because v jumps by $\asymp n^{5/4}$ per step and the branch set is a *product* of two carry lattices, not a copy of one; and the forced inner linearization $v^{3/2} = m^{9/4} - \frac{3}{2}m^{3/4}\theta_2 + E_2$ trades θ_3 for a sawtooth family at coefficient scale $kn^{45/16}$, *above* the 9/4 threshold where every method of this paper stops.

What survives at the frontier is a model problem and one theorem about it. Stripping every problem-specific structure (carries, defects, nesting) from K_3 leaves the *amplitude-product* sums

$$S = \sum_{t \le L} e\big(A(t)\,\{B(t)\}\big), \qquad 1 \ll A' \ll A,$$

with smooth monomial-type A, B (the instance above has $A \asymp P^{27/16}$, $A' \asymp P^{11/16}$). For $A' \ll 1$ partial summation makes the amplitude a tame passenger and the classical single-floor machinery applies; for $A' \gg 1$ we know of no nontrivial deterministic bound by any method. What can be proved is an L^2 identity in the shift, a one-page computation that uses no harmonic analysis:

Proposition 7.4 (shift-averaged L^2 bound). Let $A_1 < \cdots < A_L$ be reals with $|A_t - A_{t'}| \ge A'_{\min}|t - t'|$ for some $A'_{\min} \ge 1$, and let $x_1, \dots, x_L$ be arbitrary reals. For $S_\lambda = \sum_{t \le L} e(A_t\{x_t + \lambda\})$,

$$\Big|\int_0^1 |S_\lambda|^2 \, d\lambda - L\Big| \le \frac{6}{\pi}\,\frac{L}{A'_{\min}}\,(\log L + 1).$$

Consequently, for any $\varepsilon \in (0,1)$, Markov's inequality gives

$$|S_\lambda| \le \sqrt{\frac{L}{\varepsilon}\Big(1 + \frac{6}{\pi}\,\frac{\log L + 1}{A'_{\min}}\Big)}$$

outside a shift set of measure at most ε. This is square-root cancellation times $\sqrt{\log L}$ in general, and genuine square-root cancellation when $A'_{\min} \gg \log L$ — which holds in the instance above, where $A'_{\min} \asymp P^{11/16}$.

Proof. Expand the square; the diagonal gives L. For $t \ne t'$, the function $\varphi(\lambda) = A_t\{x_t + \lambda\} - A_{t'}\{x_{t'} + \lambda\}$ is piecewise linear on $[0,1)$ with *real* slope $A_t - A_{t'}$ on at most three arcs (jumps at $1 - \{x_t\}$ and $1 - \{x_{t'}\}$); on each arc $|\int e(\varphi)\,d\lambda| \le 1/(\pi|A_t - A_{t'}|)$. Summing, $\sum_{t \ne t'} 3/(\pi A'_{\min}|t - t'|) \le (6/\pi)(L/A'_{\min})(\log L + 1)$. For the second display, the measure of $\{|S_\lambda|^2 > T\}$ is at most $T^{-1}(L + (6/\pi)(L/A'_{\min})(\log L + 1))$; take $T = (L/\varepsilon)(1 + (6/\pi)(\log L + 1)/A'_{\min})$. ☐

Two cautions. The amplitude separation $A' \gg 1$ — the very property that defeats every character expansion, since a Fourier window centred at the amplitude drifts by A' harmonics per step — is what makes the shift *average* trivial; but an average over shifts says nothing about the single shift $\lambda = 0$, which is the deterministic sum. Proposition 7.4 does not make the level-3 kernel "generically cancelling" in any sense that bears on Conjecture 7.3; it only locates the conjecture's difficulty as a specific-point problem inside a metric statement.

Conjecture 7.5 (the pure amplitude-product model). $|S| \le L^{1-\delta}$ for some $\delta > 0$, for smooth monomial-type A, B with $1 \ll A' \ll A$ as above.

Three structural facts locate the difficulty of the de-randomization. No second averaging variable exists: amplitude separation forces any two sample points with $|A(p) - A(q)| \le 1$ to coincide, and every family average available in the application re-enters either the differenced-kernel class or the amplitude-product class itself. The inverse theory is self-similar: any concentration or discrepancy inverse for $A\{B\}$ mod 1 is a statement about $\sum e(jA\{B\})$ — the same class with amplitude jA — so the class is closed under its own inverse theorems and no bootstrapping is possible. And the metric statement does not transfer: $|dS_\lambda/d\lambda| \le 2\pi A_{\max} L$ almost everywhere, so S_λ decorrelates at shift scale $1/A_{\max}$, and an almost-all-λ theorem leaves $\asymp \varepsilon A_{\max}$ exceptional cells among $\asymp A_{\max}$ — no measure argument pins $\lambda = 0$. The deterministic content of Conjecture 7.5 is a specific-point-in-metric-theory problem: a class whose known successes all use special arithmetic.

Two analytic shortcuts around Conjecture 7.3 fail at recorded points, and we state them once: composing gap cells across two levels fails because on a cell where the first-level gap is constant the second-level gap still takes a new value at essentially every point, so no usable sub-cell survives; and reindexing by the image m strips one nesting level but introduces a fiber indicator of density $\asymp m^{-1/3}$, whose sawtooth requires savings beyond the exponent $1/3$ while the engine saves only $1/24$.

The open question, stated once:

It is open whether almost every odd-to-odd start has a finite descent certificate. By Proposition 7.1 that would follow from all-depth parity equidistribution, which is now a theorem through depth four for odd-rooted itineraries. The certified class through length five has certificate density 7/8. The first open counting case is the $OOOO*$ kernel of Conjecture 7.3, whose deterministic model instance is Conjecture 7.5.

8. Relation to the Juggler map

The companion paper [24] reduces the termination problem to one statement about parity words and makes precise which statistics it needs. That paper's results bear on the program of this one in two opposite directions, and we record both.

Two localization facts, stated so they cannot be confused.

1. Section 3.5 proves the depth-≤ 3 discrepancy estimates (Theorems 4.11 and 4.12) on sub-dyadic intervals of length $\geq P^{1/2}$ with a slow twist. Corollary 4.13 is the only localization this paper supplies to [24].
2. Theorem 5.3 is a dyadic-block statement. A localized kernel theorem on intervals of length $P^{23/32}$ is not proved, and the strongest even-block productions that would need it (*OOOEEE*, *OOEOEE*) remain unavailable.

What each new depth buys. Every new certificate class enters the contagion recursion of [24] as a production: a word w of fair probability P_w landing at scale x^{e_w} contributes $(P_w/e_w)\,g(e_w t)$, provided its cylinder is counted on the preimage intervals of the landing points — intervals of length x^{1-e_w}, which is why Section 3.5 localizes Theorems 4.4 and 4.7 to sub-dyadic intervals. Through Corollary 4.13 the *OOEEE* production on even blocks raises the contagion exponent from $\lambda^{**} = 0.4480$ to $\lambda^{***} = 0.5392$, the rate threshold of the almost-all reformulation from 0.552 to 0.461, and the least depth constant of its conditional theorems from 20 to 18. Those are the dividends of the depth-≤ 3 localization. A localized form of the kernel theorem (Theorem 5.3 on intervals of length $P^{23/32}$, which we have not proved; the scaling architecture is the same, with per-window absolute costs at most $P^{7/16}$) would add the words *OOOEEE* and *OOEOEE* and give 0.5561; the level-3 kernel of Conjecture 7.3 would give more. Each depth also raises the certificate density of Corollaries 4.9 and 6.4 and the constants of the Tao-type reduction. These are the quantitative dividends of the program, and they are real. The strongest of them that this paper actually proves for [24] is the depth-≤ 3 localization, not a localized Theorem 5.3.

What no depth can buy. The frontier statement of [24] is that the odd starts in $(y, 2y]$ whose orbit is still above a fixed floor after $C \log_2 \log y$ steps number at most $y(\log y)^{-e}$ for some $e > 0.508$, where $C \geq 19$. Its weakest sufficient condition is an exponential moment of the odd count on live starts — a statement about the tilted average of parity words of length $\asymp \log\log y$, which is insensitive to bias at any $o(\log\log y)$ initial depths and to any single cylinder unless it is over-populated by an exponential factor. Consequently: (i) no fixed-depth theorem, this paper's or any other, is necessary for that statement — the depth-5 split of Conjecture 7.3 is irrelevant to it; (ii) no cylinder statement of bounded depth is sufficient for it, because a word measure fair to depth k and all-O afterwards satisfies every depth-$\leq k$ statement and violates the bound; and (iii) along the per-cylinder route, an analytic method whose saving exponent loses a factor 2^c per depth reaches the required depth only if $cC < 1$, i.e. $c < 1/19$, whereas Weyl differencing loses $c \geq 1$ (this paper's chain $\frac{1}{24} \to \frac{1}{96}$ from depth three to four is $c = 2$). The differencing machinery of Sections 3–6 cannot therefore be iterated to the termination frontier, however far the kernel program is pushed; what would be needed is a saving uniform in the depth up to a factor $2^{d/19}$, or a direct treatment of the exponential moment, neither of which this paper offers.

The honest summary is the one the abstract gives: this paper solves the first genuinely nested layers of the parity process of nested floor powers — depth four completely, depth five for the two contractors — with a bound whose critical piece is the level-2 wave; it identifies the level-3 kernel as the next analytic object; and through the depth-≤ 3 localization of Section 3.5 it supplies constants to the termination reduction of [24]. It does not localize Theorem 5.3, and it does not approach the infinite-depth problem that the termination question is.

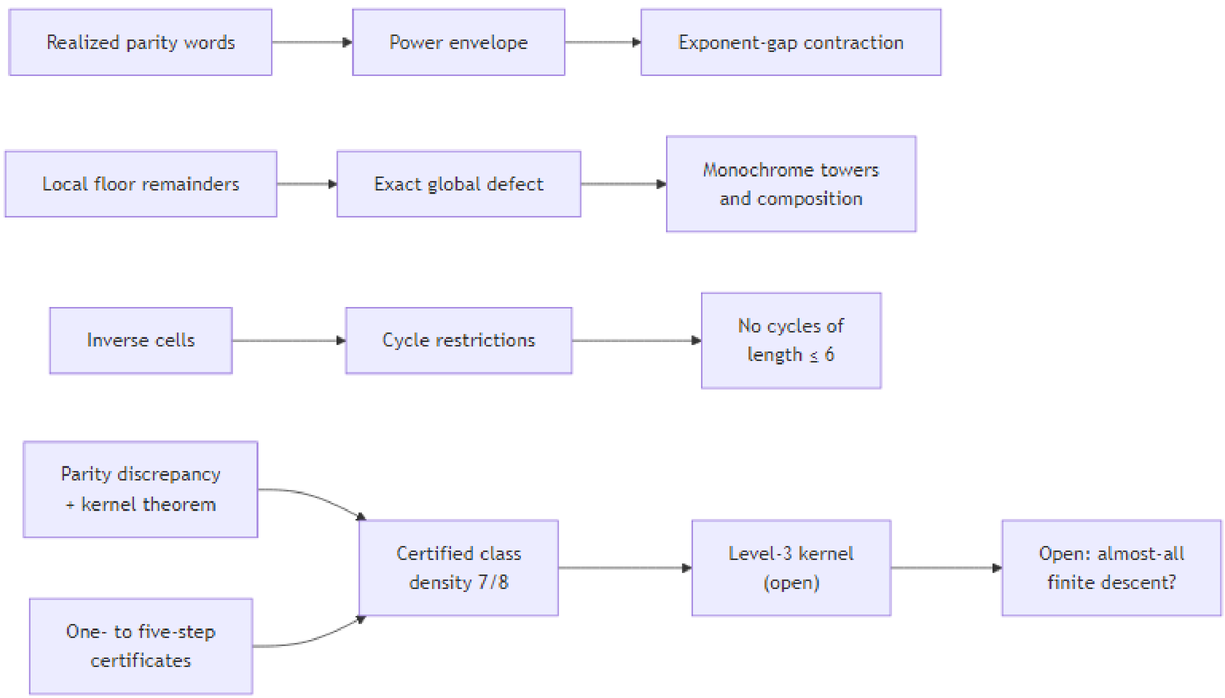

Figure 1: The theorem flow of the paper. The exact finite-itinerary calculus of the companion manuscript feeds the contraction certificates; the discrepancy calculus with the kernel theorem counts every O-rooted itinerary class through depth four and the two length-five contractors (certified-descent density 7/8), leaving the level-3 kernel — and with it almost-all descent — open.

A repository accompanies the paper (https://github.com/sneakyweasel/balanced_ternary/). It is not required to read or check any proof. Lean certificates cover the exact floor identities; the module `research.juggler_sequence.paper_b_audit` and the companion paper_b_audit_ledger.md record numerical and exponent checks. Those are not proofs and are not an independent verification of Lemma 5.2.

Acknowledgments

I used large language models extensively while drafting and revising the text, organizing companion notes, and as an interactive assistant for Lean statements, tests, and literature records. The models are not authors. The theorems of this paper are human proofs from the cited classical inequalities; the Lean certificates cover only the exact floor identities and the companion's finite-itinerary theorems. I take full responsibility for the contents.

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